S-two Whitepaper*

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Preface

This whitepaper describes *S-two*, the second generation proof system for StarkNet. S-two is a *circle STARK* [HLP24], that is, a hash-based argument of knowledge over the prime field with modulus

$$p = 2^{31} - 1,$$

a Mersenne prime which admits exceptional efficient field arithmetics on conventional computer architectures. As any present-day proof system, a circle STARK is not restricted to a specific model of arithmetic circuits used to express the computation to be proven. We consider *flat algebraic intermediate representations*, a modern arithmetization paradigm for zero-knowledge virtual machines [Ope,Irr,Suc] which is also pursued by S-two. The arithmetization paradigm breaks with the regular monolithic layout of execution traces. Instead, it facilitates a decomposition into a network of micro components, flat specific-purpose gates, which are interconnected by a universal communication channel, a bus. Flat arithmetization enables fine-grained modularity on logical level, and reduces the cost of operation multiplexing.

In a nutshell, S-two is a circle STARK for flat algebraic intermediate representations, which uses the FRI low-degree test [BBHR18] in the *multi-table* setting, adapted to the circle and serving ensembles of execution tables with different heights. The main goal of this document is the description of the protocols as they are implemented in our repositories [Stab] [Staa], backed by thorough security proofs.

The first part of the paper is of preliminary character. In Section 1 we recap basic properties of the circle curve over Mersenne prime fields, and outline the *circle FFT*, the fast interpolation algorithm by *circle polynomials*. Although often confused with related trigonometric transforms, the circle FFT is an algebraic interpolation algorithm in the style of the elliptic curve FFT [BCKL21], and it achieves less operation counts than their harmonic counterparts. Then, Section 2 introduces flat algebraic intermediate representations. We outline how

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the execution of *Cairo* programs is arranged in [Staa], and we further sketch how the paradigm can be used to describe static circuits. This section requires some background on arithmetic circuits and the design of zero-knowledge virtual machines, in order to understand the usability of our rather abstract definition.

The second part is the core of the writeup. In Section 3 and 4 we describe the underlying protocols as interactive oracle proofs of proximity, and state their security properties. Then, in Section 5 we discuss the non-interactive STARK. We go into the details of our multi-domain Merkle tree and outline the Fiat–Shamir transform, in its concrete form the Ben-Sasson–Chiesa–Spooner transform [BCS16]. We discuss soundness of the non-interactive argument, and provide example parameters for both the proven and the conjectured regime, taking into account our improved understanding of proximity gaps [BCH+25,CS25,DG25,KK25]. Finally, Section 6 provides a quick overview of additional features planned for future releases of S-two.

While security of our protocol is discussed in the main body of the paper, the proofs are postponed to the Appendix A. At this point we stress the fact, that in the multi-table setting soundness requires careful consideration, if one runs FRI beyond the unique decoding regime. The reason for this is a combinatorial blow-up of the number of proximates under regular agreement assumptions, and is explained in the introduction of Section 3.5. To mitigate this blow-up, we introduce the notion of *cross-domain correlated agreement*, and we show that multi-table FRI enforces this property. The proof, which is one of the main contributions of the writeup, is quite technical and centers on the notion of *curve-decodability* [GG25], previously dubbed "collinearity of proximates property" in [Hab24a,Hab25]. Since multi-table FRI is widely used in practice, we plan to release a non-circle variant of the soundness analysis in a separate paper [HHM25].

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Table of Contents

1 Preliminaries

In this section we introduce the notation used throughout this document, and recap elementary facts about the circle curve and circle polynomials.

1.1 Mersenne field and its extensions

We shall write $\mathsf{M31} = \mathbb{F}_q$ for the finite field[3] of prime order $q = 2^{31} - 1$. This field is the central field for S-two. It is used to model the trace via an algebraic intermediate representation over M31, and constraints between elements will be expressed by algebraic expressions using the finite field operations, i.e. multiplication and addition. (See Section 2.) However, for soundness of the proof system, it is crucial to make use of larger fields, extending M31. The current release of S-two uses a degree-4 towered extension

$$\mathbb{F}_q \subset \mathbb{F}_{q^2} \subset \mathbb{F}_{q^4},$$

as defined below. The choice of $\mathbb{F}_{q^2}$ as an intermediate field is for convenience, as it simplifies certain protocol steps, and comes with several useful properties.

Complex extension The *complex extension* $\mathsf{CM31} = \mathbb{F}_{q^2}$ is constructed as

$$\mathbb{F}_{q^2} = \mathbb{F}_q[\zeta]/(\zeta^2 + 1). \tag{1}$$

(Note that $\zeta^2 + 1$ is irreducible over $\mathbb{F}_q$, as its multiplicative group is of order $q - 1 = 2 \cdot u$ with odd u.) M31, denoted by CM31, where We write $\zeta = i = \sqrt{-1}$ and elements of the complex extension as

$$z = a + i \cdot b, \quad a, b \in \mathbb{F}_q.$$

The Frobenius automorphism $A_q(x) = x^q$ leaves the elements of $\mathbb{F}_q$ fixed, and sends i to $-i$. In other words, it is the conjugation map $a + i \cdot b \mapsto a - i \cdot b$.

Quartic extension The quartic extension $\mathsf{QM31} = \mathbb{F}_{q^4}$ is obtained from the complex extension as

$$\mathbb{F}_{q^4} = \mathbb{F}_{q^2}[\chi]/(\chi^2 - 2 - i), \tag{2}$$

by adjoining a square root $j = \sqrt{2 + i}$. (Again, the polynomial $\chi^2 - 2 - i$ is irreducible over the complex extension, since $2 + i$ is a not a square in the complex extension, which can be seen from

$$\begin{aligned}(2+i)^{\frac{q^2-1}{2}} = (2+i)^{(q+1)\cdot\frac{q-1}{2}} &= \big((2-i)\cdot(2+i)\big)^{\frac{q-1}{2}} \\ &= 5^{2^{30}-1} = -1,\end{aligned}$$

where we used the fact that $(2 + i)^q = 2 - i$.) We write elements of QM31 as

$$w = (a_0 + i \cdot b_0) + j \cdot (a_1 + i \cdot b_1),$$

with a_0, b_0 and a_1, b_1 from $\mathbb{F}_q$.

[3] M31 is also short for the Messier 31 galaxy, Andromeda. We elaborate the connection between Andromeda and the circle STARK elsewhere.

1.2 The circle curve

The circle curve over finite fields of characteristic q is the planar curve defined by

$$C : x^2 + y^2 = 1 \mod q, \tag{3}$$

or more generally

$$C : x^2 + y^2 - z^2 = 0 \mod q,$$

in homogeneous projective coordinates. Depending on the context, the curve will be either viewed over the basefield $\mathbb{F}_q = \mathsf{M31}$, the quartic extension $\mathsf{QM31}$, and occasionally also over the intermediate field $\mathsf{CM31}$. Put in formal words, we consider the set of F-rational points

$$C(F) = \{(x, y) \in F^2 \ : \ x^2 + y^2 = 1\},$$

and likewise in projective coordinates, for different extension fields F of $\mathbb{F}_q$. Projective coordinates seem artificial at first glance, since over $\mathbb{F}_q$ there are no points at infinity, i.e. points of the form $(x : y : 0)$ in homogeneous projective coordinates. That picture changes once the extension F contains the root i, the square root of -1, in which we capture the two points the curve has at infinity,

$$\infty = (1 : +i : 0), \quad \bar{\infty} = (1 : -i : 0).$$

The curve can be entirely described in univariate coordinates, which is a helpful tool in both understanding the curve, as well as the security analysis of the circle STARK.

Lemma 1. *Let F be an arbitrary extension field of $\mathbb{F}_q$. The rational function*

$$\phi(t) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2 \cdot t}{1 + t^2}\right)$$

defines a bijective map between the one-dimensional projective space $P^1(F)$ over F and the set F-rational points on the projective curve C.

The parametrization in Lemma 1 is obtained from the stereographic projection onto the y-axis and with the point $(-1, 0)$ as its center [HLP24, Lemma 1], the map $(x, y) \mapsto y/(1 + x)$.

Group structure. The circle curve is invariant under the group of rotations

$$G = SO(2, \mathbb{F}_q) = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in \mathbb{F}_q^{2\times 2} \ : \ a^2 + b^2 = 1 \right\}, \tag{4}$$

and this group plays a central role in the circle FFT. Its acts faithfully and transitively on $C(\mathbb{F}_q)$, meaning that for any point $(x, y) \in C(\mathbb{F}_q)$,

$$G \longrightarrow C(\mathbb{F}_q), \quad T \mapsto T((x, y)),$$

is injective, and the image is equal to $C(\mathbb{F}_q)$. By means of this action, the circle $C(\mathbb{F}_q)$ inherits a group structure, which is even cyclic, since it does not contain a square root of -1.

Lemma 2. *For any prime $q \equiv 3 \bmod 4$ the group $G = SO(2, \mathbb{F}_q)$ is cyclic and of order $q + 1$.*

Proof. The map

$$SO(2, \mathbb{F}_q) \longrightarrow \mathbb{F}_{q^2}, \quad \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \mapsto a + i \cdot b$$

is an injective homomorphism of G onto a multiplicative subgroup of $\mathbb{F}_{q^2}$, the unit circle $\{z \in \mathbb{F}_{q^2}\ z \cdot \bar{z} = 1\}$ in the complex extension of $\mathbb{F}_q$. Since the multiplicative group of a finite field is cyclic, so is G. The claimed order follows immediately from $|G| = |C(\mathbb{F}_q)|$ and Lemma 1. □

The group structure extends to any extension field F of $\mathbb{F}_q$. In this case $SO(2, F)$ acts faithfully and transitively on the affine points $C(F) \setminus \{\infty, \bar{\infty}\}$, with the points at infinity as fixed points.

Notation. From now on, we identify $C(\mathbb{F}_q)$ with $G = SO(2, \mathbb{F}_q)$, call it the *circle group*, and we write the group operation in multiplicative manner as

$$(x, y) \cdot (x', y') := (x \cdot x' - y \cdot y', x \cdot y' + x' \cdot y). \tag{5}$$

Under this convention, the neutral element is the point $Q_0 = (1, 0)$, and

$$C(\mathbb{F}_q) \longrightarrow \mathbb{F}_{q^2}, \quad (x, y) \mapsto x + i \cdot y$$

is the injective group homomorphism from Lemma 2, which maps the circle group onto the multiplicative subgroup of complex units in $\mathbb{F}_{q^2}$. For any $0 \leq n \leq 31$ we denote by

$$G_n \subset C(\mathbb{F}_q)$$

the unique cyclic subgroup of order $|G_n| = 2^n$, and these subgroups form a chain under the group squaring map $\pi(x, y) = (x, y) \cdot (x, y)$,

$$G_{31} \xrightarrow{\pi} G_{30} \xrightarrow{\pi} \ldots \xrightarrow{\pi} G_1, \tag{6}$$

with each of the mappings being 2-to-1 and onto. For each of these subgroups we choose Q_k, a generator of G_k, consistent with respect to π, so that

$$Q_{31} \xrightarrow{\pi} Q_{30} \xrightarrow{\pi} \ldots \xrightarrow{\pi} Q_1. \tag{7}$$

Such a projection-consistent choice of generators is useful in the implementation of the circle FFT, as well as our cross-domain Merkle tree in Section 5. S-two uses

$$Q_{31} = (2, 1268011823),$$

as a generator for the largest cyclic subgroup $G_{31} = C(\mathbb{F}_q)$ over $\mathbb{F}_q = \mathsf{M31}$.

1.3 Circle polynomials and FFT

Let F be a finite extension field of $\mathbb{F}_q = \mathsf{M31}$. The *circle polynomials* over F are the bivariate polynomials modulo the circle relation $x^2 + y^2 = 1$. Formally, they are the elements from the ring

$$\mathcal{L}(F) = F[x, y]/(x^2 + y^2 - 1), \tag{8}$$

where $(x^2 + y^2 - 1)$ denotes the ideal generated by $x^2 + y^2 - 1$, that is the set of all polynomial multiples of $x^2 + y^2 - 1$. However, we rarely make use of this ambiguity, and throughout work with the unique *canonical representative*

$$p(x, y) = p_0(x) + y \cdot p_1(x),$$

with $p_0(x), p_1(x)$ from $F[x]$, which is obtained by repeated application of the substitution rule $y^2 = 1 - x^2$. We shall call $p_0(x)$ and $p_1(x)$ the *canonical components*, and we define the degree of a circle polynomial as the total degree

$$d = \deg p(x, y) := \max \left\{ \deg p_0(x), 1 + \deg p_1(x) \right\} \tag{9}$$

of its canonical representative.

Remark 1. This definition of degree is rather uncommon, since it relies on the explicit representation of the curve. From the geometric point of view, circle polynomials are exactly those rational functions which have poles only at ∞ and $\bar{\infty}$, each of order at most d. The usual definition would be by the degree of its divisor, in our case the sum of the pole multiplicities.

Function space. The function targeted by the circle FFT are the spaces of circle polynomials of given degree bound,

$$\mathscr{L}_k(F) := \left\{ p(x, y) \in \mathcal{L}(F) \;:\; \deg p \leq 2^{k-1} \right\}, \tag{10}$$

where $k \leq 31$ is a power of two. In the context of Remark 1, $\mathscr{L}_n(F)$ is the space of all rational functions which have poles only at ∞ and $\bar{\infty}$, and each of them has order at most $2^k/2$. From the canonical representation of circle polynomials it is immediate that

$$\dim \mathscr{L}_k(F) = 2^k + 1,$$

which is similar to the univariate case.

Fast Fourier transform. The *circle FFT* is an algebraic transform in the style of the elliptic curve FFT [BCKL21], yet significantly simpler in its construction. Similar to a regular Fourier transform, the 2-to-1 group squaring homomorphism

$$\pi(x, y) = (x^2 - y^2, 2 \cdot y \cdot x) = (2 \cdot x^2 - 1, 2 \cdot y \cdot x), \tag{11}$$

plays a central role in the construction. The k-th power of the squaring map, $k \geq 0$, is of the form

$$\pi^k(x, y) = (v_{k+1}(x), y \cdot w_{k+1}(x)), \tag{12}$$

with polynomials $v_k(x), w_k(x) \in \mathbb{F}_q[x]$, $k \geq 1$, of degree

$$\deg v_k(x) = \deg y \cdot w_k(x) = 2^{k-1}. \tag{13}$$

Remark 2. In terms of the classical theory of Fourier transforms, both $y \cdot w_k(x)$ and $v_k(x)$ are real projections of the complex harmonics, i.e. trigonometric functions. These functions will be components of the specific polynomial basis of the circle FFT defined below, yet the basis will non-trigonometric.

Note that for $k = 1, \ldots, 30$, the polynomial $y \cdot w_k(x)$ has the cyclic subgroup $G_k \subset C(\mathbb{F}_q)$ as its zero set, and $v_k(x)$ the complementing coset

$$G'_k := Q_{k+1} \cdot G_k, \tag{14}$$

that is the unique coset of G_k so that $G_k \cup G'_k = G_{k+1}$. In terms of univariate coordinates, using the parametrization from Lemma 1, each of these zeros are simple.

Definition 3. For any $1 \leq k \leq 30$ the coset G'_k is called *canonical coset*, and the polynomial $v_k(x) \in \mathscr{L}_k(\mathbb{F}_q)$ is called its *vanishing polynomial*.

As a regular FFT, the circle FFT is a divide-and-conquer algorithm, which stepwise reduces the interpolation task (or, the inverse evaluation task) into two instances of half the size. However, the concrete form of reduction steps differs from those used by classical trigonometric transforms, yielding a novel "real" transform over the circle group. The first step is subject to the decomposition into canonical components,

$$p(x, y) = p_0(x) + y \cdot p_1(x),$$

where

$$\begin{aligned} p_0(x) &= \frac{p(x, y) + p(x, -y)}{2}, \\ p_1(x) &= \frac{p(x, y) - p(x, -y)}{2 \cdot y}, \end{aligned} \tag{15}$$

correspond to the "even" and "odd" parts with respect to the group inverse map,

$$J(x, y) = (x, -y). \tag{16}$$

This step is performed over a J-invariant coset, which are bound to be of the form (14), i.e. canonical cosets G'_n, with $1 \leq n \leq 30$. The algorithm then proceeds on $p_0(x)$ and $p_1(x)$ recursively, using the decomposition based on the group squaring homomorphism,

$$q(x) = q_0(2 \cdot x^2 + 1) + x \cdot q_1(2 \cdot x^2 + 1).$$

with

$$
\begin{aligned}
q_0(2\cdot x^2+1) &= \frac{q(x)+q(-x)}{2},\\
q_1(2\cdot x^2+1) &= \frac{q(x)-q(-x)}{2\cdot x}.
\end{aligned}
\tag{17}
$$

(The y-coordinates vanish by the independence of the expressions from y.) These steps are over the repeatedly squared domains, each half the size of the previous, and are performed over their projections S_{n-1}, S_{n-2}, ..., onto the x-axis, to leverage the independence of y. Recursion stops with the singleton domain S_0, over which interpolation (or, for the reverse task, evaluation) is trivial.

$$
\begin{tikzcd}
G'_n \arrow[r, "\pi"] \arrow[d, "\pi_x"] & G'_{n-1} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & G'_{n-2} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & \cdots\cdots\cdots \arrow[r, "\pi"] & G'_1 \arrow[d, "\pi_x"] \\
S_{n-1} \arrow[r, "\pi"] & S_{n-2} \arrow[r, "\pi"] & S_{n-3} \arrow[r, "\pi"] & \cdots\cdots\cdots \arrow[r, "\pi"] & S_0
\end{tikzcd}
\tag{18}
$$

Remark 4. We stress the fact that step (17) is fundamentally different to that used by classical real transforms over the circle, which essentially proceed with the regular FFT in the complex plane, optimized for real inputs.

We cite the main theorem on the circle FFT as an interpolation algorithm. There, $i_0, \ldots, i_{n-1}$ are the bits of the integer index $i \in [0, 2^n)$, and the $v_k(x)$ are as above.

Theorem 5 ([HLP24]). *Given a set of values $f \in F^{G'_n}$ over the canonical coset G'_n, the circle FFT computes an interpolating circle polynomial $p(x, y) \in \mathscr{L}_n(F)$ represented as*

$$
p(x,y) = \sum_{i=0}^{N-1} c_i \cdot y^{i_0} \cdot v_1(x)^{i_1} \cdot v_2(x)^{i_2} \cdot \ldots v_{n-1}(x)^{i_{n-1}},
\tag{19}
$$

where the coefficients $c_i \in F$, $i \in [0, 2^n)$, are the output of the circle FFT. Conversely, the evaluation FFT takes a vector of coefficients $(c_i)_{i=0}^{N-1} \in F^{2^n}$, and outputs the values of the polynomial (19) over G'_n.

An explicit description of the interpolation and evaluation FFT as an iterative algorithm, with in-place modifications of the given data vector, is given in Appendix B. Counting subtractions as additions, both algorithms[4]consume $\frac{N}{2} \cdot \log(N)$ multiplications with precomputed elements from $\mathbb{F}_q$, and $N \cdot \log(N)$ additions in F, which is on par with the operation counts of a regular multiplicative FFT.

[4] This applies to a variant of the interpolation FFT, which outputs scaled coefficients. Non-scaled interpolation requires a further normalization step. For details, see Appendix B.

Polynomial basis. The interpolating polynomial determined by the circle FFT is given with respect to the basis $\mathscr{B}_n$ comprised by the polynomials

$$b_i(x,y) = y^{i_0} \cdot v_1(x)^{i_1} \cdot v_2(x)^{i_2} \cdot \ldots v_{n-1}(x)^{i_{n-1}}, \quad i \in [0, 2^n), \tag{20}$$

where $i = i_0 + i_1 \cdot 2 + \ldots + i_{n-1} \cdot 2^{n-1}$ is the bit representation of i. Using equation (13) the degree of these polynomials is bounded as

$$\begin{aligned} \deg b_i(x,y) &= i_0 + i_1 + 2 \cdot i_2 + \ldots + 2^{n-2} \cdot i_n \\ &\le 1 + 2^{n-1} - 1 = 2^{n-1}, \end{aligned}$$

and the basis $\mathscr{B}_n$ spans the 2^n-dimensional subspace

$$\mathscr{L}'_n(F) := F[x]^{<2^{n-1}} + y \cdot F[x]^{<2^{n-1}} \tag{21}$$

of $\mathscr{L}_n(F)$, which we call the *circle FFT space*. The gap between these spaces is one-dimensional, and spanned by the vanishing polynomial of the FFT domain,

$$\mathscr{L}_n(F) = \mathscr{L}'_n(F) + \langle v_n(x) \rangle, \tag{22}$$

with $\langle v_n(x) \rangle$ being all scalar multiples of $v_n(x)$. This is easily seen from that $\deg v_n(x) = 2^{n-1}$, with its leading term not contained in $F[x]^{<N/2}$. (This decomposition is orthogonal in a suitable sense, see [HLP24, Section 4.3].)

Extension to twin-cosets. The invariance of FFT domains with respect to the inverse map J allows to operate only on a single coset of given size. To widen the notion of domain targets, [HLP24] defines the circle FFT over *twin-cosets*

$$S'_n = Q \cdot G_{n-1} \cup Q^{-1} \cdot G_{n-1},$$

with $Q^2 \notin G_{n-1}$, and where $1 \le n \le 30$. Twin-cosets consist of two disjoint cosets of G_{n-1} the subgroup half the size, which are mapped to one another under J. The chain of domains is as in (18), but with the circle domains in the upper row being throughout twin-cosets S'_k, $\pi(S'_k)$, $\pi^2(S'_k)$, $\ldots$, the sizes of which are halved with each application of the group squaring map. The Fourier transform itself remains unchanged, with the decomposition into canonical components in the first step (15), and the decomposition (17) using the group squaring map in all other steps. We note that S-two avoids twin-cosets as evaluation targets. For evaluating low-degree polynomials over larger domains, we instead implement a simple tweak of the FFT over larger canonical cosets. See Appendix B.

Related FFTs. As already mentioned in Remark 2 and 4, the circle FFT is closely related to trigonometric transforms. Restricting it to the line subsets

$$S_{n-1} \longrightarrow S_{n-2} \longrightarrow \ldots \longrightarrow S_0$$

results in the same chain of domains and even the same interpolating polynomial as the finite field variant of the *Cosine transform* and the *Chebyshev transform*.

(The line subsets are called *Chebishev nodes of the first kind.*) However, the polynomial bases used for representing the interpolating polynomial are different. Both Cosine and Chebyshev transform represent the interpolant with respect to the basis of $v_k(x)$, of all orders $k = 0, \ldots, N-1$, whereas the circle FFT basis consists of products of these functions. Compared to the real versions of the multiplicative FFT over $\mathbb{F}_{q^2}$ used in [HLN23,Dom24] and [DH24], the fast cosine number transform [Lim15], or the Galois transform [LX23] (see also [Hab24b] for a more elementary exposition) the circle FFT has the lowest arithmetic operation counts.

1.4 Circle Codes

In this section we recap basic properties of circle codes generated by circle polynomials. We assume that the reader is familiar with the notions of linear codes, minimum distance and unique decoding radius.

Definition 6 (Circle Codes)**.** Let F be any finite extension of $\mathbb{F}_q = \mathsf{M31}$, and assume that $1 \leq n \leq 30$. Given an arbitrary subset $D \subseteq C(\mathbb{F}_q)$ of size $|D| > N$, where $N = 2^n$, the *circle code of order* n is defined by the values of all circle polynomials of degree at most $N/2$,

$$\mathscr{C}_n(F, D) := \left\{ p(x,y)|_{(x,y)\in D} \,:\, p \in \mathscr{L}_n(F) \right\}.$$

The circle code is a linear code over F of length $|D|$ and dimension $N+1$, and has rate

$$\rho^+ = \frac{N+1}{|D|}. \tag{23}$$

Circle codes are essentially regular Reed-Solomon codes. This can be seen by the univariate parametrization ϕ of the circle, Lemma 1, which translates $\mathscr{L}_n(F)$ to the univariate representation

$$L_N(F) = \frac{F[t]^{\leq N}}{(1+t^2)^{N/2}},$$

and $\mathscr{C}_n(F, D)$ to the *generalized Reed-Solomon code*

$$\frac{1}{(1+t^2)^{N/2}} \cdot \mathsf{RS}[F, \phi^{-1}(D), N+1]$$

generated by $F[t]^{\leq N}$, the space of all polynomials of degree at most N, and the scaling function $1/(1+t^2)^{N/2}$, over the evaluation domain $\phi^{-1}(D)$. (For simplicity the reader might think of D not containing $(-1, 0)$, which corresponds to univariate infinity. However, all in this section also holds under this exception.)

Using the univariate representation one sees that the circle code has minimum distance

$$\delta_0(\mathscr{C}_n) = \min_{f,g\in\mathscr{C}_n} d(f,g) = 1 - \frac{N}{|D|}, \tag{24}$$

where

$$d(f,g) = \frac{1}{|D|} \cdot |\{x \in D \,:\, f(x) \neq g(x)\}| \tag{25}$$

is the fractional Hamming distance over D. (This renders circle codes as maximum distance separable.) Furthermore, the univariate representation allows for carry over all Reed-Solomon tools, which are crucial for the security of STARK to circle codes: the Guruswami–Sudan error correcting list decoder [GS99] (or, up to unique decoding radius, the Berlekamp-Welch decoder [WB86]) and the correlated agreement theorems from [BCI+20] addressed in Appendix A.2.

Theorem 7 (Circle code list decoding)**.** *Let $\mathscr{C} = \mathscr{C}_n(F,D)$ be a circle code over $F \geq \mathbb{F}_q$ with arbitrary evaluation domain $D \subseteq C(\mathbb{F}_q)$, with rate $\rho = \frac{N+1}{|D|}$ and minimum distance $\delta = 1 - N/|D|$, where $N = 2^n$. Then $\mathscr{C}$ is list decodable for distances θ up to the Johnson radius $J(\delta) = 1 - \sqrt{1-\delta}$, with list sizes at most*

$$\ell(\eta) = \frac{1}{2 \cdot \sqrt{1-\delta}} \cdot \frac{1}{\eta},$$

where $\eta = \theta - \sqrt{1-\delta}$. Moreover there is a deterministic algorithm (the Guruswami–Sudan decoder) which outputs all θ-close codewords within less than $O(|D|^4)$ field operations in F.

Proof. By the combinatorial Johnson bound, the number of θ-proximate codewords is at most

$$\ell(\eta) = \frac{\alpha - (1-\delta)}{\alpha^2 - (1-\delta)} < \frac{1}{2\sqrt{1-\delta}} \cdot \frac{1}{\eta},$$

where δ is the minimum distance of the code, and $\alpha = 1 - \theta$.

To obtain the codewords, apply the Guruswami–Sudan list decoder to the scaled word, indexed over the univariate domain $\phi^{-1}(D)$, with multiplicity parameter $m = \left\lceil \frac{1}{2\eta} \right\rceil$. It outputs all θ-close polynomials from $F[t]^{\leq N}$, represented in the monomial basis. Evaluate them over $\phi^{-1}(D)$, undo the scaling, and transform the output back to circle. Using the deterministic version of the algorithm as described in [GS99], a crude upper bound of the worst-case computational cost is $O_{\delta_0}(\frac{1}{\eta} \cdot |D|^3)$ field operations in F, independent of the field size, considered that $|F| \leq 2^{|D|}$. The few operations for the above scaling, as well as its final re-scaling, do not change the asymptotics. □

Remark 8 (Circle code unique decoding)**.** Likewise, for distances up to the unique decoding radius $\delta_0/2$, error-correcting decoding can be done by the Berlekamp-Welch decoder, which consumes at most $O(|D|^5)$ field operations.

2 Flat AIR

A *flat algebraic intermediate representation (flat AIR)* over a finite field $\mathbb{F}_q$ is the assignment of a specifically arranged arithmetic circuit over $\mathbb{F}_q$. It consists of a specified number of *component tables*

$$\mathscr{T}_t \in \mathbb{F}_q^{N_t \times M_t}, \quad t = 1, \ldots, \tau,$$

where each such component table records the wire states of multiple instances of its *flat component*, a specified arithmetic circuit used to process messages from a *virtual communication channel* $\mathcal{L}$. In other words, a flat AIR describes a message processor consisting of several components. Messages are vectors over $\mathbb{F}_q$ of specified lengths and format, and they are exchanged between the components via the virtual channel $\mathcal{L}$. Components take messages from $\mathcal{L}$, transform them into new messages according to specified arithmetic rules, and send the transformed messages back to $\mathcal{L}$. The concrete routing, which defines which message is picked up by which component, is implemented on application layer, and we will provide several examples in the course of the section. Flat AIR also support lookups via separate *lookup components*, and hence some of the tables may contain public preprocessed columns.

2.1 Components.

Each table $\mathscr{T}_t \in \mathbb{F}_q^{N_t \times M_t}$ of the flat AIR is subject to its own flat *component*, an arithmetic *row gate*

$$\mathsf{Comp}_t[Y_1, \ldots, Y_{R_t}, X_1, \ldots, X_{M_t}, Z_1, \ldots, Z_{S_t}],$$

in the *witness variables* $X_1, \ldots, X_{M_t}$, *input variables* $Y_1, \ldots, Y_{R_t}$, and *output variables* $Z_1, \ldots, Z_{S_t}$, dedicated to a specified *component relation*

$$\mathcal{R}_t \subseteq \mathbb{F}_q^{R_t + M_t + S_t},$$

between inputs, outputs and witnesses, by the means explained below. The inputs and output variables are *virtual*, that is, determined from the witness variables by specified polynomials

$$\begin{aligned}(Y_1, \ldots, Y_{R_t}) &= \vec{U}_t(X_1, \ldots, X_{M_t}),\\ (Z_1, \ldots, Z_{S_t}) &= \vec{V}_t(X_1, \ldots, X_{M_t}),\end{aligned}$$

with $\vec{U}_t, \vec{V}_t \in \mathbb{F}_q[X_1, \ldots, X_{M_t}]^*$. (Here, and in the sequel, we write $\mathbb{F}_q[X_1, \ldots, X_{M_t}]^*$ for the collection of all polynomial vector expressions $\bigcup_{k=0}^{\infty} \mathbb{F}_q[X_1, \ldots, X_{M_t}]^k$.) Each row of $\mathscr{T}_t$ corresponds to a separate instance of the row gate, with assigment $(X_1, \ldots, X_{M_t}) = (\mathscr{T}_t[i,1], \ldots, \mathscr{T}_t[i, M_t])$, and

$$\begin{aligned}\vec{u}_t[i] &= \vec{U}_t(\mathscr{T}_t[i,1], \ldots, \mathscr{T}_t[i, M_t]),\\ \vec{v}_t[i] &= \vec{V}_t(\mathscr{T}_t[i,1], \ldots, \mathscr{T}_t[i, M_t]),\end{aligned}$$

as the messages exchanged with $\mathcal{L}$. Each input message $\vec{u}_t[i]$ is called a *used* message, and each output $\vec{v}_t[i]$ a *yielded* message, and we write

$$\vec{u}_t[i] \leftarrow \mathcal{L}, \quad \vec{v}_t[i] \rightarrow \mathcal{L}.$$

to distinguish used from yielded messages. Membership of

$$\vec{u}_t[i] \parallel (\mathscr{T}_t[i,1], \ldots, \mathscr{T}_t[i, M_t]) \parallel \vec{v}_t[i] \quad \in \mathcal{R}_t$$

for each i, is achieved by a combination of arithmetic constraints described as follows.

Arithmetic constraints. The arithmetic constraints are specified polynomials in the gate variables $X_1, \ldots X_{M_t}$,

$$C_{t,k} \in \mathbb{F}_q[X_1, \ldots, X_{M_t}], \quad k = 1, \ldots, K_t, \tag{26}$$

which are imposed on every row of the component table,

$$C_{t,k}(\mathscr{T}_t[i,1], \ldots, \mathscr{T}_t[i, M_t]) = \vec{0},$$

for $i \in [1, N_t]$ and $k = 1, \ldots, K_t$.

Enabler multiplicities. For the sake of padding, all uses and yields of a component are equipped with binary multiplicities

$$m_t[i] \in \{0, 1\},$$

taken from an *enabler variable* m_t, a specified witness variable of the gate. We assume that the arithmetic constraints (26) of the component cover the Boolean constraint $e_t \cdot (1 - e_t) = 0$. Enablers are used to block, or let pass, the messages from and to $\mathcal{L}$, and we write

$$e_t[i] * \vec{u}_t[i] \leftarrow \mathcal{L}, \quad e_t[i] * \vec{v}_t[i] \rightarrow \mathcal{L}.$$

2.2 Lookup components.

Lookup components are helper components that are used as a plug-in within other components. They are components with no inputs,

$$\mathsf{Comp}_t[X_1, \ldots, X_{M_t}, Z_1, \ldots, Z_{S_t}],$$

and their yielded messages $\vec{v}_t[i]$, called *lookup messages*, allow unconstrained multiplicities

$$m_t[i] \in [0, p-1],$$

which declare how often the message is replicated in $\mathcal{L}$, again denoted by $m_t[i] * \vec{v}_t[i] \rightarrow \mathcal{L}$. Often, lookup components consist only of public preprocessed columns and do not have arithmetic constraints. Their yielded messages are used by other components via *auxiliary gate inputs*, which are introduced in the same way as ordinary inputs. (Our abstract definition below will not distinguish between the two type of inputs.) Also lookup components may have auxiliary inputs. We allow lookup components to rely on other lookup components, as long as cyclic dependencies are avoided.

2.3 Public inputs and channel consistency

The inputs of the flat AIR are a set of specified messages $\vec{I}_1, \ldots, \vec{I}_{k_{in}}$ yielded to the channel (each of them of separate specified length), while its outputs $\vec{O}_1, \ldots \vec{O}_{k_{out}}$ are a specified messages used from the channel,

$$\vec{I}_1, \ldots \vec{I}_{k_{in}} \rightarrow \mathcal{L}, \quad \vec{O}_1, \ldots \vec{O}_{k_{out}} \leftarrow \mathcal{L}.$$

As before, the concrete routing of the messages to their target component is implemented on application layer.

We say that the communication channel $\mathcal{L}$ is *consistent*, if every message that is yielded is also used, and vice-versa, taking their enabler/yield multiplicities into account.

2.4 Extensions

Components may have several input and output messages specified by further virtual input and output variables, primary or auxiliary.

Moreover, there is no reason to restrict the communication with lookup components to a single lookup message which is consumed by an component via its auxiliary inputs. For example, S-two specifies *chainable* lookup components for the round functions of certain hashes. Chainable lookups are similar to regular components, having inputs and outputs which carry over the state of an iterative computation. They are called via an auxiliary output message yielded to $\mathcal{L}$, and the returned result is used by auxiliary inputs. Their constraints of the chainable component assure the correct routing and that the number of iterative calls is as specified by the caller.

2.5 Formal definition

Let us summarize our discussion by a concise abstract definition, which covers all the above features, including the extensions outlined in Section 2.4.

Definition 9. A *flat AIR* $\mathscr{A}$ over the finite field $\mathbb{F}_q$ is comprised of $\tau \geq 1$ triples of non-negative integer parameters (M_t, r_t, s_t), $1 \leq t \leq \tau$, and collections of polynomials

$$\mathsf{Comp}_t = \left\{\vec{C}_t, \vec{U}_{t,1}, \ldots \vec{U}_{t,r_t}, \vec{V}_{t,1}, \ldots, \vec{V}_{t,s_t}\right\} \subset \mathbb{F}_q[X_1, \ldots, X_{M_t}]^*,$$

where $t = 1, \ldots, \tau$. The use polynomials $\vec{U}_{t,j}$ are assigned to one and the same enabler variable e_t from $\{X_1, \ldots, X_{M_t}\}$, and each of the polynomials $\vec{V}_{t,k}$ is assigned a multiplicity variable $m_{t,k}$ from $\{X_1, \ldots, X_{M_t}\}$.

Definition 9 does not differentiate between primary and auxiliary gate inputs (both are covered by the same set of use polynomials), and likewise for the outputs. It does not mention public inputs of the AIR, public variables and their preprocessed columns, and moreover treats ordinary and lookup components equally. Overall, the definition hides important context of the concrete computation. Yet it allows a compact treatment of our main protocol, which is reason for our choice of abstraction.

Definition 10. Given public inputs $\vec{I}_1, \ldots, \vec{I}_{k_{in}}$ and outputs $\vec{O}_1, \ldots, \vec{O}_{k_{out}}$, an ensemble of tables $\mathscr{T}_t \in \mathbb{F}_q^{N_t \times M_t}$, $1 \leq t \leq \tau$, with $N_t \geq 0$, is said to *satisfy* the flat AIR $\mathscr{A}$, if for every t,

$$\vec{C}_t(\vec{x}_i^{(t)}) = \vec{0},$$

over all rows $\vec{x}_i^{(t)} = (\mathscr{T}_t[i,1], \dots, \mathscr{T}_t[i, M_t])$, and if their multi-sets[5]of used and yielded elements

$$\mathcal{U}_t = \left\{ e_t[i] * \vec{U}_{t,1}(\vec{x}_i^{(t)}), \dots, e_t[i] * \vec{U}_{t,r_t}(\vec{x}_i^{(t)}) \; : \; i = 1, \dots, N_t \right\}, \tag{27}$$

$$\mathcal{V}_t = \left\{ m_{t,1}[i] * \vec{V}_{t,1}(\vec{x}_i^{(t)}), \dots, m_{t,s_t}[i] * \vec{V}_{t,s_t}(\vec{x}_i^{(t)}) \; : \; i = 1, \dots, N_t \right\}, \tag{28}$$

satisfy the multi-set equality

$$\{\vec{O}_1, \dots, \vec{O}_{k_{out}}\} \cup \bigcup_{t=1}^{\tau} \mathcal{U}_t = \{\vec{I}_1, \dots, \vec{I}_{k_{in}}\} \cup \bigcup_{t=1}^{\tau} \mathcal{V}_t. \tag{29}$$

We refer to Section 2.6 for examples, and conclude this section with the definition of the degree of a flat AIR.

Definition 11. Given a flat AIR $\mathcal{A} = \{\mathsf{Comp}_1, \dots, \mathsf{Comp}_\tau\}$ we define the degree of the component

$$\deg(\mathsf{Comp}_t) = \max_{j,k} \left\{ \deg \vec{C}_t, 1 + \deg \vec{U}_{t,j}, 1 + \deg \vec{V}_{t,k} \right\}, \tag{30}$$

where the degree of a vector expression is the maximum total degree of its entries. The degree of the flat AIR is the maximum of the degree of its components,

$$\deg(\mathcal{A}) = \max_t \deg(\mathsf{Comp}_t). \tag{31}$$

The plus one in equation (30) for the degrees of the use and yield polynomials is for technical reasons only, and becomes clear in Protocol 1. It attributes that within logUp lookup argument, which is our choice for proving the multi-set equality (29), the used and yielded expressions are multiplied by their multiplicity variable.

2.6 Examples

Proving Cairo over M31. *S-two Cairo* [Staa], in the following simply called *S-two*, is a *zero-knowledge virtual machine (zkVM)* which proves the correct execution of programs written in *Cairo assembly (CASM)* [GPR21], a zero-knowledge-friendly machine language over $\mathsf{F252} = \mathbb{F}_q$, with prime modulus

$$q = 2^{252} + 2^{196} + 2^{192} + 1.$$

This large prime field is the native arithmetic environment of *Stone*, the first generation prover for Starknet. S-two instead runs over $\mathsf{M31}$ and is backed by the circle STARK described in Section 3 and 5. It emulates the execution of the first generation Cairo virtual machine with native words in $\mathsf{F252}$.

[5] We use the asterisk notation to denote multiplicities. For example $m * \vec{v}$ is the m-fold duplication of the message $\vec{v}$.

The architecture of S-two as a zkVM is the same as described in [GPR21], and we assume that the reader is roughly familiar with that paper. In a nutshell, S-two is a von-Neumann machine, tailored for zero-knowlege proofs: It operates directly on memory cells, and has merely a minimal set of specific purpose registers for flow control,

$$\mathsf{pc}, \mathsf{ap}, \mathsf{fp},$$

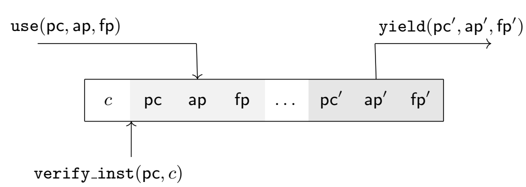
the program pointer pc, allocation pointer ap and frame pointer fp. These three pointers, together with the memory (an in-advance prepared "read-only" tape) define the state of the virtual machine. (See [GPR21] on how Cairo leverages non-determinism.) The arithmetization of S-two is a flat AIR over $\mathsf{M31}$, of cubic degree

$$\deg(\mathscr{A}_{\text{Cairo}}) = 3,$$

which makes heavy use of lookup tables, yet the task of emulating CASM with word size $\mathsf{F252}$ required careful optimization of the arithmetic circuits. In this section we will only give an overview of its components and their metrics. Certain selected components and their constraints will be covered in a future version of the whitepaper.

Opcode components. CASM has a restricted instruction set, which consists of few instructions for integer arithmetic in $\mathsf{F252}$, and few instructions for flow control. The flat AIR of S-two uses one component per instruction, the inputs and outputs of which consist only of the three $\mathsf{F252}$ registers pc, ap, fp, each represented by 28 $\mathsf{M31}$ elements in the 9 bit range $[0, 2^9 - 1]$. Proper routing is ensured by matching the instruction at memory address pc with the opcode of the component, a public constant. In this way, only one component can use a triple $(\mathsf{pc}, \mathsf{ap}, \mathsf{fp})$ from the communication channel $\mathcal{L}$, process it according its constraints, and yield an updated triple of registers $(\mathsf{pc}', \mathsf{ap}', \mathsf{fp}')$ back to $\mathcal{L}$. Some of the components have separate implementations for smaller argument sizes, which are used as replacements whenever possible. See Table 2 for all opcode components and their width.

Fig. 1. The interfaces of an opcode component. A state $(\mathsf{pc}, \mathsf{ap}, \mathsf{fp})$ is picked-up by a use from $\mathcal{L}$, and the instruction at memory address pc is enforced to match the opcode constant c via a lookup-call to the `verify_inst` component.



Extended instructions. S-two supports two additional instructions for recursive proofs. First, `blake_compress` for the BLAKE2s [SA15] compression function,

Table 1. Opcode components, their width and number of messages exchanged with the communication channel.

	cols	msgs		cols	msgs
`add`	102	9	`add_small`	38	9
`mul`	129	37	`mul_small`	36	12
`assert_eq`	11	5	`assert_eq_deref`	18	7
`assert_eq_imm`	8	5	`add_ap`	16	7
`call`	24	9	`call_rel_imm`	23	9
`jnz_taken`	46	7	`jnz_non_taken`	8	5
`jmp_rel`	15	5	`jmp_rel_imm`	12	5
`jmp_abs`	13	5	`jmp_deref`	20	7
`ret`	15	7			
`blake_compress`	168	74	`qm_31_add_mul`	72	12

which hashes an input of 512 bit to a 256 bit output. Second, the `qm_31_add_mul` instruction bundle implements QM31 arithmetics. See Table 2 for the width of their components.

Memory components. The memory is an instance-specific read-only lookup table. Its F252 entries are mapped to their M31 representation via *identifiers*, which are listed in an auxiliary lookup table `mem_id_to_big`. For efficiency, that lookup table has an alternative variant `mem_id_to_small` handling smaller entries up to 72 bits, which replaces `mem_id_to_big` whenever possible. Both tables make heavy use of pairwise nine-bit range checks, provided by eight duplicates of the preprocessed lookup component `range_check_9_9`.

Built-in components. Built-ins are the traditional way to extend the instruction set of Cairo by manually defined coprocessors. They are custom circuits with a memory-mapped I/O, and their inputs and outputs are simply uses from the memory component. S-two supports built-ins for bitwise aritmetics on F252 elements (`bitwise` computes simultaneously AND, XOR and OR of the operands), built-ins for modular 384 bit integer arithmetics with specifiable modulus, and built-ins for the two hash functions over F252 used in Starknet: The Pedersen hash [Stac] in the elliptic curve [Stad], and the Poseidon hash [GKR+21] over F252, as specified in [Stae]. See Table 2 for all built-ins and their component width.

Lookup components. These are the helper components for opcodes and builtins. They cover non-preprocessed lookups (also chainable ones as outlined in Section 2.4), and make heavy use of preprocessed tables: Range checks in all variants (on pairs, triples, quadtruples of M31 elements, etc.), bitwise XOR in all variants, and round key material for hashes. See Table 3 for all lookup components and their width.

The whole execution chain is roughly as follows: Given the prepared memory trace, which hosts the CASM program together with its claimed inputs and

Table 2. Builtin components, their width and number of messages exchanged with the communication channel. The components for Poseidon and Pedersen hash are only wrappers; their real complexity are hidden in the lookup components in Table 3.

	cols	msgs		cols	msgs
`add_mod`	251	53	`bitwise`	89	38
`mul_mod`	410	187	`poseidon`	6	7
`range_check`	17	2	`pedersen`	3	4
`range_check96`	12	3	`pedersen_narrow_windows`	3	4

Table 3. The lookup components used by `blake_compress`, `poseidon`, `pedersen`, `pedersen_narrow_window`, and the integer arithmetics opcodes, together with the preprocessed components they depend on. Chainable components are marked with an asterisk, and preprocessed tables are gray.

	cols	msgs		cols	height
`triple_xor_32`	20	9	`blake_round_sigma`	16	2^4
`blake_round`*	211	59	`bitwise_xor_{4,7,8,9,10}`	3	$2^8 - 2^{20}$
`blake_g`	52	17	`range_check_7_2_5`	3	2^{14}
`poseidon_agg`	341	28	`poseidon_round_keys`	30	2^6
`poseidon_full_round`*	125	12	`rng_chk_3_3_3_3_3`	5	2^{15}
`poseidon_3_part_round`*	168	18	`rng_chk_4_4_4_4`	4	2^{16}
`range_chk_252_width_27`	19	15	`rng_chk_4_4`	2	2^8
`cube_252`	140	99			
`pedersen_agg_window_18`	205	12	`pedersen_points_window_18`	56	2^{23}
`part_ec_mul_window_18`*	296	128	`rng_chk_8`	1	2^8
`pedersen_agg_window_9`	233	12	`pedersen_points_window_9`	56	2^{15}
`part_ec_mul_window_9`*	310	128	`rng_chk_8`	1	2^8
integer arithm.			`rng_chk_11,12,18,20`	1	2^{11}–2^{20}
(also `cube_252`,`part_ec_mul_window_*`)			`rng_chk_3_6_6_3`	4	2^{18}
			`rng_chk_9_9`	2	2^{18}

outputs in a well-defined public address range (the *public memory*), the channel is initialized by the input of the AIR, the single triple

$$\vec{I}_1 = (\mathsf{pc}_I, \mathsf{ap}_I, \mathsf{fp}_I) \to \mathcal{L},$$

which points at the entry point of the program, and the current beginning of the free part of the memory. Starting from that triple, the further sequence of states

$$\mathsf{st}_i = (\mathsf{pc}_i, \mathsf{ap}_i, \mathsf{fp}_i), \quad i = 1, \dots, s,$$

are output by the one and only opcode component, which is able to pick-up the previous state as input, as indicated in Figure 1. The output of the AIR is the final state of the execution,

$$\vec{O}_0 = \mathsf{st}_s = (\mathsf{pc}_F, \mathsf{ap}_F, \mathsf{fp}_F) \leftarrow \mathcal{L},$$

which points at the exit point of the program. The verifier matches the content of the public memory with the expected content $\{c_j\}$, by looking up the latter through memory calls $\vec{O}_j = \texttt{memory}(add_j, c_j)$ over the entire public address range.

Static circuits. Flat AIRs can be also used to depict static arithmetic circuits consisting of several types of gates. This is done in a similar manner as in Plonk [GWC19]. For simplicity we consider gates with a single output

$$\mathsf{Comp}_t[\vec{Y}_1, \ldots, \vec{Y}_{r_t}, X_1, \ldots, X_{M_t}, \vec{Z}].$$

Each gate output has its own unique *output address*, a value from $\mathbb{F}_q$ specified by a public variable

$$A_t,$$

which in most applications can be treated entirely virtually, no preprocessing required. Output addresses are used in the output messages for routing, for example $Z_1 = A_t$, and the remaining entries of $\vec{Z}$ for the payload. Each gate input is connected to one of the gate outputs by dedicated *routing variables*

$$B_{t,1}, \ldots B_{t,R_t} \in \{X_1, \ldots, X_{M_t}\},$$

which store the desired output addresses, and which are used in the input messages in the same way as the output address A_t is encoded into output messages. The yield multiplicity

$$m_t \in \{X_1, \ldots, X_{M_t}\}$$

for the output message corresponds to the fan-out number of the row gate instance, and the enabler variable e_t is used for padding. Routing variables, yield and enabler multiplicities are public preprocessed columns, they specify the complete wiring of the circuit.

Again, the components may be helped with auxiliary inputs and lookup components.

3 The circle IOPP for flat AIRs

Interactive oracle proofs [BCS16], in short *IOP*, are interactive proof systems with an ideal model for vector commitments, the *oracles*. They are multi-round generalizations of probabilistic checkable proofs [FRS88,BFLS91,AS98,ALM$^+$98], and as such, an IOP consists of a well-specified number of interactions between two algorithms, the *prover* and the *verifier*. In each round, the verifier sends a random challenge, on which the prover answers with the commitments of vectors of well-specified length and alphabet[6]. Once a vector is committed the verifier is able to read them at locations of its choice. In other words, the verifier is given *oracle access* to the committed vectors, it queries the oracles at its chosen location, and receives the committed value as an answer. After the last round the verifier outputs a verdict, representing acceptance or rejection.

[6] Both vector length and alphabet may vary within and from round to round.

In this section we describe the interactive oracle proof for showing an ensemble of tables

$$\mathscr{T}_t \in \mathbb{F}_q^{N_t \times M_t}, \quad t = 1, \dots, \tau,$$

encoded and committed as circle code words with sufficient redundancy, to satisfy a given flat AIR $\mathcal{A} = \{\mathsf{Comp}_1, \dots, \mathsf{Comp}_\tau\}$ as specified in Definition 9 and 10. By the nature of probabilistic checkable proofs, which query oracles at a few random locations, the resulting protocol will assure merely proximity of the committed vectors to a solution of the AIR, rather than being a solution by themselves. The described oracle proof, Protocol 1 below, relies on common techniques for proving univariate polynomials to satisfy the AIR, and then applies multi-table FRI, as described in Section 4.2, to guarantee aforementioned proximity.

3.1 Setup

We restrict ourselves to $\mathbb{F}_q = \mathsf{M31}$, and stress the fact that the protocol, as well as its security analysis generalizes to any *circle-FFT friendly* prime field $\mathbb{F}_q$, in the sense of [HLP24]. We use the notation introduced in Section 1.

The protocol refers to the unique family of canonical cosets in the circle group over $\mathbb{F}_q$,

$$G'_n = Q_{n+1} \cdot G_n = \left\{ Q_{n+1} \cdot Q_n^i \; : \; i = 0, \dots, 2^n - 1 \right\}, \tag{32}$$

where $1 \leq n \leq 30$, parametrized by a projection-consistent choice of generators Q_n as in Section 1.3. The cosets form a chain under the group squaring map,

$$G'_{30} \xrightarrow{\pi} G'_{29} \xrightarrow{\pi} \dots \xrightarrow{\pi} G'_1,$$

in which each G'_k is mapped onto G'_{k-1} in a 2-to-1 manner. This chain of cosets will serve both witness as well as evaluation domains for the circle code words.

We assume the height of each table $\mathscr{T}_t$ is adapted to a power of two,

$$N_t = 2^{n_t}, \quad 1 \leq n_t \leq 29,$$

matching one of the domains in the chain. (Recall the padding strategy via enabler or multiplicity columns.) We fix a blow-up factor $B := 2^\beta$, with integer $\beta \geq 1$, which defines the redundancy

$$\rho := \frac{1}{B} = 2^{-\beta} \tag{33}$$

used in the encoding, and assume that

$$\deg(\mathcal{A}) \leq B + 1, \tag{34}$$

and

$$N \cdot B^2 \leq 2^{30}, \tag{35}$$

where $N = 2^n$, $n = \max_t n_t$, to ensure the existence of sufficiently large evaluation domains as required by the protocol. (The squaring of B is due to that we do not split the overall composition polynomial into smaller ones, see below. This can be optimized.) The verifier challenges will be drawn from the quartic extension $F = \mathsf{QM31}$ of $\mathsf{M31}$, but we emphasize that any other extension field can be treated in the same manner.

3.2 Encoding

Using the enumeration of the canonical cosets by their generators, Equation (32), the columns of each table $\mathscr{T}_t \in \mathbb{F}_q^{N_t \times M_t}$, $t = 1, \ldots, \tau$, are placed over the *witness domain* for $\mathscr{T}_t$,

$$H_t := G'_{n_t}, \tag{36}$$

in the order given by the parameterization (32), and interpolated by circle polynomials

$$p_{t,1}, \ldots, p_{t,M_t} \in \mathscr{L}'_{n_t}(\mathbb{F}_q), \tag{37}$$

using the circle FFT described in the introduction of the paper. The values of the polynomials are committed over the B times larger and disjoint domain, the *evaluation domain*

$$D_t := G'_{n_t+\beta}, \tag{38}$$

as words from the circle code $\mathscr{C}'_{n_t}(\mathbb{F}_q, D_t)$ introduced in Section 1.4. This code has redundancy $\rho = 2^{-\beta}$, regardless of the table size.

Solution of circle polynomials. Recall Definition 9 and 10 for a flat AIR $\mathcal{A} = \{\mathsf{Comp}_1, \ldots, \mathsf{Comp}_\tau\}$ and a satisfying ensemble of component tables.

Definition 12. With the above encoding in mind, we say that an *ensemble of circle polynomials*

$$\{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \tag{39}$$

satisfies the flat AIR $\mathcal{A}$ on inputs $\{\vec{I}_1, \ldots \vec{I}_{k_{in}}\}$ and $\vec{O} = \{\vec{O}_1, \ldots, \vec{O}_{k_{out}}\}$, if their tables $\mathscr{T}_t \in \mathbb{F}_q^{N_t \times M_t}$, $t = 1, \ldots, \tau$, column-wise defined by the values of the polynomials over the respective witness domain, do.

For future reference, let us make the constraints of $\mathcal{A}$ explicit: For each component $t = 1, \ldots, \tau$, the row-constraints

$$\vec{C}_t(p_{t,1}(P), \ldots, p_{t,M_t}(P)) = \vec{0}, \tag{40}$$

are satisfied for every $P \in H_t$, and the multi-sets of used and yielded elements

$$\mathcal{U}_t = \bigcup_{P \in H_t, j} \left\{ e_t(P) * \vec{U}_{t,j}(p_{t,1}(P), \ldots, p_{t,M_t}(P)) \right\}, \tag{41}$$

and

$$\mathcal{V}_t = \bigcup_{P \in H_t, k} \left\{ m_{t,k}(P) * \vec{V}_{t,k}(p_{t,1}(P), \ldots, p_{t,M_t}(P)) \right\}, \tag{42}$$

respect the multi-set equality (29), where each of the enablers e_t and yield multiplicities $m_{t,k}$ are assigned to one of the polynomials $p_{t,1}, \ldots, p_{t,M_t}$.

Remark 13. We note that in certain variants of the protocol, Definition 12 needs to be extended to polynomials of slightly larger degree bound, which moreover are not necessarily in the image of the circle FFT. However, for the main variant as described below, the stricter requirement (39) will be enforced by FRI with degree-correction, Protocol 4.

3.3 The protocol

The interactive oracle proof takes the common approach for proving univariate constraints, the component constraints (40), and uses logUp [Eag22,Hab22] for proving the multi-set identity (29). In order to prove that a sequence of field elements $\{u_i\}$ is contained in a given set $\mathcal{V} \subseteq \mathbb{F}_q$, logUp shows the equality of the rational functions

$$\sum_i \frac{1}{X - u_i} = \sum_{v \in \mathcal{V}} \frac{m(v)}{X - v}, \tag{43}$$

where each multiplicity $m(v)$ counts how often the element v occurs on the left-hand side. If the number of occurrences does not exceed the characteristic of the field, then equality of these functions is in fact equivalent to the claimed set membership. The prover commits to the multiplicities, and (43) is point-checked at a random point β_1, leading to the claim that

$$\sum_{u \in \mathcal{U}} \frac{1}{\beta_1 - u} = \sum_{v \in \mathcal{V}} \frac{m(v)}{\beta_1 - v},$$

which is then proven by standard sumcheck techniques. (The current version of our protocol uses running sums.) To apply logUp to the messages used and yielded to the communication channel $\mathcal{L}$ of the flat AIR, the vector-valued messages are reduced to scalar-valued ones, via

$$\left\langle \vec{\beta}_0, \vec{U}_{t,j} \right\rangle := \rho_{t,j} + \beta_0 \cdot U_{t,j,1} + \beta_0^2 \cdot U_{t,j,2} + \ldots + \beta_0^{\rho_{t,j}} \cdot U_{t,j,\rho_{t,j}}, \tag{44}$$

where β_0 is random, and $\rho_{t,j}$ is the vector dimension of $\vec{U}_{t,j}$, interpreted as element in $\mathbb{F}_q$. (Hashing the vector dimension is a generic measure against trivial collisions by length extension, and can be also handled on application layer.[7]) Likewise we define $\langle \vec{\beta}_0, \vec{V}_{t,k} \rangle$ for the yield polynomials $\vec{V}_{t,k}$, as well as the input and output messages $\vec{I}_k$, $\vec{O}_k$ of flat AIR.

The component constraints, including the constraints for logUp, are proven table-wise, each with its separate domain quotient, which are then combined altogether to a single cross-domain polynomial.

Protocol 1 (IOPP for flat AIR)**.** *Given proximity parameter $\theta \in (0,1)$, and $F \geq \mathbb{F}_{q^2}$. The prover has committed to the ensemble of words $\{f_{t,j}\}$ from the table polynomials $\{p_{t,j}\}$ satisfying the flat AIR $\mathcal{A} = \{\mathsf{Comp}_1, \ldots, \mathsf{Comp}_\tau\}$ on given inputs $\{\vec{I}_1, \ldots, \vec{I}_{k_{in}}\}$ and outputs $\{\vec{O}_1, \ldots, \vec{O}_{k_{out}}\}$.*

1. *(logUp.) Receiving $\beta_0, \beta_1 \leftarrow\$ F$ from the verifier, the prover computes for each table $t = 1, \ldots, \tau$ the helper polynomials $\bar{u}_{t,1}, \ldots, \bar{u}_{t,r_t} \in \mathscr{L}'_{n_t}(F)$ and*

[7] We thank Gregor Mitscha-Baude for pointing out a gap in a previous version of the document.

$\bar{v}_{t,1}, \ldots, \bar{v}_{t,s_t} \in \mathscr{L}'_{n_t}(F)$ *defined by*

$$\bar{u}_{t,j} = \frac{e_t}{\beta_1 + \langle \vec{\beta}_0, \vec{U}_{t,j}(p_{t,1}, \ldots, p_{t,M_t}) \rangle}, \quad j = 1, \ldots, r_t, \tag{45}$$

$$\bar{v}_{t,k} = \frac{m_{t,k}}{\beta_1 + \langle \vec{\beta}_0, \vec{V}_{t,k}(p_{t,1}, \ldots, p_{t,M_t}) \rangle}, \quad k = 1, \ldots, s_t, \tag{46}$$

over H_t, where e_t and $m_{t,k}$ are the assigned enabler and multiplicity polynomials. It moreover computes the "centered" running sum polynomial

$$\bar{w}_t \in \mathscr{L}'_{N_t}(F)$$

for the total $\nu_t \in F$ of its uses and yields, determined by the cyclic constraint

$$T_{Q_{n_t}} \bar{w}_t - \bar{w}_t = -\frac{\nu_t}{N_t} - \sum_{j=1}^{r_t} \bar{u}_{t,j} + \sum_{k=1}^{s_t} \bar{v}_{t,k} \tag{47}$$

over H_t. (Recall $N_t = 2^{n_t}$ is the size of H_t.) It commits to the values of all these polynomials over the corresponding domains D_t, gives the verifier oracle access to them, and claims the table sums ν_t, $t = 1, \ldots, \tau$.

2. *(Composition.) The verifier sends $\gamma \leftarrow\!\!\$\, F$ to the prover, which for each table $t = 1, \ldots, \tau$ determines the quotient*

$$q_t = \frac{p_t}{v_{n_t}} \in \mathscr{L}'_{n_t+\beta}(F),$$

of the composition polynomial

$$p_t = \mathsf{comp}_{t,\gamma}(p_{t,1}, \ldots, p_{t,M_t}, \bar{u}_{t,1}, \ldots, \bar{u}_{t,r_t}, \bar{v}_{t,1}, \ldots, \bar{v}_{t,s_t}, \bar{w}_t, \nu_t)$$

for the component constraints (40), (45), (46), (47), *defined below. It then builds their cross-domain linear combination,*

$$q = \sum_{t=1}^{\tau} \gamma^{\alpha_t} \cdot q_t \in \mathscr{L}'_{n+\beta}(F), \tag{48}$$

with the α_t are chosen as the smallest exponents, such that no power of γ occurs twice. It commits to the values of q over $G'_{n+2\cdot\beta}$, and provides the verifier oracle access.

3. *(Out-of-domain query.) The verifier samples a random point $Q \leftarrow\!\!\$\, C(F) \setminus C(\mathbb{F}_q)$, sends it to the prover. For each $t = 1, \ldots, \tau$, the prover reveals the values*

$$p_{t,i}(Q),\ \bar{u}_{t,j}(Q),\ \bar{v}_{t,k}(Q)$$

for each $i = 1, \ldots, M_t$, $j = 1, \ldots, r_t$, $k = 1, \ldots, s_t$, as well as

$$\bar{w}_t(Q),\ \bar{w}_t(Q_{n_t} \cdot Q),$$

together with $q(Q)$.

After that, both prover and verifier run the circle FRI batch evaluation proof, Protocol 4 below, for the above claims on the ensemble of polynomials, with proximity parameter θ.

If the prover passes Protocol 4, and if the opening claims from Round 3 satisfy the equality (119)*, together with*

$$\sum_{k=1}^{k_{in}} \frac{1}{\langle \vec{\beta}_0, \vec{I}_k \rangle} + \sum_{t=1}^{\tau} \nu_t = \sum_{k=1}^{k_{out}} \frac{1}{\langle \vec{\beta}_0, \vec{O}_k \rangle}$$

then the verifier accepts. (Otherwise, it rejects.)

We point out the the current implementation in [Staa,Stab] sligthly deviates from the protocol as described above, see Section 3.4 below.

Composition polynomial. The composition polynomial of a component is the linear combination

$$\begin{aligned}
&p_t = \mathsf{comp}_{t,\gamma}(p_{t,1}, \ldots, p_{t,M_t}, \bar{u}_{t,1}, \ldots, \bar{u}_{t,r_t}, \bar{v}_{t,1}, \ldots, \bar{v}_{t,s_t}, \bar{w}_t, \nu_t) \\
&= \sum_{i=1}^{k_t} \gamma^{i-1} \cdot C_{t,i}(p_{t,1}, \ldots, p_{t,M_t}) \\
&+ \sum_{j=1}^{r_t} \gamma^{k_t+j-1} \cdot \left(\bar{u}_{t,j} \cdot \left(\beta_1 + \left\langle \vec{\beta}_0, \vec{U}_{t,j}(p_{t,1}, \ldots, p_{t,M_t}) \right\rangle \right) - e_t \right) \\
&+ \sum_{k=1}^{s_t} \gamma^{k_t+r_t+k-1} \cdot \left(\bar{v}_{t,k} \cdot \left(\beta_1 + \left\langle \vec{\beta}_0, \vec{V}_{t,k}(p_{t,1}, \ldots, p_{t,M_t}) \right\rangle \right) - m_{t,k} \right) \\
&+ \gamma^{k_t+r_t+s_t} \cdot \left(T_{Q_{n_t}} \bar{w}_t - \bar{w}_t + \frac{\nu_t}{N_t} + \sum_{j=1}^{r_t} \bar{u}_{t,j} - \sum_{k=1}^{s_t} \bar{v}_{t,k} \right).
\end{aligned}$$

By our assumption the degree of the AIR $\deg(\mathcal{A}) = B+1$ is an odd number, and the quotient polynomial $q^{(t)} = p^{(t)}/v_{H_t}$ is again in the FFT space $\mathscr{L}'_{n_t+\beta}(F)$. This follows from the limit-at-infinity calculus used in [HLP24, Section 5.3]. Since $\mathscr{L}'_{n_t+\beta} \subset \mathscr{L}'_{n+\beta}$, the final cross-domain combination is again in the claimed FFT space.

3.4 Implementation.

S-two takes the quartic extension $\mathsf{QM31}$ for the extension field F, the verifier challenges are drawn from. This choice of extension degree yields reasonable security levels both in the unique decoding and list decoding regime (cf. Section 5.5). However, the current release differs from Protocol 1 in the following aspects.

- *Domain separation by randomness.* S-two does channel separation on the level of the proof system, instead of the AIR, by drawing separate challenges

β_0, β_1 for each of the channel. Summation is done via a single cross-channel running sum, and requires a slightly more careful soundness argument. See Remark 36.

– *Grouping of logUp columns.* The degree of the AIR for Cairo, $\deg(\mathcal{A}_{CASM}) = 3$, and hence allows for pairwise, instead of individual, helper functions for the logUp inverses. (The constraints for these helpers are then cubic, too.) This halves the number of polynomials $\bar{u}_{t,j}, \bar{v}_{t,k}$ in Round 1 of the protocol, and yields a significant decrease of the proof sizes.

Furthermore, the FRI opening proof as implemented treats extension field polynomials as vector of basefield polynomials, and hence requires four evaluation claims instead of a single one in Round 3. See Section 4.4 for details.

3.5 Cross-domain correlated agreement and soundness

The purpose of Protocol 1 is to prove that the committed ensemble $\{f_j^{(t)}\}$ is proximate to a solution $\{p_j^{(t)}\}$ of the AIR, with distance less than the proximity parameter θ. However, the concrete notion of distance has to be chosen carefully, as we will explain now.

Cross-domain correlated agreement. For interactive oracle proofs with several oracles per round, the notion of *correlated agreement* [BCI+20] plays a central role in their soundness analysis. Correlated agreement demands a sufficiently large set, over which the functions *jointly* coincide with the solution, or in terms of disagreement,

$$d(\{f_j^{(t)}\}, \{p_j^{(t)}\}) < \theta, \tag{49}$$

where d is the fractional Hamming distance between the vector-valued functions (often viewed as *interleaved* words). Since vector-valued Reed-Solomon codes are Reed-Solomon codes over a finite extension field, with the same rate and minimum distance δ, one can limit the number of such correlated proximates by the Johnson bound $\ell(\theta)$ for the given distance θ.

However, the situation becomes more delicate in the context of multiple domains, since correlated agreement over each of the domains separately is not good enough for a reasonable soundness errors of the overall protocol. A security argument that relies on such separate correlated agreement would require to take into account all combinations of such proximates *between* the batches (arranged by domain), which are

$$\ell(\theta)^r,$$

causing a soundness error exponential in r, the number of used domains. In order to avoid this combinatorial blow-up, the soundness analysis needs to link the proximates across the separate domains.

We achieve such a linking by *lifting* all functions to the largest evaluation domain used by the ensemble, $D^* = G_{n+\beta}$, with $n = \max_t n_t$. Given a function $f_i^{(t)}$ from a domain D_t, we define its lifting to D^* as

$$(f_i^{(t)})^* = f_i^{(t)} \circ \pi^{n-n_t}, \tag{50}$$

and we define *cross-domain correlated agreement* as correlated agreement between the lifted functions.

Definition 14 (Cross-domain proximate solution)**.** Given a proximity parameter $\theta \in (0,1)$, we say that an ensemble of functions

$$\left\{f_j^{(t)}\right\} \in (\mathbb{F}_q^{D_1})^{M_1} \times \ldots \times (\mathbb{F}_q^{D_\tau})^{M_\tau}$$

is θ-proximate to a solution of the flat AIR, if it belongs to the set

$$\mathscr{R}_{IOPP,\theta} =$$

$$\left\{ \{f_j^{(t)}\} \in (\mathbb{F}_q^{D_1})^{M_1} \times \ldots \times (\mathbb{F}_q^{D_\tau})^{M_\tau} \; : \; \begin{array}{c} \exists \{p_j^{(t)}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \\ d\left(\{(p_j^{(t)})^*\}, \{(f_j^{(t)})^*\}\right) < \theta, \\ \{p_j^{(t)}|_{H_t}\} \text{ satisfies (40) for } t = 1, \ldots, \tau, \\ \text{and (29) with (41) and (42)} \end{array} \right\},$$

where the distance is taken with respect to the functions, lifted to the largest domain D^*. Conversely, each such ensemble of polynomials $\{p_j^{(t)}\}$ is called a *θ-proximate solution* of $\{f_j^{(t)}\}$.

Soundness of our IOPP. Without going into the security notions for interactive oracle proofs, we state round-by-round soundness of Protocol 1. A rigorous discussion postponed to Appendix A. For brevity, we shall focus on the *Johnson regime*, i.e. for distances

$$\theta \in \left[\frac{\delta}{2}, 1 - \sqrt{1-\delta}\right), \tag{51}$$

where

$$\delta = 1 - 2^{-\beta}, \tag{52}$$

the minimum distance of the circle codes. (Recall that all the involved circle codes are of the same minimum distance.) Security in the *unique decoding regime*, with distances $\theta < \delta/2$, is outlined in a line of remarks.

In the following we use the combinatorial Johnson bound

$$\ell(\theta) = \frac{\alpha - (1-\delta)}{\alpha^2 - (1-\delta)}, \tag{53}$$

with $\alpha = 1 - \theta$, as an upper bound for the number of θ-proximate circle polynomials.

Theorem 15 (IOPP Soundness). *For proximity parameter $\theta \in [\frac{\delta}{2}, 1-\sqrt{1-\delta})$. Assume that the height N_t of all non-zero tables are so that*

$$1 - \theta > (1-\delta) \cdot \left(1 + \frac{2}{N_t}\right), \tag{54}$$

and assume that the multiplicities use messages of the flat AIR are ensured to be less than the characteristic of $\mathbb{F}_q$.

Then Protocol 1 is a round-by-round knowledge-sound interactive oracle proof for the relation $\mathcal{R}_{IOPP,\theta}$ as in Definition 14, with the following round-by-round errors:

1. *Logup. Round 1 of Protocol 1 has knowledge soundness error at most*

$$\varepsilon_1 \leq \ell(\theta) \cdot r \cdot \frac{k_{in} + k_{out} + \sum_{t=1}^{\tau}(r_t + s_t) \cdot N_t}{|F|},$$

   where $r = \max_{t,j,k}\{\rho_{t,j}, \sigma_{t,k}\}$ is the maximum vector dimension of all use and yields.
2. *Composition. Round 2 of Protocol 1 has knowledge soundness error at most*

$$\varepsilon_2 \leq \ell(\theta) \cdot \frac{2 + \max_t k_t + r_t + s_t}{|F|}.$$

3. *Out-of-domain query. Round 3 of Protocol 1 has knowledge soundness error at most*

$$\varepsilon_3 \leq \ell(\theta) \cdot \frac{1 + \deg(\mathcal{A}) \cdot N}{|C(F) \setminus C(\mathbb{F}_q)|},$$

   with $N = \max_t N_t$.

The soundness errors of the subsequent rounds of the circle FRI batch evaluation proof are given in Theorem 19 and Theorem 21.

Proof. Round-by-round soundness of Protocol 1 is proven in Appendix A, where the protocol is viewed as a sequence of randomized reductions, which reduces the main claim, that the committed ensemble belongs to $\mathscr{R}_{IOPP,\theta}$, to $\mathcal{V}_S$, synonymous for that all verifier checks pass. For the first three rounds of Protocol 1, the concrete specifications of the reductions are provided in Appendix A.4, and their soundness errors are covered by Lemma 5, 6 and 7 therein. In the same manner, round-by-round soundness of circle FRI is proven in Appendix A.3.

Dealing with error-correcting codes, round-by-round soundness is as good as (round-by-round) knowledge soundness. Straight-line extraction of the oracle contents, together with the Berlekamp–Welch or Guruswami–Sudan error correcting decoder, allows to extract a solution of the flat AIR in deterministic polynomial time. This is outlined in Appendix A.1. □

Remark 16. Inequality (54) is a technical constraint on the minimal non-trivial table height, and is required when moving from quotients to non-quotients in the proof of Theorem 21. It attributes the fact that over small domains, multiplication by a linear polynomial (the denominator of the quotients) significantly impacts the distance of the corresponding circle codes, whereas over large domains the change is mostly negligible. This constraint can be dropped under certain protocol changes, see Section 6.

Remark 17 (Bounding multiplicities in S-two)**.** To ensure the requirement that each use message occurs with multiplicity larger less than the characteristic of $\mathbb{F}_q$, S-two is aware of the number of lookup component calls from each main component of the flat AIR. (These are the opcode and built-in components.) These parameters, together with the table heights of the main components, allow to deduce a bound for the number of calls of each lookup component, which is checked by the verifier. The sum of these table heights of the opcode components (the length of the main execution trace) bounds the number of primary uses of the opcode components, and is also checked to be below the characteristic bound.

Although we target the non-interactive variant of the protocol, we quickly mention that the overall soundness error of the interactive protocol is bounded by the sum of the roundwise errors,

$$\varepsilon_{IOPP} \leq \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_{FRI}(\theta).$$

This overall error sum expresses the following. Given the committed ensemble of functions, if an arbitrary algorithm passes the verifier of Protocol 1 with a probability larger than ε_{IOPP}, then there exists a θ-proximate solution of circle polynomials to the flat AIR, in the sense of Definition 14. Moreover, the solution can be extracted efficiently from the committed ensemble of function values.

Remark 18 (Unique decoding regime)**.** The statement of the theorem holds also for distances $\theta < \delta/2$, taking $\ell(\theta) = 1$, and with the correspondingly smaller soundness errors for circle FRI, as listed in Remark 20.

4 Circle FRI

In this section we describe the key ingredient of our IOPP, the FRI low degree test for circle polynomials, adapted to the multi-table setting. This setting supports heterogeneous ensembles of different table heights, while avoiding excessive padding overhead: The evaluation domains of the ensemble are chosen in alignment with those of the FRI folding cascade, and their functions are injected into the corresponding folding step. While this procedure is common sense (and in fact frequently used in practice), proving it sound with respect to *cross-domain correlated agreement* (as required for tight soundness errors of our IOPP, see Section 3.5) is surprisingly technical.

Our security analysis concentrates on the circle case, with its own peculiarities due to the algebraic nature of the circle FFT. However, we emphasize that it can be translated to regular FRI without much effort, thereby providing a theoretical underpinning for a commonly used practice.[8]

[8] A detailed discussion of the non-circle case will appear in [HHM25].

The chapter is organized as follows. First, in Section 4.1, we recap circle FRI for a single function. We then move on to the multi-table case, Section 4.2, and summarize the main soundness result in Section 4.3, including a short outline of the technical subtleties of its proof. (The detailed analysis is postponed to Appendix A.3.) Finally, in Section 4.4, we consider circle FRI as a "polynomial commitment scheme", by discussing the batch evaluation proof for out-of-domain queries. We keep with the notation of the previous chapters.

4.1 The single domain case

Circle FRI is an adjustment of the celebrated *Fast Reed–Solomon Code Interactive Proof of Proximity* [BBHR18] to the circle curve. In this section we describe circle FRI [HLP24] in a slightly restricted setting, as a proof of proximity to a polynomial from the circle FFT space

$$\mathscr{L}'_n(F) = F[x]^{<2^{n-1}} + y \cdot F[x]^{<2^{n-1}},$$

instead of $\mathscr{L}_n(F)$ the full space of polynomials of same degree bound. (Recall that $\dim \mathscr{L}_n(F)/\mathscr{L}'_n(F) = 1$, see Section 1.3.)

Given a function $f \in F^D$ over a canonical coset D of size $|D| = 2^{n+\beta}$, with $\beta \geq 1$, and proximity parameter $\theta \in (0,1)$. We wish to prove that there exists a polynomial $p \in \mathscr{L}'_n(F)$ which is at most θ-close to the function, in other words, that f belongs to

$$\mathcal{R}_\theta = \left\{ f \in F^D \;:\; \exists p \in \mathscr{C}'_n(F, D) \text{ s.t. } d(p, f) \leq \theta \right\}, \tag{55}$$

where $\mathscr{C}'_n(F, D)$ is the linear code generated by $\mathscr{L}'_n(F)$ and d is the fractional Hamming distance. In terms of agreement,

$$\mathcal{R}_\theta = \left\{ f \in F^D \;:\; \begin{array}{c} \exists p \in \mathcal{L}'_N(F), \\ \exists A \subseteq D, |A|/D| \geq 1 - \theta, \\ \text{s.t. } p|_A = f|_A \end{array} \right\}. \tag{56}$$

Circle FRI is aligned with the mechanics of the circle FFT. By repeatedly folding "even" and "odd" parts, in the meaning of the circle FFT, the proximity claim is reduced to instances of half the size, again and again, until one ends up with a function small enough to be checked in plain. The protocol has $r + 1$ rounds, where $0 \leq r < n$, and goes along the chain of FFT domains

$$\begin{tikzcd} D \arrow[d, "\pi_x"] \\ S_0 \arrow[r, "\pi"] & S_1 \arrow[r, "\pi"] & S_2 \arrow[r, "\pi"] & \cdots\cdots\cdots \arrow[r, "\pi"] & S_r \end{tikzcd} \tag{57}$$

where in the first step D is projected in a 2-to-1 manner onto its image S_0 on the x-axis, which is then repeatedly halfed in size (again, in a 2-to-1 manner) by the projected group squaring map $\pi(x) = 2 \cdot x^2 - 1$. The initial proximity claim is gradually reduced to a proximity claim with respect to the polynomial spaces

$$\begin{tikzcd}
\mathscr{L}'_n(F) \arrow[d, "\text{round } 0"] & & & & \\
L_0 \arrow[r, "\text{rnd. } 1"] & L_1 \arrow[r, "\text{rnd. } 2"] & L_2 \arrow[r, "\text{rnd. } 3"] & \cdots\cdots \arrow[r, "\text{rnd. } r"] & L_r
\end{tikzcd} \tag{58}$$

where we write

$$L_i := F[x]^{<2^{n-1-i}} \tag{59}$$

for the spaces of univariate polynomials (which we sometimes call *line polynomials*). The first step reduces the proximity claim between $f \in F^D$ and $\mathscr{L}'_n(F)$, to proximity of the random linear combination

$$g_0(x) = \frac{f(x,y) + f(x,-y)}{2} + \lambda \cdot \frac{f(x,y) - f(x,-y)}{2 \cdot y} \in F^{S_0},$$

to the space $L_0 = F[x]^{<2^{n-1}}$ instead. The further steps $i = 1, \ldots, r$ are entirely univariate. Given $g_{i-1} \in F^{S_{i-1}}$, its proximity to L_{i-1} is reduced to proximity of a random folding $g_i \in F^{S_i}$, defined via

$$g_i(2 \cdot x^2 + 1) = \frac{g_{i-1}(x) + g_{i-1}(-x)}{2} + \lambda_{i-1} \cdot \frac{g_{i-1}(x) - g_{i-1}(-x)}{2 \cdot x},$$

where $\lambda_{i-1} \leftarrow\!\!\$\ F$, to the space L_i. The final $g_r \in F^{S_r}$, assumed to be a polynomial from L_r, is revealed in plain. After these rounds, the verifier samples random points from D, and checks consistency of the provided oracles along their traces under the projection chain (57).

Protocol 2 (Circle FRI IOPP for $\mathscr{L}'_n(F)$)**.** *The prover is given $f \in F^D$, the values of a circle polynomial from $\mathscr{L}'_n(F)$ over a canonical coset D as above, and the verifier has oracle access to it. Both prover and verifier engage in the following protocol.*

1. *Commit phase, consists of $r+1$ rounds, where $0 \leq r < n$.*
   (a) *In the first round, corresponding to $\pi_x : D \longrightarrow S_0$, the prover receives a random challenge $\lambda \leftarrow\!\!\$\ F$ from the verifier and uses it to determine*

   $$g_0 = f_0 + \lambda \cdot f_1 \in F^{S_0}, \tag{60}$$

   where $f_0, f_1 \in F^{S_0}$ are defined via

   $$\begin{aligned} f_0(x) &= \frac{f(x,y) + f(x,-y)}{2}, \\ f_1(x) &= \frac{f(x,y) - f(x,-y)}{2 \cdot y}. \end{aligned} \tag{61}$$

   The prover commits to g_0 and gives the verifier oracle access to it.

(b) In each of the further rounds $j = 1, \dots, r$, the prover holds a function $g_{j-1} \in F^{S_{j-1}}$ from the previous round, receives $\lambda_{j-1} \leftarrow\$ F$ from the verifier, and determines

$$g_j = g_{j-1,0} + \lambda_{j-1} \cdot g_{j-1,1} \in F^{S_j}, \tag{62}$$

where $g_{j-1,0}, g_{j-1,1}$ over $S_j = \pi(S_{j-1})$ are defined via

$$\begin{aligned} g_{j-1,0}(2 \cdot x^2 - 1) &= \frac{g_{j-1}(x) + g_{j-1}(-x)}{2}, \\ g_{j-1,1}(2 \cdot x^2 - 1) &= \frac{g_{j-1}(x) - g_{j-1}(-x)}{2 \cdot x}. \end{aligned} \tag{63}$$

The prover commits to g_j, and gives the verifier oracle access to it. (In the last round, the prover sends g_r in plain.)

2. Query phase, consists of $s \geq 1$ rounds.

(a) In each round the verifier samples a random $(x_0, y_0) \leftarrow\$ D$. Let $x_i \in S_i$, $i = 0, \dots, r$, be the trace of (x_0, y_0) under the projection chain (57). The verifier queries for the values $f(x_0, y_0)$ and $f(x_0, -y_0)$, and $g_i(x_i)$ and $g_i(-x_i)$ for $i = 0, \dots, r$, and uses them to point-check the folding relations

$$g_0(x_0) = \frac{f(x_0, y_0) + f(x_0, -y_0)}{2} + \lambda \cdot \frac{f(x_0, y_0) - f(x_0, -y_0)}{2 \cdot y_0}, \tag{64}$$

$$g_i(x_i) = \frac{g_{i-1}(x_{i-1}) + g_{i-1}(-x_{i-1})}{2} + \lambda_{i-1} \cdot \frac{g_{i-1}(x_{i-1}) - g_{i-1}(-x_{i-1})}{2 \cdot x_{i-1}}, \tag{65}$$

for every $i = 1, \dots, r$.

If the oracle answers satisfy these checks in each of the s rounds, and if $g_r \in L_r$, then the verifier accepts. (Otherwise, it rejects.)

The soundness analysis of the protocol is almost verbatim to that of regular FRI [BCI+20], see [HLP24, Appendix B]. We skip the details and directly proceed with the more general multi-domain case.

4.2 Over multiple domains

We now describe the extension of circle FRI to multi-domain ensembles $\{f_{i,j}\}$, consisting of batches of functions

$$f_{i,1}, \dots, f_{i,M_i} \in F^{D_i},$$

over different domains D_i, where the latter form a chain of canonical cosets

$$D_0 \xrightarrow{\pi} D_1 \xrightarrow{\pi} \dots \xrightarrow{\pi} D_r,$$

and the function values stem from polynomials from the respective circle FFT spaces

$$\mathscr{L}'_{n-i}(F).$$

Without loss of generality, we assume that all batches are of equal size $M_i = M \geq 1$. (One can always add zero-words to achieve equal size.)

Cross-domain batching step. In a first step, each of the batches are compressed into a separate random linear combination,

$$h_i = \sum_{j=1}^{M} \lambda^{j-1} \cdot f_{i,j} \quad \in F^{D_i}, \tag{66}$$

where we use the *same* random λ for all the batches. (We will see that this does not affect soundness.)

Folding with insertion. The cross-domain linear combinations $h_i \in F^{D_i}$ are absorbed by the corresponding step of the folding cascade, along the following diagram of domains.

$$\begin{tikzcd} D_0 \arrow[r, "\pi"] \arrow[d, "\pi_x"] & D_1 \arrow[r, "\pi"] \arrow[d, "\pi_x"] & \cdots\cdots \arrow[r, "\pi"] & D_r \arrow[d, "\pi_x"] \\ S_0 \arrow[r, "\pi"] & S_1 \arrow[r, "\pi"] & \cdots\cdots \arrow[r, "\pi"] & S_r \end{tikzcd} \tag{67}$$

In each step, except the first one, given a previous folding $g_{i-1} \in F^{S_{i-1}}$ over the previous "line domain" S_{i-1}, the line components $h_{i,0}, h_{i,1} \in F^{S_i}$ of $h_i \in F^{D_i}$ are combined together with the "even" and "odd" parts of g_{i-1}, that is,

$$g_i = g_{i-1,0} + \lambda_i \cdot g_{i-1,1} + \lambda_i^2 \cdot (h_{i,0} + \lambda_i \cdot h_{i,1}). \tag{68}$$

(In the first folding step $i = 0$, there is no previous line domain, and we think of $g_{-1,0}$, $g_{-1,1}$ being omitted.[9])

Protocol 3 (Multi-domain circle FRI)**.** *The prover holds the values of* $\{f_{i,j}\} \in \prod_i (F^{D_i})^M$ *of circle polynomials from the respective spaces* $\mathscr{L}'_{n-i}(F)$*. The verifier is given oracle access to each set of values.*

1. *Commit phase, consisting of a cross-domain batching step, and* $r+1$ *folding steps.*
   (a) *(Batching.) The prover receives* $\lambda \leftarrow\$ F$ *from the verifier, commits to the linear combinations* $h_i \in F^{D_i}$ *as defined in* (66)*, for* $i = 0, \dots, r$*, and gives the verifier oracle access to them. Let* $h_{i,0}, h_{i,1} \in F^{S_i}$ *denote the line components of* $h_i \in F^{D_i}$*, as defined in equation* (61)*.*
   (b) *(Folding cascade.) In each round* $i = 0, \dots, r$*, the prover holds a previous oracle* $g_{i-1} \in F^{S_{i-1}}$*. (In the first round* $i = 0$ *we take* $g_{-1} = 0$*.)*
   The verifier sends $\lambda_i \leftarrow\$ F$ *and the prover commits to* $g_i \in F^{S_i}$ *as defined in* (68)*, where* $g_{i-1,0}, g_{i-1,1} \in F^{S_i}$ *are circle FFT "even" and "odd" parts of* g_{i-1} *as defined via equation* (63)*. The prover gives the verifier oracle access to* g_i*, where in the final round* g_r *is sent in plain.*

[9] The current implementation in [Stab] takes the terms in (68) in a different order.

2. *Query phase. Consists of $s \geq 1$ rounds. In each of the rounds, the verifier samples $P_0 = (x_0, y_0) \leftarrow\!\!\$\ D_0$, and queries the oracles for their values to spot-check the folding identities along the projection trace $P_i = (x_i, y_i) = \pi^i(x_0, y_0)$. That is, it checks whether*

$$h_i(P_i) = \sum_{j=1}^{M} \lambda^{j-1} \cdot f_{i,j}(P_i), \tag{69}$$

$$\begin{aligned} g_i(x_i) = &\frac{g_{i-1}(x_{i-1}) + g_{i-1}(-x_{i-1})}{2} + \lambda_i \cdot \frac{g_{i-1}(x_{i-1}) - g_{i-1}(-x_{i-1})}{2 \cdot x_{i-1}} \\ &+\lambda_i^2 \cdot \left(\frac{h_i(x_i, y_i) + h_i(x_i, -y_i)}{2} + \lambda_i \cdot \frac{h_i(x_i, y_i) - h_i(x_i, -y_i)}{2 \cdot y_i} \right), \end{aligned} \tag{70}$$

for all $i = 0, \ldots, r$.

If in each of the query rounds all spot-checks hold, and if $g_r \in L_{n-1-r}$, then the verifier accepts. (Otherwise, it rejects.)

4.3 Soundness

We are able to show that for distances up to the Johnson radius , Protocol 3 enforces cross-domain correlated agreement in the following sense. We prove that the liftings to the largest domain D_0,

$$f^*_{i,j} := f_{i,j} \circ \pi^i, \tag{71}$$

agree with respective lifted polynomials from $\mathscr{L}'_{n-i}(F)$, over a joint set $A \subseteq D_0$ of target density. That is, the committed ensemble $\{f_{i,j}\}$ belongs to the set

$$\mathcal{R}_\theta := \left\{ (f_{i,j}) \in \prod_{i=0}^{r} (F^{D_i})^M \ : \begin{array}{c} \exists (p_{i,1}, \ldots, p_{i,M}) \in \mathscr{L}'_{n-i}(F)^M, 0 \leq i \leq r, \\ \exists A \subseteq D_0, |A|/|D_0| \geq 1 - \theta, \\ (f^*_{i,1}, \ldots, f^*_{i,M})|_A = (p^*_{i,1}, \ldots, p^*_{i,M})|_A, \\ \text{for all } i = 0, \ldots, r. \end{array} \right\}. \tag{72}$$

We again restrict to the Johnson regime, that is to proximity parameters $\theta \in \left[\frac{\delta}{2}, 1 - \sqrt{1-\delta}\right)$, where δ is the minimal distance of the circle codes. (Recall that all the involved circle codes have the same distance.) The corresponding result in the unique decoding regime are deferred to remarks. Unlike for the analysis of the main rounds of our IOPP, we use the Guruswami–Sudan bound

$$\ell_{GS}(\theta) = \frac{m(\theta) + \frac{1}{2}}{\sqrt{1-\delta}}, \tag{73}$$

with multiplicity parameter

$$m(\theta) = \max\left\{ \left\lceil \frac{\sqrt{1-\delta}}{2(\alpha - \sqrt{1-\delta})} \right\rceil, 3 \right\}, \tag{74}$$

where $\alpha = 1 - \theta$. This bound is slightly larger than the combinatorial bound from equation (53), but naturally shows up in the algebraic analysis of the Guruswami–Sudan list decoder [BCI+20]. We state soundness using the recently improved bounds from [BCH+25].

Theorem 19 (IOPP soundness multi-domain circle FRI)**.** *For proximity parameter $\theta \in \left[\frac{\delta}{2}, 1 - \sqrt{1-\delta}\right)$, where $\delta = 1 - \rho$ is the minimum distance of the circle codes. Protocol 3 is a round-by-round knowledge-sound IOP for $\mathcal{R}_\theta$ as defined in* (72)*, with the following round-by-round errors.*

1. *(Batching step.) Round 1a of Protocol 3 has knowledge soundness error at most*
$$\varepsilon_{FRI,batch} = (M-1) \cdot \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta) + 1\right) \cdot \frac{|D_0|}{|F|},$$
with $M = \max_i M_i$, and $\ell_{GS}(\theta)$ the Guruswami-Sudan list size bound for the circle code with minimum distance δ.
2. *(Folding steps.) Round $k = 0, \ldots, r$ in Phase 1b of Protocol 3, has knowledge soundness error at most*
$$\varepsilon_{FRI,fold,k} = 3 \cdot \ell_k(\theta) \cdot \left(\frac{2\ell_k(\theta)^4}{3} \cdot (1-\delta_k) + 1\right) \cdot \frac{|S_k|}{|F|},$$
where $\ell_k(\theta)$ is the Guruswami–Sudan list size bound for the line code over S_k, a Reed–Solomon code with minimum distance $\delta_k = 1 - \rho + \frac{1}{|S_k|}$.
3. *(Query phase.) Round 2 of Protocol 3 has knowledge soundness error at most*
$$\varepsilon_{FRI,query} = (1-\theta)^s.$$

Proof. We only summarize the statements proven in Appendix A.3. The soundness errors of the cross-domain batching step and the folding steps are covered by Lemma 3 and 4, and the soundness error of the query phase is a direct consequence of the reduced claim after the final folding step.

Again (as in the proof of Theorem 15) the claimed *knowledge* soundness follows from round-by-round soundness, with no change of the error bounds. □

The technical challenges in proving Theorem 19. The soundness analysis in Appendix A.3 takes the one from [BCI+20] and [BGKS20] as a blue-print. Nevertheless, it is not a straight-forward generalization. For cross-domain agreement, it turns out that the agreement theorems as stated in [BCI+20] are not sufficient, and certain strengthenings are required:

- Our proof of the cross-domain batching step relies on the correlated agreement theorem for linear subspaces of Reed–Solomon codes [Hab24a]. This type of agreement theorem has been already useful in the round-by-round analysis of Basefold, and helps us to show that the proximates of the lifted functions are again *lifted* polynomials. We recap the theorem in Appendix A.2, Theorem 31, where we adapt it to the circle.

– Second, for inductively "carrying" correlated agreement from a domain to the next larger one, we require an agreement theorem with respect to specified agreement sets, Appendix A.2, Theorem 28. This theorem helps us to conclude that whenever a random folding is proximate to a low-degree polynomial, jointly with some other "attached" functions, then the same is true for the folded components.

Both theorems rely on the collinearity of a sufficiently large fraction of proximates, see Appendix A.2.

Remark 20 (Unique decoding regime)**.** The entire analysis of Protocol 3 translates verbatim to distances $\theta \in [0, \delta/2)$, using bounds from [BCI$^+$20, Theorem 4.1].[10] The round-by-round soundness errors are

$$\varepsilon_{FRI,batch} = (M-1) \cdot \frac{|D_0|}{|F|},$$

and

$$\varepsilon_{FRI,fold,k} = 3 \cdot \frac{|S_k|}{|F|},$$

for the folding rounds $k = 0, \ldots, r$. The query phase error remains unchanged. For details, see Remark 32, 30, and 20 in Appendix A.

4.4 Batch evaluation proof

Evaluation proofs for out-of-domain queries use quotients to enforce proximates which evaluate to the claimed value at the query. In the regular univariate case, if v is the claim for the out-of-domain point z, these quotients are simply of the form

$$\frac{f(x) - v}{x - z},$$

using the single point vanishing polynomial $v_z(x) = x - z$. However, over the circle curve one has to be careful in the choice of the vanishing polynomial, since linear circle polynomials with coefficients in $\mathbb{F}_q$ have *two* zeros, due to the curvature of the circle.

Single-point zerofier. We demand our extension field F to contain the root $i = \sqrt{-1}$. In this case there are linear circle polynomials over F with a *single pole*, hence a single zero:

$$v_0(x, y) = 1 - (x + i \cdot y), \tag{75}$$

[10] Although leading to slightly smaller bounds, we avoid the lossy setting from [BCH$^+$25, Theorem 1.3].

or in univariate representation

$$v_0(t) = 2 \cdot \frac{t}{t - i},$$

using the circle parameterization introduced in Section 1.2. This circle polynomial has $(x, y) = (1, 0)$ as the only zero on the curve, and the only pole at the projective point $\infty = (1 : i : 0)$, both with multiplicity one.[11] For any point $Q \in C(F) \setminus \{\infty, \bar{\infty}\}$, we use the group action and define

$$v_Q = v_0 \circ T_Q^{-1}, \tag{76}$$

where T_Q is the translation with respect to the group operation.

With degree correction. For simplicity we describe the evaluation proof for a single query point, and leave a generalization to multiple points to the reader. In the protocol, we demand both the quotient *and* the non-quotient to be subject to the low-degree test. While this requirement is rather unusual in the context of single-domain IOPP, it is important in the multi-domain setting since we might have small domains in our ensemble: Over very small domains the degree difference between the quotient and the non-quotient is significant.

Protocol 4 (Circle FRI batch evaluation proof, degree corrected)**.** *Given the values $\{f_{i,j}\}$ of an ensemble of circle polynomials $\{p_{i,j}\}$ from the respective spaces $\mathscr{L}'_{n-i}(F)$, and an out-of-domain affine query point $Q \in C(F) \setminus C(\mathbb{F}_q)$. Let $\{v_{i,j}\} = \{p_{i,j}(Q)\}$ be the values of the polynomials at Q.*

1. *Both prover and verifier run circle FRI, Protocol 3, on the multi-domain ensemble comprised of the functions*

$$f_{i,j}, \ \frac{f_{i,j} - v_{i,j}}{v_Q}, \tag{77}$$

for all i, j, where v_Q is as in (76)*. If the prover passes circle FRI, then the verifier accepts. (Otherwise, it rejects.)*

Based on Theorem 19, we can easily deduce that Protocol 4 is a round-by-round knowledge-sound interactive oracle proof for the relation

$$\mathcal{R}_{\theta,Q} := \left\{ \{f_{i,j}\} \in \prod_{i=0}^{r} (F^{D_i})^M \ : \ \begin{array}{c} \exists (p_{i,1}, \ldots, p_{i,M}) \in \mathscr{L}'_{n-i}(F)^M \text{ with} \\ (p_{i,1}, \ldots, p_{i,M})(Q) = (v_{i,1}, \ldots, v_{i,M}), \\ \text{for } i = 0, \ldots, r, \text{ such that} \\ d(\{f^*_{i,j}\}, \{p^*_{i,j}\}) < \theta. \end{array} \right\}, \tag{78}$$

as long as the agreement density enforced by FRI is large enough for uniquely determining polynomials of slightly larger degree,

$$\mathscr{L}^+_{n-i}(F) = \left\{ p(x, y) \in F[x, y]/(x^2 + y^2 - 1) \ : \ \deg p \leq 2^{n-i-1} + 1 \right\}. \tag{79}$$

[11] Multiplicities are defined by means of the univariate representation.

For this it is sufficient that

$$1-\theta > 1-\delta+\frac{2}{|D_i|} = (1-\delta)\cdot\left(1+\frac{2}{N_i}\right), \tag{80}$$

for every nontrivial batch i, which contains non-zero functions. (Here, δ is again the minimum distance of the circle codes.) This is exactly the condition required in Theorem 15.

Theorem 21 (Soundness batch evaluation proof, degree corrected)**.** *Under the above condition* (80) *on the proximity parameter θ, Protocol 4 is a round-by-round knowledge-sound interactive oracle proof for $\mathcal{R}_{\theta,Q}$ as defined in* (78) *with the same round-by-round soundness errors as in Theorem 19.*

Proof. This is a black-box application of Theorem 19. Given $\{f_{i,j}\}$ we need to show that circle FRI on the extended batch of functions $\{f_{i,j}, (f_{i,j}-v_{i,j})/v_Q\}$ proves that $\{f_{i,j}\}$ belongs to $\mathcal{R}_{\theta,Q}$.

Suppose we have an ensemble of circle polynomials $\{p_{i,j},\, q_{i,j}\}$ from the respective spaces $\mathscr{L}'_{n-i}(F)$, and a joint set $A\subseteq D_0$ of density $1-\theta$, over which we have agreement of the liftings. That is,

$$f^*_{i,j}\big|_A = p^*_{i,j}\big|_A\,, \quad \wedge \quad \left.\frac{f^*_{i,j}-v_{i,j}}{v^*_Q}\right|_A = q^*_{i,j}\big|_A\,,$$

and hence

$$p^*_{i,j}\big|_A = v_{i,j}+v^*_Q\cdot q^*_{i,j}\big|_A\,,$$

for all i,j. Then on the projected set, which has at least the same density,

$$p_{i,j}|_{\pi^i(A)} = v_{i,j}+v_Q\cdot q_{i,j}|_{\pi^i(A)}\,.$$

We now conclude from (80) that $p_{i,j}=v_{i,j}+v_Q\cdot q_{i,j}$ as elements from $\mathscr{L}^+_{n-i}(F)$. This shows that $p_{i,j}$, a polynomial from $\mathscr{L}'_{n-i}(F)$, evaluates to $v_{i,j}$ at the point Q, showing that $\{f_{i,j}\}$ is in $\mathcal{R}_{\theta,Q}$. $\square$

Without degree correction. Let us quickly discuss the consequences of dropping the degree correction in Protocol 4, and running circle FRI only on the single-point quotients. In this case the modified protocol is a oracle proof for the relation

$$\mathcal{R}_{\theta,Q} := \left\{ \{f_{i,j}\}\in\prod_{i=0}^{r}(F^{D_i})^M \;:\; \begin{array}{c} \exists (p_{i,1},\ldots,p_{i,M})\in\mathscr{L}^+_{n-i}(F)^M \text{ with} \\ (p_{i,1},\ldots,p_{i,M})(Q)=(v_{i,1},\ldots,v_{i,M}), \\ \text{for } i=0,\ldots,r, \text{ such that} \\ d(\{f^*_{i,j}\},\{p^*_{i,j}\})<\theta. \end{array} \right\}, \tag{81}$$

with proximates from the larger polynomial spaces $\mathscr{L}^+_{n-i}$ as defined in (79). The main issue is that over small domains D_i, the proximity parameter θ might not be sufficient to ensure uniqueness of the functions $\mathscr{L}^+_{n-i}$. In extreme cases one might even leave the Johnson regime. To avoid the latter, we constrain the proximity parameter with respect to

$$\rho^+ := (1-\delta) \cdot \max_i \left(1 + \frac{2}{N_i}\right), \tag{82}$$

where the max ranges over all domains with a non-zero table. We require that

$$\rho^+ < (1-\theta)^2 \tag{83}$$

and obtain the increased list size bounds

$$\ell^+(\theta) = \frac{1-\theta}{1-\theta-\sqrt{\rho^+}} \tag{84}$$

for the number of proximates from $\prod_i (\mathcal{L}^+_{n-i}(F) \circ \pi^i)^M$. This larger bound replaces $\ell(\theta)$ in Theorem 15.

4.5 Implementation

The current release of S-two uses the circle FRI batch evaluation proof without degree correction, but differs from the described protocols in the following aspects.

Out-of-domain quotients via conjugates. Instead of using the single-point zerofier (75) the current release considers two-point quotients for the values at Q and its conjugate under the Frobenius isomorphism A_{q^2}, the q^2-th power map which leaves $\mathsf{CM31} = \mathbb{F}_{q^2}$ element-wise fixed. For each circle polynomial over $\mathbb{F}_q$, the verifier can deduce the value at the conjugate via $p(A_q^2(Q)) = A_q^2(p(Q))$, which amounts to a simple sign-switch of half of the coordinates of $p(Q)$ written in the basis $\{1, i, j, j \cdot i\}$ described in Section 1.1. Polynomials over $\mathsf{QM31}$ are written as

$$p_0 + i \cdot p_1 + j \cdot (p_2 + i \cdot p_3)$$

with $p_i \in \mathcal{L}(\mathbb{F}_q)$, and treated component-wise, for which the prover needs to provide the four values $p_0(Q), \ldots, p_3(Q)$. However, to assure that Q and its conjugate form a two-point set, the query point is drawn uniformly from $C(F) \setminus C(\mathbb{F}_{q^2})$. We omit further details of this approach, and only remark that the entire soundness analysis of Protocol 4 carries over verbatim to two-point quotients.

Different circle domain sampling. Our implementation chooses the y-coordinates of the circle domain queries P_i in Protocol 3 in a non-squaring consistent manner. This is due to a different commit procedure across circle domains, which is a variant of the one described in Section 5.1 below. (See Remark **??**.) While this minor difference does not impact security in the unique decoding regime, it complicates the soundness analysis in the list decoding regime, making it additionally challenging to relate correlated agreement across the circle domains. While we believe that an appropriate weakening of cross-domain correlated agreement allows a formal soundness proof, we do not provide such a proof in this writeup, mainly due to the fact that a future release will implement *lifted FRI*. See Section 6 for details.

5 The non-interactive STARK

The non-interactive STARK is obtained from Protocol 1 by means of the Ben-Sasson–Chiesa–Spooner transform [BCS16]: The oracles are replaced by Merkle commitments [Mer88], and interactivity is removed by the Fiat–Shamir transform [FS87]. In this section we briefly summarize the specific form of Merkle commitments, the Fiat–Shamir transform, and the security of the resulting non-interactive argument.

5.1 Cross-domain Merkle trees

In each round of Protocol 1, the words committed by the prover are hashed via a single cross-domain Merkle tree. (The same is done for the presumed ensemble, the ensemble subject to the AIR, including pre-processed columns.) That tree is arranged in such a manner, so that it respects the way that multi-table circle FRI, Protocol 3, is asking for openings, serving a single shared authentication path for a query.

Cross circle domain indexing. We introduce a group-squaring consistent index scheme on the canonical cosets G'_k, $1 \leq k \leq n_{max}$. To this end, any reverse index along the following chain of 2-to-1 mappings is a possible candidate.

$$\begin{tikzcd}
G'_{n_{max}} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & G'_{n_{max}-1} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & G'_{n_{max}-2} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & \cdots\cdots \arrow[r, "\pi"] & G'_1 \arrow[d, "\pi_x"] \\
S_{n_{max}-1} \arrow[r, "\pi"] & S_{n_{max}-2} \arrow[r, "\pi"] & S_{n_{max}-3} \arrow[r, "\pi"] & \cdots\cdots \arrow[r, "\pi"] & S_0
\end{tikzcd}$$

Our choice is adapted to our FFT implementation, which works with bit-reverse circle FFT coordinates, described in Appendix B. Given the π-consistent choice of subgroup generators $Q_1, \ldots, Q_{n_{max}+1}$ as in Section 1.2, Equation (7), a canonical coset

$$G'_k = Q_{k+1} \cdot G_k = G_{k+1} \cdot G_{k-1} \cup G_{k+1}^{-1} \cdot G_{k-1},$$

is indexed via

$$\begin{aligned} P'_{i_1,i_2,\ldots,i_k} &= \left(Q_{k+1} \cdot Q_{k-1}^{i_k + \ldots + i_2 2^{k-2} + i_1 2^{k-1}}\right)^{1-2i_k} \\ &= \left(Q_{k+1} \cdot Q_{k-1}^{i_{k-1}} \cdot \ldots Q_2^{i_2} \cdot Q_1^{i_1}\right)^{1-2i_k} \end{aligned} \tag{85}$$

with $(i_1, i_2, \ldots, i_k)$ ranging over $\{0,1\}^k$. (We use the same notation for the cosets of different sizes, and assume that the reader is aware of the number of bits in the subscript of P'.) We call this scheme *transposed bit-reverse order*, since it is obtained from the circle FFT bit-reverse order via swapping the least significant bit with the block of remaining bits. However, we stress that fact that this transposition that is only done in our minds, never explicitly on the data, see Algorithm 1.

In the transposed order the *most significant* bit i_k chooses between the two half-sized cosets $Q_{k+1} \cdot G_{k-1}$ and $Q_{k+1}^{-1} \cdot G_{k-1}$, and the remaining bits $(i_1, \ldots, i_{k-1})$ enumerate each of the cosets in a regular bit-reverse manner. By our choice of generators,

$$\begin{aligned}\pi(P'_{i_1,i_2,\ldots,i_k}) &= \left(Q_{k+1} \cdot Q_{k-1}^{i_{k-1}} \cdot \ldots \cdot Q_2^{i_2} \cdot Q_1^{i_1}\right)^{(1-2i_k)\cdot 2} \\ &= \left(Q_k \cdot Q_{k-2}^{i_{k-1}} \cdot \ldots \cdot Q_1^{i_2} \cdot Q_0^{i_1}\right)^{(1-2i_k)} = P'_{i_2,\ldots,i_k}, \quad (86)\end{aligned}$$

since $Q_0 = (1, 0)$ is the group unit. Thus, the images of a point $P'_{i_1,\ldots,i_n} \in G'_n$ under repeated application of the group-squaring map is

$$P'_{i_1,i_2,\ldots,i_n} \xrightarrow{\pi} P'_{i_2,\ldots,i_n} \xrightarrow{\pi} \ldots \xrightarrow{\pi} P'_{i_n},$$

and for each $1 \leq k \leq n-1$, the preimages of $P'_{i_2,\ldots,i_k} \in G'_{k-1}$ are the two points $P'_{0,i_2,\ldots,i_k}$ and $P'_{1,i_2,\ldots,i_k}$.

The hashing procedure. Given an ensemble of words over the domains from the chain of canonical cosets, we construct the hash tree as follows. Assume that $|D^*| = 2^n$ (where $n \leq n_{max}$) is the largest domain size used of the ensemble. The tree is a binary tree of height n, and each level of the tree, $k = 1, \ldots, n$ from top to bottom (root is level 0), addresses the points of the canonical coset G'_k in transposed bit-reverse manner: Any path of length k, starting from the root rt and ending at a node in level k,

$$\mathsf{rt} \xrightarrow{i_k} P'_{i_k} \xrightarrow{i_{k-1}} P'_{i_{k-1},i_k} \ldots \xrightarrow{i_1} P'_{i_1,\ldots,i_{k-1},i_k}$$

corresponds to the point as defined in equation (85). Whenever there are functions present over the domain G'_k, the hashing procedure inserts them into the compression of the children hashes,

$$h_{i_1,\ldots,i_k} = H(H(h_{0,i_1,\ldots,i_k} \| h_{1,i_1,\ldots,i_k}) \| \mathrm{vals}_{i_1,\ldots,i_k}), \quad (87)$$

where $h_{0,i_1,\ldots,i_k}$, $h_{1,i_1,\ldots,i_k}$ are the hashes of left and right children of the node $P'_{i_1,\ldots,i_k}$, and $\mathrm{vals}_{i_1,\ldots,i_k}$ are the values of the functions at that point.

A simple implementation of the algorithm transposes the hashes $\mathrm{vals}_{i_1,\ldots,i_k}$ given in bit-reverse circle FFT coordinates over each domain, and then computes a regular binary hash tree. Algorithm 1 avoids such a transposition and directly works in bit-reverse FFT coordinates. Without transposition, the computation of the hash tree amounts to computing two half-sized Merkle trees in interleaved manner, one for the even indices and one for the odd indices, which are eventually merged at the top layer of the tree, corresponding to the smallest circle domain G'_1.

Algorithm 1 Cross circle domain Merkle hash. Over each canonical coset G'_k, $1 \leq k \leq n$, we have given the evaluation hashes $(v_{k,i})_i = (H(f_{k,1}(P_k(i)), \dots, f_{k,M_k}(P_k(i)))_i$ in circle FFT bit-reverse order, where $M_k \geq 0$.

$\triangleright$ $P_k(i) = P_{i_1,\dots,i_k}$ as in Equation (125), with $i = [i_1, \dots, i_k] = i_1 + 2i_2 + \ldots + 2^{k-1} i_k$.
$\triangleright$ Returns interleaved hash tree $\bigcup_{k=0}^{n} \{h(P_k(i)) \;:\; i \in [0, 2^k - 1)\}^k\}$ of height n, with $h_\varnothing$ as the root hash.

function MerkleHash($\{(v_{k,i})_i\}$)
    $n \leftarrow \max\{k \in [0, n_{max}] \;:\; M_k \neq 0\}$
    for $k = n$ down to 0 **do**
        for $i = 0, \dots, 2^k - 1$ **do** $\triangleright$ Compute hashes over level k domain
            if $k = n$ **then**
                $h' \leftarrow 0$ $\triangleright$ No children for bottom level domain
            else
                $h' \leftarrow H(h_{k+1,(i \bmod 2)+4\lfloor i/2 \rfloor} \| h_{k+1,(i \bmod 2)+4\lfloor i/2 \rfloor + 1})$ $\triangleright$ Merge of children hashes
            if $M_k = 0$ **then**
                $h_{k,i} \leftarrow h'$ $\triangleright$ No functions present
            else $\triangleright$ Otherwise, insert function values
                $h_{k,i} \leftarrow H(h' \| v_{k,i})$ $\triangleright$ This is rt for $k = 0$.
    return $\bigcup_{k=0}^{n} \{h_{k,i}\}$

Authentication paths. The authentication path is as in the single-domain case. Given a point $P \in G'_n$ as sampled in the query phase of Protocol 3, with bit-reverse circle FFT coordinates $(j_1, \dots, j_n)$. In transposed bit-reverse coordinates, P corresponds to

$$P = P'_{i_1,\dots,i_{n-1},i_n},$$

with $(i_1, \dots, i_{n-1}, i_n) = (j_2, \dots, j_n, j_1)$. The prover looks up the path $\mathsf{rt} \xrightarrow{i_n} P'_{i_n} \xrightarrow{i_{n-1}} P'_{i_{n-1},i_n} \dots \xrightarrow{i_1} P'_{i_1,\dots,i_{n-1},i_n}$, and reveals: For each layer $1 \leq k \leq n$, the values

$$\{f_{k,j}(P'_{i_{n-k+1},\dots,i_n})\} = \{f_{k,j}(\pi^{n-k}(P))\},$$

and for each $0 \leq k \leq n-1$ the missing children hash

$$h_{1-i_{n-k}, i_{n-k+1},\dots,i_n},$$

for the verification of the root hash $\mathsf{rt} = h_\varnothing$. This is done by recomputing the hashing procedure, along the given path defined by $(j_1, \dots, j_n)$. Note that the size of the authentication path is the same as for a single-domain Merkle tree.

Over line domains. Line domains are indexed in ordinary bit-reverse FFT manner, see Appendix B, which coincides with

$$x'_{i_2,\dots,x_k} = \pi_x(P'_{0,i_2,\dots,i_k}) = \pi_x(P'_{1,i_2,\dots,i_k}),$$

for $(i_2, \dots, x_k) \in \{0,1\}^{k-1}$. The cross-domain Merkle tree with respect to that order achieves π-consistency without any transposition.

5.2 Fiat–Shamir transform

The Fiat–Shamir transform [FS87] removes interaction by replacing the (public coin) verifier with a hash function

$$H : \{0,1\}^* \longrightarrow \{0,1\}^\lambda,$$

where the output length λ is chosen sufficiently large so that collisions are unlikely. Briefly, the i-th verifier randomness ρ_i is drawn deterministically according to

$$\rho_i = H(m_0 \| m_1 \| \dots \| m_{i-1}), \tag{88}$$

where $m_0, m_1, \dots, m_{i-1}$ are the prover messages sent in the previous rounds. Given the presumed ensemble of codewords, the prover simulates the verifier via (88), and outputs a proof string comprised of:

- the prover messages of each of the rounds (mostly, but not only Merkle roots),
- the openings of the Merkle commitments, together with their authentication path, for each of the verifier queries.

This proof string is the certificate for the claim of the protocol, in our context the STARK, and it is verified again by simulating the interactive verifier via (88), using the prover messages, and validating each authentication path.

5.3 Implementation.

Different cross-domain Merkle tree. The current release of S-two implements a variant of the cross-domain Merkle tree over circle domains, which does not use the transpose of the circle FFT bit-reverse index scheme. Instead the Merkle tree is a regular binary hash tree, addressing each canonical coset in (non-transposed) bit-reverse coordinates. A bit-reverse index $(i_1, \dots, i_n)$ as sampled in the FRI query phase, therefore asks for the following points to be opened,

$$\mathsf{rt} \xrightarrow{i_n} P_{i_n} \xrightarrow{i_{n-1}} P_{i_{n-1},i_n} \xrightarrow{i_{n-2}} \dots \xrightarrow{i_1} P_{i_1,\dots,i_{n-1},i_n},$$

where $P_{i_1,\dots,i_k}$ is as in Equation (125). We stress the fact that these points are not group-squaring consistent, since $\pi(P_{i_1,i_2,i_3,\dots,i_k}) = P_{i_1,i_3,\dots,i_k}$ and not $P_{i_2,i_3,\dots,i_k}$. However, their x-coordinates are consistent with respect to the projected group squaring map, since

$$0 \xleftarrow{\pi} x_{i_n} \xleftarrow{\pi} x_{i_{n-1},i_n} \xleftarrow{\pi} \dots \xleftarrow{\pi} x_{i_2,\dots,i_{n-1},i_n}.$$

In other words, and thinking of the query sampler in Protocol 3 as a bottom up sampling process, the implementation samples the circle points above the line domain points $x_{i_n}, \dots, x_{i_2,\dots,i_{n-1},i_n}$ in a different manner than in Protocol 3.

BCS transform. The current release of S-two takes BLAKE2s [AMPH14,SA15] with output size $\lambda = 256$ bit for both the Merkle tree and the Fiat–Shamir pseudo-random number generator. The latter is implemented as an iterative hash chain, which is initialized with the context of the proof (the FRI parameters, the flat AIR, and the preprocessed columns) and which is incrementally updated with the prover messages of each round. Elements are uniformly sampled from $\mathsf{M31}$ by increasing a counter until all 8 chunks of consecutive 32 bits of the hash output are in $[0, 2 \cdot q) = [0, 2^{32} - 2)$, and then reduced modulo q. (The probability of failure is less than 2^{-28}, so in most cases no repetition is needed.) Extension field elements are sampled coordinate-wise, and curve points by using the univariate parametrization. Grinding, explained below, is currently only applied for Round 1 in Protocol 1, and for the query phase of FRI, Round 2 of Protocol 3.

5.4 Soundness

In the non-interactive setting, security is a subtle matter. First of all, by removing interaction, the prover is allowed to replay certain rounds of the protocol by re-drawing the verifier randomness, whenever there is a certain degree of freedom in the prover messages. (This is particularly the case in protocols that provide zero-knowledge). The additional flexibility is captured by the notion of *state-restoration attacks* on interactive oracle proofs [BCS16]. Round-by-round soundness of our IOPs, as proven by Theorem 15 and Theorem 19 implies soundness against state restoration attacks [CMS19], yet the overall soundness error is the *maximum* of the round errors, rather than the stated union bound. Intuitively, the maximum stands for the weakest link in the chain of reductions, and this is the only round the attacker needs to break.

The second, much more delicate issue is pseudo-randomness: The security of a hash function relies on the unpredictability of its output. Optimally, the hash function is intractable for cryptanalysis, so that the attacker does not gain any advantage from its explicit description. This property is embraced by the *random oracle model (ROM)*, an idealized model in which the hash is a random function sampled uniformly from the space of all functions with λ bits output. The attacker is given only black-box, i.e. oracle access to it, thus not learning anything about its internals beyond what it has queried. Its computational power is measured in T, the *maximum number of accesses* to the random oracle, on which the concrete (adaptive) soundness error of the protocol depends on. That error is the chance that the attacker may produce a valid proof on an invalid statement, on a flat AIR it chooses, where the probability is taken over all randomly sampled functions. We postpone a formal treatment to a future version of the whitepaper, and only state the following theorem.

Theorem 22 ([BCS16,CMS19])**.** *Given an interactive oracle proof with $R \geq 1$ rounds and round-by-round soundness errors $\varepsilon_1, \ldots, \varepsilon_R$. Then the non-interactive random oracle proof obtained by the BCS transform of the IOP is (adpative)* *knowledge sound against T-query attackers, with a soundness error*

$$\varepsilon(T) \leq (T+R) \cdot \max_{i=1,\ldots,R} \varepsilon_i \ + 3 \cdot \frac{T^2+1}{2^\lambda}. \tag{89}$$

Remark 23. The generalized Valiant extractor of [BCS16] constructs a partial decommitment from the answers of the oracles of an ordinary Merkle tree. Without giving a formal proof, we note that [BCS16] can be adapted to mixed-domain Merkle trees. Future versions of S-two will implement the *memoized Merkle tree* from [HHM25], which is again a regular Merkle tree over the largest domain.

The first term in (89) essentially represents the above mentioned weakest link of the reduction chain, on which the attacker might focus all of its queries. The second term essentially captures the probability of finding collisions of the hash function, a bound for breaking the binding property of hash commitments.

How to set the security level. We use Theorem 22 to quantify the security of the non-interactive proof as follows: Given a T-query attacker, we say that the non-interactive random oracle proof has σ bits of security against T-query attackers, if such an attacker requires on average $2^\sigma/T$ accesses to the random oracle to produce a valid proof on an invalid statement. This leads to $1/\varepsilon(T) \geq 2^\sigma/T$, or equivalently, that the success probability per query

$$\frac{\varepsilon(T)}{T} \leq 2^{-\sigma}. \tag{90}$$

We henceforth choose our example parameters so that

$$\left(1+\frac{R}{T}\right) \cdot \max_{i=1,\ldots,R} \varepsilon_i \ + 3 \cdot \frac{T^2+1}{T} \cdot 2^{-\lambda} \leq 2^{-\sigma}.$$

In terms of the maximum attacker queries, we assume T clearly below 2^{128}, so that $\lambda = 256$ guarantees collisions with a probability less than 2^{-128}.

Grinding. The roundwise soundness errors can be improved by adding a relatively small proof of work to a round, demanding that the verifier randomness has a certain number of leading zeros. In the interactive setting, grinding is simply modeled by the additonal requirement that the verifier accepts only those $\rho_i \leftarrow\!\$\ \{0,1\}^\lambda$ for which

$$\rho_i \in \{0\}^{z_i} \times \{0,1\}^{\lambda - z_i},$$

where $z_i \geq 0$ is the grinding parameter. This reduces the soundness error ε_i of that round down to

$$\varepsilon'_i = 2^{-z_i} \cdot \varepsilon_i,$$

at the cost of completeness, which however does not harm in the non-interactive setting. There, grinding requires the prover message m_{i-1} to be extended by an additional *salt* $s_{i-1} \in \{0,1\}^\lambda$, the prover is free to choose, until

$$\rho_i = H(m_0 \| \ldots \| m_{i-1} \| s_{i-1})$$

has the demanded number of leading zeros. We stress the fact that salts do not affect Theorem 22, since the BCS state-restoration analysis presumes them per default.

A note on security in the ROM. As an idealized model, the random oracle model is only a testbed for the security of real-world protocols. Random oracles are known for their shortcomings, and limitations of the model have been pointed out since its introduction in [BR93]. A line of research, for example [Bar02,GK03,CGH04,BBH+19] is devoted to challenge the Fiat–Shamir transform on explicit constructions which are secure in the random oracle model, but replacing the random oracle by any concrete hash function renders the protocol entirely insecure. While these examples are contrived and designed to fail, the recent work [KRS25] instead breaks Fiat–Shamir of a protocol actively used in practice: GKR [GKR08], an interactive proof system for layered circuits, which celebrates a revival in modern systems for its key feature, that intermediate computations are not committed. Based on this feature, the authors construct a circuit which encapsulates an emulation of the entire first round of the Fiat–Shamir transformed protocol, and use it to trick false input-output statements into acceptance. (They construct a circuit polynomial which always tests to zero at the first Fiat–Shamir challenge.) This type of attack is more concerning than it seems at first glance, since it is hard to reason that certain circuits are not hiding the logic of the Fiat–Shamir round, in particular for universal circuits. While [KRS25] underscores the fundamental issue with idealized models, we note that it is currently unknown if the attack on GKR generalizes to proof systems which commit the entire computation trace.

5.5 Example parameters

In this section we provide example parameters for a security level of

$$\lambda = 100$$

bit, in both the Johnson regime (covered by the proofs from Section A) and beyond, under Conjecture 1 and 2 established in Appendix A.5.

Recall the proof parameters $\rho = 2^{-\beta}$, and the minimum distance $\delta = 1 - \rho$ of the circle codes. The AIR consists of τ components and has k_{in} input messages, and k_{out} output messages. The *main trace* consists of the component tables

$$\mathscr{T}_t \in \mathsf{M31}^{N_t \times (k_t+1)}, \quad t = 1, \ldots, \tau,$$

where a table $N_t = 2^{n_t}$ rows and $k_t + 1$ columns (including multiplicity/enabler). The logUp *interaction trace* consists of the tables

$$\mathscr{T}_t' \in \mathsf{QM31}^{N_t \times (k_t+s_t)}, \quad t = 1, \ldots, \tau,$$

where k_t and s_t are the numbers of used and yielded messages of a component gate. For simplicity, we use $r = \max_{t,j,k}\{\rho_{t,j}, \sigma_{t,k}\}$ the maximum vector dimension of any of the used and yielded messages, and we assume that

$$1 - \theta > (1 - \delta) \cdot \max_t \left(1 + \frac{2}{N_t}\right),$$

recall Remark 16. Let us summarize the roundwise soundness errors from Theorem 15 and Theorem 19 in Table 4.

Table 4. Roundwise soundness errors of Protocol 1 and its main component, multi-table FRI, Protocol 4, with a multilinear batching step. The first column is for proven security, with distances $\theta \in [\frac{\delta}{2}, 1 - \sqrt{1-\delta})$. The second column is for larger distances up to the Elias radius r_E, and based on the conjectures from Appendix A.5 on list-decodability and line-decodability up to r_E. The concrete expressions for list sizes and line-decodability parameters are summarized in the middle block of the table.

<table>
<tr><td colspan="3">Main rounds</td></tr>
<tr><td>logUp</td><td colspan="2">$\ell(\theta) \cdot r \cdot \frac{k_{in}+k_{out}+\sum_{t=1}^{T}(r_t+s_t)\cdot N_t}{\lvert F\rvert}$</td></tr>
<tr><td>comp.</td><td colspan="2">$\ell(\theta) \cdot \frac{\sum_t k_t+r_t+s_t}{\lvert F\rvert}$</td></tr>
<tr><td>DEEP</td><td colspan="2">$\ell(\theta) \cdot \frac{1+\deg(\mathcal{A})\cdot N}{\lvert C(F)\setminus C(\mathbb{F}_q)\rvert}, \quad N = \max_t N_t$</td></tr>
<tr><td></td><td>proven</td><td>conjectured ($\theta < r_E$)</td></tr>
<tr><td>$\ell(\theta)$</td><td>$\frac{1}{(1-\theta)^2-(1-\delta)} \leq \ell_{GS}(\theta) = \frac{m(\theta)+\frac{1}{2}}{\sqrt{1-\delta}}$</td><td>$2^{H(\rho)/(\theta-1-\rho)}$</td></tr>
<tr><td>a/b</td><td>$\log(L) \cdot \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta)+1\right)$</td><td>$\log(L) \cdot \ell(\theta)$</td></tr>
<tr><td colspan="3">FRI</td></tr>
<tr><td>batching</td><td>$\log(L) \cdot \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta)+1\right) \cdot \frac{N\cdot\rho^{-1}}{\lvert F\rvert}$</td><td>$\ell(\theta) \cdot \log(L) \cdot \frac{N\cdot\rho^{-1}}{\lvert F\rvert}$</td></tr>
<tr><td>$k = 0, \ldots, r$</td><td>$3 \cdot \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta_k)+1\right) \cdot \frac{N\cdot\rho^{-1}\cdot 2^{-(k+1)}}{\lvert F\rvert}$</td><td>$\ell(\theta) \cdot 3 \cdot \frac{N\cdot\rho^{-1}\cdot 2^{-(k+1)}}{\lvert F\rvert}$</td></tr>
<tr><td>query</td><td colspan="2">$(1-\theta)^s$</td></tr>
</table>

Since it is hard to give example parameters for general ensembles, we constrain ourselves to $\deg(\mathcal{A}) = 3$, and we assume M_{M31} the number of main trace columns, and M_{QM31} the number of the logUp interaction trace columns to relate as $M_{\mathsf{M31}} : M_{\mathsf{QM31}} \approx 4 : 1$; and we further assume that L the maximum batch size over a domain is

$$L \approx (M_{\mathsf{M31}} + M_{\mathsf{QM31}})/4,$$

which we often found in practice. The maximum vector length of uses and yields is taken as $r = 4$, and for simplicity we assume that FRI is running multilinear combinations, instead of powers of a single randomness.

Table 5 shows the parameters in the Johnson regime, for medium-sized and large-sized ensembles. Table 6 does the same in the conjectured regime, for distances θ up to the *Elias radius* $r_E(\rho)$, the maximum distance one can hope for. (We refer to Appendix A.5 for the definition of the Elias radius, and a discussion of the used conjectures.) In both tables we use 26 bits of grinding to reduce the number of FRI queries and hence the proof size. However, to meet the security target, grinding is also required in other rounds of the protocol, foremost the batching step of FRI, and the logUp round, Round 1 in the main protocol, which reduces the equality of the extensive pole sum to a point-check.

Table 5. Example parameters in the Johnson regime, for medium sized and large ensembles, where M_{M31} and M_{QM31} are the total number of columns over the respective field. All configurations are for 100 bit soundness, with maximum grinding difficulty of 26 bits. For comparison, we added an extra row for $s_{1-\sqrt{\rho}}$ the number of FRI queries at Johnson radius. The last ensembles take the complete component set for a Cairo AIR, including all built-ins and preprocessed tables, as a reference. All proof sizes are without compression, and without the optimizations from Section 6.

		small ensemble			CASM + precomp.		
	M_{M31}	594			3769		
	M_{QM31}	84			577		
	n	20			26		
	β	3	2	1	4	2	1
	$\log_2(1-\theta)$	1.46	0.97	0.48	1.83	0.93	0.45
	$\ell(\theta)$	32	25	13	13	10	6
	$\ell_{GS}(\theta)$	49	51	45	18	21	14
grind	round 1	15	15	14	14	14	13
	round 2*	0	0	0	0	0	0
	round 3	3	3	2	8	8	7
	batch**	26	26	26	25	26	26
	fold	$24-3$	$24-3$	$23-0$	$22-0$	$23-0$	$23-0$
	query	26	26	26	26	26	26
prf. size	s_θ	51	77	155	41	80	165
	$s_{1-\sqrt{\rho}}$	50	74	148	37	74	148
	DEEP	9.5 kiB			69.5 kiB		
	per query	13.8 kiB			40.1 kiB		
	total	694 kiB	1.04 MiB	2.1 MiB	1.68 MiB	3.20 MiB	6.54 MiB

* For the composition step, we used the generous bound $\sum(k_t + r_t + s_t) \leq 4 \cdot M$.

** We assume a multilinear batching step, so that the soundness error is essentially uniform across the different batch sizes.

Comparison with the legacy conjectures. Let us quickly discuss the impact of the new Conjecture 1 and 2 on the proof sizes. The conventional line of conjectures [BBHR18,BGKS20,BCI+20,Sta23] assumed correlated agreement for distances up to capacity $1-\rho$, and they estimated the security of FRI with

$$\beta \text{ bits per query},$$

for a blow-up factor of 2^β. However, Crites and Stewart [CS25] showed that correlated agreement can hold at most up to the Elias radius

$$r_E \approx 1 - \rho - \frac{1}{\log p},$$

which is slightly below capacity. Moreover, both the list sizes and the soundness error for correlated agreement can at best depend exponentially on the gap to r_E [KK25]. (See Table 4 for the concrete formula.) That exponential behavior degrades security down to

$$\approx 0.83\beta - 0.91\beta \text{ bits per query,}$$

given the limit we put on grinding, see the rows for $\log(1-\theta)$ in Table 6. Compared to proven security, Table 5, this is still less than 55% of the proof size, approaching 50% the higher the blow-up.

Table 6. Example parameters for 100 bit conjectured security, with grinding difficulty of at most 26 bits. For comparison, also $s_{1-\rho}$ the number of samples under the conventional conjecture (β bit per query) are given. All proof sizes are without compression, and without the optimizations from Section 6.

		small ensemble			CASM + precomp.		
	M_{M31}	510			3769		
	M_{QM31}	84			577		
	n	20			26		
	β	3	2	1	4	2	1
	$\log_2(1-r_E)$	2.79	1.85	0.91	3.74	1.85	0.91
	$\log_2(1-\theta)$	2.65	1.73	0.83	3.56	1.73	0.83
	$\log_2(\ell(\theta))$	15.8	15.8	16.0	15.3	15.8	16.0
grind	round 1	26	26	26	26	26	26
	round 2*	2	1	2	4	4	5
	round 3	14	14	14	20	20	20
	batch**	19	17	17	25	24	23
	fold	$16-1$	$15-0$	$14-0$	$22-1$	$21-0$	$20-0$
	query	26	26	26	26	26	26
prf. size	s_θ	28	43	90	21	43	90
	$s_{1-\rho}$	25	37	74	20	37	74
	DEEP	9.5 kiB			69.5 kiB		
	per query	13.8 kiB			40.1 kiB		
	total	386 kiB	587 kiB	1.22 MiB	0.89 MiB	1.75 MiB	3.60 MiB

* For the composition step, we used the generous bound $\sum(k_t + r_t + s_t) \le 4 \cdot M$.
** We assume a multilinear batching step, so that the soundness error is essentially uniform across the different batch sizes.

6 Future releases

There are several optimizations and extensions considered for future releases of S-two.

Lifted FRI. The not yet published manuscript [HHM25] describes several alternatives of FRI in the multi-table context. In lifted FRI, the prover commits to the function ensemble lifted to largest evaluation domain. Remarkably, this can be done without paying the price of duplication, by using a hashing strategy which branches along the bit-reverse indexed tree. (The *memoized Merkle tree* in [HHM25].) Lifted FRI has a more uniform verifier, and it liberates the FRI folding cascade from the ensemble domains. It furthermore greatly simplifies the soundness analysis, since cross-domain correlated agreement is given per default.

Entirely flat constraints. Currently, the sumcheck for logUp consistency uses running sums which require constraints between adjacent table rows. The flat sumcheck from [BCR$^+$19] can be carried over to the circle by using the Laurent representation of circle polynomials [DH24], yielding a trigonometric decomposition which proves the sum of a column.

Zero-knowledge. Zero-knowledge guarantees that a valid proof reveals no information on the execution trace, except is explicitly intended. While in practice it is often hard to reconstruct private witness data from proofs, a guaranteed information barrier is introduced by additional randomization mechanism in the prover algorithm. Depending on the implementation effort, such randomization does not cause much prover overhead, see the discussion in [HK24]. We note that randomization is significantly easier in the lifted setting, since it allows a single one-for-all column for the logUp bus consistency.

logUp GKR. Using the Goldwasser–Kalai–Rothblum [GKR08] protocol for layered circuits significantly reduces the trace for logUp, since no helper columns for fractions need to be committed [PH23]. The circuit for computing the sums of fraction is elementary and homogeneous, rendering it resistant against the recent attacks [KRS25] on the non-interactive GKR protocol using Fiat–Shamir. There is already a prototype implemented in [Stab], yet optimizations of the multivariate logic are still pending.

Quantum security. In a future version of the whitepaper, we will explicitly discuss the security of S-two in the quantum random oracle model.

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A Soundness of the interactive oracle proofs

In this section we analyze soundness of our interactive oracle proofs, Protocol 1 and 3. The introductory Section A.1 clarifies the connection between our preferred view of a (public coin) IOP as a chain of randomized reductions, and the notions of round-by-round soundness and round-by-round knowledge soundness. In Section A.2 we survey certain strengthened statements on correlated agreement for Reed-Solomon codes, which are needed for the soundness analysis of multi-table FRI in Section A.3. Finally, the round-wise analysis of the complete Protocol 1 is then given in Section A.4.

A.1 Notation

We assume that the reader is aware of interactive oracle proofs [BCS16] and their security notions soundness, proof of knowledge and state restoration soundness [BCS16], as well as the closely related notions of round-by-round soundness and round-by-round knowledge soundness [CCH+18,CMS19][12]. However, our notation slightly deviates from the one commonly used in literature, in the following respect: Firstly, we consider the functions subject to the oracle proof statement (i.e., the ensemble committed in the first place) as part of the *presumptions* of the protocol, excluded from the first round, which under our convention starts with the first verifier challenge. That is, we think of the exchanged messages arranged as

$$m_0; \quad \underbrace{\rho_1, m_1}_{\text{round } 1}, \underbrace{\rho_2, m_2}_{\text{round } 2}, \ldots, \underbrace{\rho_r, m_r}_{\text{round } r}, \quad v,$$

where m_0 is the presumed ensemble, (ρ_i, m_i) are verifier challenge and prover answer of the rounds $i = 1, \ldots, r$, and v is the final decision by the verifier. With this stylistic convention, the main relation to be proven is a statement on the committed collection of functions $m_0 = \{f_j\}$ of the form

$$\mathcal{R} = \left\{ \{f_j\} \ : \ \begin{array}{c} \exists \text{ proximate polynomials } \{p_j\} \\ \text{such that } \ldots \end{array} \right\},$$

for example "there exist a proximate polynomial which solves the AIR", rather than any second-hand implication (e.g. the input-output relation of the concrete

[12] An excellent self-contained reference is the textbook [CY].

arithmetic circuit). This convention allows us to treat oracle proofs in a modular manner (subprotocols can be seen as separate IOPs, with their own target relation) and it emphasizes the nature of protocol itself, a proof of proximity, with proximates being present in the target relation of the proof system.

We stress the fact that "taking m_0 out of the game" does not affect security, since our statements will be with respect to any presumed m_0, regardless if was prepared by a malicious prover, or not.

Round-by-round soundness. When it comes to round-by-round security, we pursue a notation which is closer to the mechanics of the proof and its enforced relations. We consider IOPs which are *public coin*, and the rounds of which form a chain of randomized reductions

$$\mathcal{R} = \mathcal{R}_0 \xrightarrow{\rho_1} \mathcal{R}_1 \xrightarrow{\rho_2} \ldots \xrightarrow{\rho_r} \mathcal{R}_r,$$

where each verifier message ρ_i is sampled uniformly from a specified domain (in our case $\rho_i \leftarrow\!\!\$\ F$), and each relation $\mathcal{R}_i = \mathcal{R}_i(m_0, \rho_1, m_1, \ldots, \rho_i)$ depends on the messages exchanged so far. The soundness error of a reduction step is the threshold probability under which the desired reverse conclusion,

$$m_i \in \mathcal{R}_i(m_0, \rho_1, m_1, \ldots, \rho_i) \Rightarrow m_{i-1} \in \mathcal{R}_{i-1}(m_0, \rho_1, m_1, \ldots, \rho_{i-1}),$$

might not be true. Concretely, if $m_{i-1} \notin \mathcal{R}_{i-1}(m_0, \rho_1, m_1, \ldots, \rho_{i-1})$ then

$$\Pr_{\rho_i}\left[\exists m_i \in \mathcal{R}_i(m_0, \rho_1, m_1, \ldots, \rho_i)\right] \leq \varepsilon_i,$$

independent of the prior messages $m_0, \rho_1, m_1, \ldots, m_{i-1}$, where the probability is taken over the space of all verifier messages of the i-th round. In other words, if any (computationally unbounded) prover strategy $\mathcal{A}$ is able to answer the verifier challenge ρ_i with a message m_i compliant with $\mathcal{R}_i$, with a probability

$$\Pr\left[m_i \in \mathcal{R}_i(m_0, \rho_1, m_1, \ldots, \rho_i) \middle| \begin{array}{c} \rho_i \leftarrow\!\!\$\ F \\ m_i \leftarrow \mathcal{A}(m_0, \rho_1, m_1, \ldots, m_{i-1}, \rho_i, \mathsf{aux}) \end{array}\right] > \varepsilon_i,$$

then the previous message $m_{i-1} \in \mathcal{R}_{i-1}(m_0, \rho_1, m_1, \ldots, \rho_{i-1})$.

We stress the fact that our definition via randomized reductions is merely a different way to express *round-by-round soundness* [CCH+18,CMS19] of the interactive oracle proof, which uses a state function to tell whether $m_i \in \mathcal{R}_i$ or not. Further, the soundness errors $\varepsilon_1, \ldots, \varepsilon_r$ from above coincide with the separate round-by-round soundness errors, proving $(\varepsilon_1, \ldots, \varepsilon_r)$-round-by-round soundness in the sense of [CY].

Round-by-round knowledge soundness. Likewise, the soundness errors of the reduction chain equal the *round-by-round knowledge errors* in [CMS19], thanks to straight-line extraction: Whenever the prover message m_i happens to be in $\mathcal{R}_i$ with a probability larger than

$$\Pr[m_i \in \mathcal{R}_i(m_0, \rho_1, m_1, \ldots, \rho_i)] > \varepsilon_1 + \ldots + \varepsilon_i,$$

then also $m_0 \in \mathcal{R}_0$ (by the usual union bound of "bad" events) and we may apply the Guruswami–Sudan (or Berlekamp–Welch) decoder for circle codes, Theorem 7 and Remark 8, to extract a witness from m_0, the proximate claimed by the main relation $\mathcal{R}_0$.

A.2 Correlated agreement under constraints

In this section we derive the key tools for the soundness analysis of multi-table FRI, Protocol 3. These are two strengthened statements on correlated agreement, Theorem 28 and 31 below, which are a byproduct of the proof in [BCI+20], yet require an explicit discussion.

We focus on the Johnson regime, with proximity parameter

$$\theta \in \left[\frac{\delta}{2},\, 1 - \sqrt{1-\delta}\right),$$

where δ is the miminum distance of the Reed–Solomon code $\mathcal{C} = \mathsf{RS}(F, D, k)$, generated by the space of polynomials $L = F[X]^{<k}$ of degree less than k, over the finite field F, with evaluation domain $D \subseteq F$. (The unique decoding regime will be discussed in a line of remarks.) Instead of the combinatorial Johnson bound (53), we use

$$\ell_{GS}(\theta) = \frac{m(\theta) + \frac{1}{2}}{\sqrt{1-\delta}},$$

the Guruswami–Sudan bound with multiplicity parameter

$$m(\theta) = \max\left\{ \left\lceil \frac{\sqrt{1-\delta}}{2(\alpha - \sqrt{1-\delta})} \right\rceil, 3 \right\},$$

where $\alpha = 1-\theta$, to bound the number of θ-close codewords. This bound emerges naturally in the algebraic analysis of the Guruswami–Sudan list decoder. (Recall that it is slightly larger than the combinatorial bound.)

Collinearity of proximates and curve-decodability. Given two functions $f_0, f_1 \in F^D$, and assume that sufficiently many linear combinations $f_z = f_0 + z f_1$ are close to the code $\mathcal{C}$. A *correlated agreement theorem* says that then, there exist codewords $c_0, c_1 \in \mathcal{C}$ which agree with f_0 and f_1 on a *joint* set $A \subseteq D$ of large density.

The study of correlated agreement is tightly related to showing that a noticable fraction of proximate codewords are collinear, meaning they again lie on a line. This is the key property behind all known agreement theorems, starting with the works on general linear codes [AHIV17,RZ18] and [BBHR18,BKS18,BGKS20], as well as the later improvements for Reed–Solomon codes [BCI+20,BCH+25], including the recent [GG25] analysis of random ones. The latter frames the property as *line-decodability*, or in the more general case of curves

$$f(z) = f_0 + z f_1 + \ldots + z^M f_M, \tag{91}$$

with $f_0, \ldots, f_M \in F^n$, as *curve-decodability.*

Definition 24 (Curve-decodability [GG25])**.** A linear code $\mathcal{C} \subseteq F^n$ is *degree-M curve-decodable* at distance θ, with *threshold* $a > 0$ and *concurrency number* $b > 0$, if the following holds for any degree-M curve $f : F \longrightarrow F^n$: *For any selection of codewords $\{p_z\}_{z \in S}$, with $d(f(z), p_z) \leq \theta$ for every $z \in S$, and $|S| > a$, there exists a degree-M curve of codewords $P : F \longrightarrow \mathcal{C}$, so that $|\{z \in S : p_z = P(z)\}| > b$.*

In the above definition, the degree-M curve of codewords can be again represented as in (91), spanned by codewords $P_0, \ldots, P_M \in \mathcal{C}$. These are the codewords which then agree with $f_0, \ldots, f_M$ on a joint set of large density, which is shown via a well-known argument that goes back to the early works of [AHIV17,RZ18] and [BBHR18]. We will reproduce it in our proofs of Theorem 28 and Theorem 31.

Curve-decodability up to Johnson radius. Let us frame the main result of [BCI+20] in the context of curve-decodability, and with the improved error bounds from [BCH+25].

Theorem 25 ([BCI+20,BCH+25])**.** *Let $\mathcal{C} = \mathsf{RS}[F, D, k]$ be any Reed–Solomon code over an arbitrary finite field F, generated by $F[X]^{<k}$ and with evaluation domain $D \subseteq F$. Then $\mathcal{C}$ is curve-decodable for distances $\theta \in \left[\frac{\delta}{2}, 1 - \sqrt{1-\delta}\right)$, where δ is the minimum distance of the code, for threshold*

$$a = \ell_{GS}(\theta) \cdot b$$

and concurrency number

$$b \geq \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1 - \delta) + 1 \right) \cdot M \cdot |D|.$$

In words, the theorem states the following. Given a degree-M curve of functions spanned by $f_0, \ldots, f_M$ as in (91), then, whenever we have a collection of more than

$$a = \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1 - \delta) + 1 \right) \cdot M \cdot |D|$$

proximates, then at least a $(1/\ell_{GS}(\theta))$-fraction of them are lying on a degree-M curve

$$P(X, Z) = P_0(X) + Z \cdot P_1(X) + \ldots + Z^M \cdot P_M(X),$$

with polynomials $P_0(X), \ldots P_M(X) \in F[X]^{<k}$.

Remark 26. For simplicity, our bounds in Theorem 25 are taken slightly larger than in [BCH+25], which in full length are

$$a = \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1 - \delta) + \frac{\theta|D| + 1}{|D|} \right) \cdot M \cdot |D|,$$

with $b = a/\ell_{GS}(\theta)$. Our modified statement is a convenience choice, and easily obtained by replacing the term $\theta|D| + 1$ by $|D|$ in the analysis of [BCH+25, Section 3.2]. We stress the fact, that all statements in this Section also hold for the slightly smaller bound, and can be obtained by a minor modification of their proofs.

Remark 27 (Unique decoding regime)**.** For distances $\theta \in (0, \delta/2)$, the proof of [BCI+20, Theorem 6.1] shows that Theorem 25 holds for any

$$a = b \geq M \cdot |D|,$$

saying that all proximates are lying on the same curve. This bound further improved by [BCH+25, Theorem 4.1] to $M \cdot (\theta \cdot |D| + 1)$, holding up to $\theta = \frac{\delta}{2} - \frac{3}{n\delta}$, and given that the minimum distance of the code is not too small.

Correlated agreement over given sets. The following strengthening relates the agreement sets of the linear combinations (with their proximate polynomials) with the set of joint agreement. Concretely, given a sufficiently large selection of θ-proximates, then at least over one of its agreement sets we must have correlated agreement. The proof just takes a closer look into the aforementioned finalization argument, for showing that collinearity implies correlated agreement.

Theorem 28 (CA over given sets)**.** *For $\theta \in [\delta/2, 1 - \sqrt{1-\delta})$ and functions $f_0, f_1, \ldots, f_M \in F^D$. Let $\mathcal{A} = \{A_z\}_{z \in S}$ be any family of sets of density $1 - \theta$, over which $f(z) = f_0 + z \cdot f_1 + \ldots + z^M \cdot f_M$ coincides with some word from $\mathcal{C}$. If*

$$|S| > \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1 - \delta) + 1 \right) \cdot M \cdot |D|,$$

with $\ell_{GS}(\theta)$ as above, then there exists $z_0 \in S$ and $(p_0, p_1, \ldots, p_M) \in \mathcal{C}^{M+1}$, so that $(p_0, p_1, \ldots, p_M)|_A = (f_0, f_1, \ldots, f_M)|_A$ over $A = A_{z_0}$.

Proof. Select polynomials $\{p_z\}_{z \in S}$ from $F[x]^{<k}$ satisfying $p_z|_{A_z} = f_z|_{A_z}$ for every $z \in S$. By Theorem 25 there is a curve of polynomials

$$P(X, Z) = P_0(X) + Z \cdot P_1(X) + \ldots + Z^M \cdot P_M(X),$$

spanned by $P_0(X), \ldots, P_M(X) \in F[X]^{<k}$ so that $p_z(X) = P(z, X)$ for more than $b > M \cdot |D|$ values of $z \in S$. We denote this subset of values as S'.

Now, consider the joint agreement set

$$A = \left\{ x \in D \,:\, (f_0(x), \ldots, f_M(x)) = (P_0(x), \ldots, P_M(x)) \right\}.$$

Obviously, for every $x \in A$ we also have $f_z(x) = P(x, z)$ for every $z \in F$. On the other hand, at each $x \in D \setminus A$, we have disagreement between $(f_0, \ldots, f_M)$ and $(p_0, \ldots, p_M)$. If a scalar $z \in S'$ does not resolve any of the disagreements, meaning that

$$f_0(x) + z \cdot f_1(x) + \ldots + z^M \cdot f_M(x) \neq p_0(x) + z \cdot p_1(x) + \ldots + z^M \cdot p_M(x),$$

for every $x \in D \setminus A$, then $A_z = \{x \in D \,:\, f_z(x) = P_z(x)\} = A$. We show that by the size of S', it must contain such a scalar. By the degree of the curve, the number of z which resolve at least one of these disagreements, meaning that the above inequality becomes an equality at some point $x \in D \setminus A$, is bounded by

$$M \cdot |D \setminus A| \leq M \cdot |D|,$$

M for each point of disagreement. Since $|S'| > M \cdot |D|$ there must be at least one $z_0 \in S'$ which does not resolve any of the disagreements. For this scalar we have $A_{z_0} = A$, proving the claim of the theorem. □

Weighted correlated agreement. A interesting consequence of Theorem 28 is that it immediately implies the *weighted* agreement theorem from [BCI+20] in a simpler form than therein, and with no restriction on the weights. Assume a sub-probability measure on D of the form

$$\mu(\{x\}) = \frac{\delta(x)}{|D|},$$

for some function δ with values in $[0, 1]$. We again relate to the improved bounds from [BCH+25].

Theorem 29 (CA with weights)**.** *For $\theta \in [\delta/2, 1 - \sqrt{1-\delta})$, and μ as above. Given functions $f_0, f_1, \ldots, f_M \in F^D$, let*

$$S = \big\{z \in F \,:\, \exists p_z \in \mathcal{C}, \mu(\{x : f_z(x) = p_z(x)\}) \geq 1 - \theta\big\}.$$

Then, if

$$|S| > \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1 - \delta) + 1\right) \cdot M \cdot |D|,$$

there exist polynomials $(p_0, \ldots, p_M) \in \mathcal{C}^{M+1}$, and a set A of weight $\mu(A) \geq 1 - \theta$ such that $(p_0, \ldots, p_M)|_A = (f_0, \ldots, f_M)|_A$.

Proof. Since μ is dominated by the regular density, the agreement sets A_z, $z \in S$, satisfy $|A_z|/|D| \geq 1 - \theta$. We now apply Theorem 28 to see that, over one of these agreement sets we have correlated agreement. □

The weighted agreement theorem is an essential tool in the security analysis of interactive oracle proofs, which query the oracles across the rounds in a correlated manner. It is useful for FRI [BCI+20] or Basefold [Hab24a], and for the same reason it plays an important role in our cross-domain analysis of multi-table FRI.

Remark 30 (Unique decoding regime)**.** Based on Remark 27, the proof of both Theorem 28 and Theorem 29 carry over to the unique decoding regime, with the smaller bound $|S| > M \cdot |D|$.

Correlated agreement with linear constraints. Another corollary of Theorem 25 relates to linear subspaces of Reed–Solomon codes [Hab24a]. We state the theorem in a slightly more general form, and with the improved bounds from [BCH+25].

Theorem 31 (Correlated agreement under linear constraints)**.** *For finite extensions $\mathbb{F}_q \leq F \leq K$, and $\mathcal{C}'$ an F-linear subspace of the Reed-Solomon code $\mathcal{C} = \mathsf{RS}[K, D, k]$. Given $\theta \in [\delta/2, 1-\sqrt{1-\delta})$ and functions $f_0, f_1, \ldots, f_M \in K^D$. If*

$$\left|\left\{z \in F : d(f(z), \mathcal{C}') \leq \theta\right\}\right| > \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta) + 1\right) \cdot M \cdot |D|,$$

where $f(z)$ is as in (91)*, then there exist polynomials $p_0, \ldots, p_M$ belonging to the subspace $\mathcal{C}'$, and a set A of density $1-\theta$, over which $(f_0, \ldots, f_M)|_A = (p_0, \ldots, p_M)|_A$.*

Proof. Any F-linear subspace L' of $L = K[x]^{<k}$ is the intersection of the nullspaces of finitely many F-linear functionals $\Lambda : L \longrightarrow F$. Let

$$P(Z, X) = P_0(X) + ZP_1(X) + \ldots + Z^M P_M(X)$$

be the curve of polynomials given by Theorem 25, with more than $b > M \cdot |D|$ proximates lying on it. By F-linearity we see that, if $\Lambda P(z, X) = 0$ for more than M different $z \in F$, then also $\Lambda P_i = 0$ for each $i = 0, \ldots, M$. Since Λ was arbitrary, each $P_i(X)$ must be in the subspace L'.

The remainder of the proof is as in for Theorem 28, showing that these polynomials constitute the codewords from $\mathcal{C}'$ with the claimed correlated agreement with $f_0, \ldots, f_M$. □

Remark 32 (Unique decoding regime)**.** Again, using Remark 27, the proof of Theorem 31 carries over verbatim to the unique decoding regime, with the smaller bound $|S| > M \cdot |D|$.

Over the circle. Recall that using a univariate parametrization of the circle curve, circle codes translate to Reed-Solomon codes (Section 1.4), and Theorem 31 can be directly stated as follows.

Corollary 1 (Circle CAT under linear constraints)**.** *For finite extensions F and K, with $\mathbb{F}_q \leq F \leq K$, and $\mathcal{C}'$ an F-linear subspace of $\mathcal{C}_N(K, D)$ the circle code over K with evaluation domain $D \subseteq C(\mathbb{F}_q)$ and minimum distance $\delta = 1 - N/|D|$. Given $\theta \in [\delta/2, 1-\sqrt{1-\delta})$ and functions $f_0, f_1, \ldots, f_M \in K^D$, if*

$$|\{z \in F\ : d(f_z, \mathcal{C}') \leq \theta\}| > \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta) + 1\right) \cdot M \cdot |D|,$$

there exist polynomials $p_0, \ldots, p_M$ from the subspace $\mathcal{C}'$, and $A \subseteq D$ of density $|A|/D| \geq 1 - \theta$, over which $(f_0, \ldots, f_M)|_A = (p_0, \ldots, p_M)|_A$.

Again, the result holds in the unique decoding regime, with the smaller error bound, compare Remark 32.

A.3 Multi-domain circle FRI

In this section we prove that multi-table FRI, Protocol 3, enforces cross-domain correlated agreement. The analysis makes use of the two refinements of correlated agreement from Appendix A.2, the set-specific Theorem 28, and the Theorem for linearly constrained codes, Corollary 1. The proofs are written for the Johnson regime, that is for proximity parameters

$$\theta \in \left[\frac{\delta}{2}, 1 - \sqrt{1-\delta}\right),$$

where $\delta = 1-\rho$ is the minimum distance of the circle codes. Their adaptation to the unique decoding regime is discussed separately in Remark 20. We assume the reader is familiar with the notation introduced in Section 4, and for convenience we use

$$\mathcal{L}_i := \mathscr{L}'_{n-i}(F)$$

for the polynomial spaces behind the circle codes over the respective domains.

Liftings. As for the lifting to the largest circle domain introduced in Section 4, we define for any $0 \leq k \leq i \leq r$, and function $f_{i,j} \in F^{D_i}$ its *lifting* to D_k as

$$f_{i,j}^{(k)} := f_{i,j} \circ \pi^{i-k}, \tag{92}$$

and likewise we do for the function spaces

$$\mathcal{L}_i^{(k)} := \mathcal{L}_i \circ \pi^{i-k}, \tag{93}$$

with $\mathcal{L}_i = \mathscr{L}'_{n-i}(F)$ as above. We use the same convention for functions over line domains, writing $h_{i,j}^{(k)} = h_{i,j} \circ \pi^{i-k}$ for the lifting of $h_{i,j} \in F^{S_i}$ to the line domain S_k, and

$$L_i^{(k)} = L_i \circ \pi^{i-k}, \tag{94}$$

where $L_i = F[x]^{<N/2^{i+1}}$ as before. (Again, we use the same notation for the group squaring map and its projection onto the line.) When viewing at all functions from smaller domains, together with odd and even parts of the previous folding, the joint spaces

$$L^{(k)} := L_k^2 \times \prod_{i=k}^{r} (L_i^{(k)})^2, \tag{95}$$

$0 \leq k \leq r$, will be the appropriate function spaces. In the edge case $k = 0$, the L_0^2 is omitted in the definition of $L^{(0)}$.

Anatomy of a query round. For the analysis it is useful to think of a single query round as a probabilistic process, which checks consistency of the foldings in the a bottom-up manner, while postponing the verification of the cross-domain batching step to the very last.

We start by drawing $x_r \leftarrow\!\!\$\, S_r$ uniformly from the smallest line domain, check the collinearity relation (70) at x_r,

$$\begin{aligned}
g_r(x_r) = &\frac{g_{r-1}(x) + g_{r-1}(-x)}{2} + \lambda_r \cdot \frac{g_{r-1}(x) - g_{r-1}(-x)}{2 \cdot x} \\
&+\lambda_r^2 \cdot \left(\frac{h_r(x_r, y) + h_r(x_r, -y)}{2} + \lambda_r \cdot \frac{h_r(x_r, y) - h_r(x_r, -y)}{2 \cdot y} \right),
\end{aligned}$$

using the values of g_{r-1} at the points from the fiber $\pi^{-1}(x_r) = \{\pm x\}$, and the values of h_r at the circle points from $\pi_X^{-1}(x_r) = \{(x_r, \pm y)\}$. If consistent, we choose with which of the two preimages from $\{\pm x\}$ to continue, by drawing $x_{r-1} \leftarrow\!\!\$\, \pi^{-1}(x_r)$, and checking (70) at x_{r-1}. Continuing in this way, sampling the preimages uniformly from the corresponding fibers, we obtain correlated $x_0, x_1, \ldots, x_r$, each uniformly distributed over the corresponding line domains $S_0, S_1 \ldots, S_r$,

$$x_0 \xrightarrow{\pi} x_1 \xrightarrow{\pi} \ldots \xrightarrow{\pi} x_r,$$

with consistent folding checks (70), $i = 0, \ldots, r$. (If one of the checks does not succeed, the verifier aborts.)

In the final step we sample the circle points above the x_i in a correlated manner. We draw $P_0 = (x_0, y_0) \leftarrow\!\!\$\, \pi_X^{-1}(x_0)$, and do the correlated check of the batching relations (69),

$$h_i(P_i) = \sum_{j=1}^{M} \lambda^{j-1} \cdot f_{i,j}(P_i),$$

at $P_i = \pi^i(P_0)$. Since π_X and the group squaring map π commute, the obtained points P_i are in fact preimages of x_i,

$$\begin{array}{ccccccc}
P_0 & \xrightarrow{\pi} & P_1 & \xrightarrow{\pi} & \cdots\cdots & \xrightarrow{\pi} & P_r \\
\downarrow{\scriptstyle \pi_x} & & \downarrow{\scriptstyle \pi_x} & & & & \downarrow{\scriptstyle \pi_x} \\
x_0 & \xrightarrow{\pi} & x_1 & \xrightarrow{\pi} & \cdots\cdots & \xrightarrow{\pi} & x_r
\end{array}$$

and we thus have selected a point above each x_i in uniformly distributed but correlated manner. The obtained distribution of P_0, and its trace under π and π_X, is as in Step 2 of Protocol 3.

Consistency weights. To capture the consistency of what has been committed so far, from round to round, we introduce the conditional success probabilities

$\delta \in [0,1]^{D_0}$, and $\delta_i \in [0,1]^{S_i}$, $i = 0, \ldots, r$, of a query round that would start with x from the respective domain, and then works upwards as described above.

For the cross-domain batching step, the conditional probability is equal to one, if the consistency (70) holds along the trace of $P \in D_0$, and zero otherwise. In terms of the liftings,

$$\delta(P) := \begin{cases} 1 & \text{if } h_i^{(0)}(P) = \sum_{j=1}^{M_i} \lambda^j \cdot f_{i,j}^{(0)}(P) \text{ for all } i = 0, \ldots, r, \\ 0 & \text{otherwise.} \end{cases} \tag{96}$$

The success probability of a minimal query phase, which starts over D_0 and only checks consistency of the batching step, would be equal to $\mu(D_0)$, with μ being the subprobability measure over D_0 with weights δ, i.e.

$$\mu(\{P\}) = \frac{\delta(P)}{|D_0|}.$$

The other probabilities are the conjunction of the corresponding folding step with the remaining average consistency. That is,

$$\delta(x) := \begin{cases} \delta^*(x) & \text{if } g_0(x) = h_{0,0}(x) + \lambda_0 \cdot h_{0,1}(x) \\ 0 & \text{otherwise,} \end{cases} \tag{97}$$

with $\delta^*(x) = \frac{1}{2} \cdot (\delta(x,y) + \delta(x,-y))$ the expected consistency over the fiber $\pi_X^{-1}(x)$, and for $i = 1, \ldots, r$,

$$\delta_i(x) := \begin{cases} \delta_{i-1}^*(x) & \text{if } g_i(x) = g_{i-1,0}(x) + \lambda_i \cdot g_{i-1,1}(x) \\ & \qquad\qquad +\lambda_i^2 \cdot (h_{i,0}(x) + \lambda_i \cdot h_{i,1}(x)) \\ 0 & \text{otherwise,} \end{cases} \tag{98}$$

where $\delta_{i-1}^*(2 \cdot x^2 - 1) = \frac{1}{2} \cdot (\delta_{i-1}(x) + \delta_{i-1}(-x))$. Their sub-probability measures μ_i over S_i are defined by

$$\mu_i(\{x\}) := \frac{\delta_i(x)}{|S_i|}. \tag{99}$$

As for the densities, we define the projected measures $\mu^* = \mu \circ \pi_X^{-1}$ and $\mu_i^* = \mu_i \circ \pi^{-1}$, $i = 0, \ldots, r-1$. Note that μ^* has density δ^*, and μ_i^* has the density δ_i^*.

Round-by-round soundness. Given proximity parameter $\theta \in (0,1)$, the multi-domain variant of circle FRI proves correlated agreement of all lifted functions to lifted circle polynomials of corresponding degree,

$$\mathcal{R}_\theta := \left\{ (f_{i,j}) \in \prod_{i=0}^{r} (F^{D_i})^{M_i} \ : \ \begin{array}{c} \exists (p_{i,1}, \ldots, p_{i,M_i}) \in \mathcal{L}_i^{M_i}, 0 \le i \le r, \\ \exists A \subseteq D_0, |A|/|D_0| \ge 1 - \theta, \\ (f_{i,1}^{(0)}, \ldots, f_{i,M_i}^{(0)})|_A = (p_{i,1}^{(0)}, \ldots, p_{i,M_i}^{(0)})|_A, \\ \text{for all } i = 0, \ldots, r. \end{array} \right\}. \tag{100}$$

In the cross-domain batching step, Step 1a in Protocol 3, the verifier challenge $\lambda \leftarrow\!\!\$\ F$ reduces $\mathcal{R}_\theta$ to correlated agreement of the lifted domain combinations $(h_i^{(0)})_{i=0}^r$, more concretely to correlated agreement of their lifted line components,

$$\mathcal{R}_0 := \left\{ (h_i)_i \in \prod_{i=0}^{r} F^{D_i} \;:\; \begin{array}{c} \exists (p_{i,0}, p_{i,1}) \in L_i^2, 0 \leq i \leq r, \\ \exists A \subseteq S_0, \mu^*(A) \geq 1 - \theta, \\ (h_{i,0}^{(0)}, h_{i,1}^{(0)})|_A = (p_{i,0}^{(0)}, p_{i,1}^{(0)})|_A, \\ \text{for all } i = 0, \ldots, r. \end{array} \right\}, \tag{101}$$

which we directly write as

$$\mathcal{R}_0 = \left\{ (h_{i,j}) \;:\; d_{\mu^*}((h_{i,j}^{(0)}), L^{(0)}) \leq \theta \right\}.$$

In this notation, which will be used in the sequel, the weighted distance d_{μ^*} is synonymous for agreement over a set A of measure $\mu^*(A) \geq 1 - \theta$. The other steps, Step 1b in Protocol 3, correspond to the folding along the chain of line domains from S_0 down to S_r. Each random $\lambda_k \leftarrow\!\!\$\ F$, $k = 0, \ldots, r$, reduces $\mathcal{R}_k$ to $\mathcal{R}_{k+1}$, where

$$\mathcal{R}_{k+1} := \left\{ \begin{array}{c} (h_{i,j})_{i \geq k+1} \\ (\lambda_0, g_0, \ldots, \lambda_k, g_k) \end{array} \;:\; d_{\mu_k^*}\left(\left[(g_{k,j}), (h_{i,j}^{(k+1)})_{i \geq k+1} \right], L^{(k+1)} \right) \leq \theta \right\}, \tag{102}$$

except for the final case $k = r$, where

$$\mathcal{R}_{r+1} := \left\{ (\lambda_0, g_0, \ldots, \lambda_r, g_r) \;:\; g_r \in L_r \;\wedge\; \mu_r(D_r) \geq 1 - \theta \right\}. \tag{103}$$

The relation $\mathcal{R}_{r+1}$ is the final relation of the commit phase, which is eventually reduced by the query phase, Step 2 in Protocol 3, to the verifier relation over the set $S \subseteq D_0$ of $s \geq 1$ independent and uniformly sampled points from D_0,

$$\mathcal{V}_S = \left\{ \begin{array}{c} (f_{i,j})_{i \geq 0} \\ (\lambda_0, g_0, \ldots, \lambda_r, g_r) \end{array} \;:\; \begin{array}{c} g_r \in \mathcal{L}_r \text{ and all collinearity checks} \\ \text{hold along the trace of each of the} \\ \text{points from } S \end{array} \right\}. \tag{104}$$

Overall, Protocol 3 consists of the following random reductions,

$$\begin{array}{l} \mathcal{R}_\theta \\ \Big\downarrow \lambda \\ \mathcal{R}_0 \xrightarrow{\lambda_0} \mathcal{R}_1 \xrightarrow{\lambda_1} \cdots\cdots\cdots \xrightarrow{\lambda_r} \mathcal{R}_{r+1} \xrightarrow{S} \mathcal{V}_S \end{array}$$

each with its own soundness error, quantifying the threshold probability above which one can do the desired reverse conclusion, clarified in Lemma 3 and Lemma 4 below.

Recall that the circle codes over the domains $D_0, \dots, D_r$ are of the same minimum distance $\delta = 1 - \rho$, where $\rho = 2^{-\beta}$ is the rate of the circle IOPP. (The Reed-Solomon codes over the line domains $S_0, \dots, S_r$ have slightly larger distance.) For proximity parameter $\theta \in [\delta/2, 1 - \sqrt{1-\delta})$, where δ is the minimum distance, recall the Guruswami–Sudan bound

$$\ell_{GS}(\theta) = \frac{m(\theta) + \frac{1}{2}}{\sqrt{1-\delta}},$$

with

$$m(\theta) = \max\left\{ \left\lceil \frac{\sqrt{1-\delta}}{2 \cdot (\alpha - \sqrt{1-\delta})} \right\rceil, 3 \right\}$$

being the smallest multiplicity parameter $m \geq 3$ that supports agreement $\alpha = 1 - \theta$.

Lemma 3 (Soundness cross-domain batching)**.** *For $\theta \in [\delta/2, 1 - \sqrt{1-\delta})$. Step 1a in Protocol 3 is a random reduction $\mathcal{R}_\theta \xrightarrow{\lambda} \mathcal{R}_0$ with soundness error*

$$\varepsilon \leq \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot \rho + 1 \right) \cdot (M - 1) \cdot \frac{|D_0|}{|F|},$$

where $M = \max_i M_i$ and $\ell_{GS}(\theta)$ is the Guruswami-Sudan bound for the circle code with minimum distance $\delta = 1 - \rho$. In words, if a (computationally unbounded) algorithm is able to answer with $(h_i)_{i=0}^r \in \mathcal{R}_0$ on $\lambda \leftarrow\!\!\$\ F$, with a probability larger than ε, then the ensemble $(f_{i,j})$ belongs to $\mathcal{R}_\theta$.

Proof. Without loss in generality we assume that the batches are of the same height M. (Otherwise, fill up with zero-words to obtain the same height.)

We say $\lambda \in F$ is *good*, if the algorithm is able to answer with functions $(h_i)_{i=0}^r$ from $\mathcal{R}_0$, meaning that their lifted line components $(h_{i,0}^{(0)}, h_{i,1}^{(0)})_i$ agree with liftings of line polynomials $(p_{i,0}, p_{i,1}) \in L_i^2$, $0 \leq i \leq r$, over a joint set $A_\lambda \subseteq S_0$ of measure $\mu^*(A_\lambda) \geq 1 - \theta$. We write $p_i(x, y) = p_{i,0}(x) + y \cdot p_{i,1}(x) \in \mathcal{L}_i$ for the composed circle polynomials. Note that $h_i = h_{i,0} + y \cdot h_{i,1}$ is lifted to

$$h_i^{(0)}(x, y) = h_{i,0}^{(0)}(x) + y \cdot w_{i+1}(x) \cdot h_{i,1}^{(0)}(x),$$

and likewise for the circle polynomials, where $w_{i+1}(x) \in \mathbb{F}_q[x]^{<2^i}$ is the vanishing polynomial as defined in (12). Thus correlated agreement between the lifted line components implies correlated agreement between the circle liftings,

$$(h_0^{(0)}, \dots, h_r^{(0)})\big|_{\pi_X^{-1}(A_\lambda)} = (p_0^{(0)}, \dots, p_r^{(0)})\big|_{\pi_X^{-1}(A_\lambda)},$$

over the preimage $\pi_X^{-1}(A_\lambda)$. Restricting to the support of the consistency measure, i.e.

$$A'_\lambda = \left\{ P \in \pi_X^{-1}(A_\lambda) \ : \ \delta(P) > 0 \right\},$$

we obtain a set of density $|A'_\lambda|/|D_0| \geq \mu(A'_\lambda) = \mu^*(A_\lambda) \geq 1-\theta$, over which

$$(p_0^{(0)}, \ldots, p_r^{(0)})\big|_{A'_\lambda} = \sum_{j=1}^{M} \lambda^{j-1} \cdot (f_{0,j}^{(0)}, \ldots, f_{r,j}^{(0)})\big|_{A'_\lambda},$$

as vector-valued equality. In words, for every good λ the random linear combinations of each of the batches agree with lifted polynomials from the respective space $\mathscr{L}_i$, over a joint set of density $1-\theta$. To conclude that the same holds also for the components of the linear combinations, we will use the correlated agreement theorem for subspaces, Corollary 1.

To this end, we use an extension field K of degree $[K : F] = r+1$ to encode vectors as field elements,

$$f_j := f_{0,j}^{(0)} + a \cdot f_{1,j}^{(0)} + \ldots + a^r \cdot f_{r,j}^{(0)} \in K^{D_0},$$

$j = 1, \ldots, M$ (the element a is a primitive root of K) and obtain that over the set A'_λ, we have agreement of the linear combination $\sum_{j=1}^{M} \lambda^{j-1} \cdot f_j$ with the polynomial

$$p := p_0^{(0)} + a \cdot p_1^{(0)} + \ldots + a^r \cdot p_r^{(0)},$$

from the

$$\mathscr{L} := \sum_{i=0}^{r} a^i \cdot \mathscr{L}_i^{(0)},$$

which is an F-linear subspace of $\mathcal{L}_{\rho\cdot|D_0|}(K)$ the space of all circle polynomials over K of total degree at most $\rho \cdot |D_0|/2$. The corresponding circle code has distance $\delta = 1-\rho$, and by Corollary 1, we arrive at the following conclusion: There exist polynomials from the subspace,

$$p_j = p_{j,0}^{(0)} + a \cdot p_{j,1}^{(0)} + \ldots + a^r \cdot p_{j,r}^{(0)} \in \mathscr{L},$$

$j = 1, \ldots, M$, and a set $A \subseteq D_0$ of density $1-\theta$, over which

$$p_{j,0}^{(0)} + a \cdot p_{j,1}^{(0)} + \ldots + a^r \cdot p_{j,r}^{(0)}\Big|_A = f_{0,j}^{(0)} + a \cdot f_{1,j}^{(0)} + \ldots + a^r \cdot f_{r,j}^{(0)}\Big|_A.$$

Translated to the F-components, we arrive at that $(f_{i,j})$ belongs to $\mathcal{R}_\theta$. $\square$

Lemma 4 (Soundness folding steps)**.** *For any $k = 0, \ldots, r$ and proximity parameter $\theta \in [\delta_k/2, 1-\sqrt{1-\delta_k})$, where $\delta_k = 1-\rho+1/|S_k|$ is the minimum distance of the Reed–Solomon code over S_k. Round k in Step 1b of Protocol 3, is a randomized reduction $\mathcal{R}_k \xrightarrow{\lambda_k} \mathcal{R}_{k+1}$, with soundness error*

$$\varepsilon_k \leq \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta_k) + 1\right) \cdot 3 \cdot |S_k|,$$

where $\ell_{GS}(\theta)$ is the Guruswami-Sudan bound with minimum distance δ_k.

In words, if an arbitrary algorithm is able to answer on $\lambda_k \leftarrow\!\$\, F$ with $g_k \in F^{S_k}$ so that $(\lambda_0, g_0, \ldots, \lambda_k, g_k)$ and $(h_i)_{i\geq k+1}$ belong to $\mathcal{R}_{k+1}$, with a probability larger than ε_k, then the given $(\lambda_0, g_0, \ldots, \lambda_{k-1}, g_{k-1})$ and $(h_i)_{i\geq k}$ belong to $\mathcal{R}_k$.

Proof. Assume that $0 \leq k \leq r-1$. (The last step is similar, and postponed to the end of the proof.) For the sake of a uniform treatment, we assume that in the first round $k=0$ we have given the previous oracle $g_{-1}=0$. Given $g_0, \ldots, g_{k-1}$ and the not yet absorbed $(h_i)_{i \geq k}$, we say that $\lambda_k \in F$ is *good*, if the prover is able to respond with $g_k \in F^{S_k}$ so that the next-domain ensemble

$$g_{k,0}, g_{k,1},\ (h_{i,j}^{(k+1)})_{i \geq k+1}$$

agrees with

$$q_0, q_1,\ (p_{i,j}^{(k+1)})_{i \geq k+1}$$

from $L^{(k+1)}$ over a joint set $A_{\lambda_k} \subseteq S_{k+1}$ of measure $\mu_k^*(A_{\lambda_k}) \geq 1-\theta$. The preimage $\pi^{-1}(A_{\lambda_k})$ is a set of weight $\mu_k(\pi^{-1}(A_{\lambda_k})) = \mu_k^*(A_{\lambda_k})$, over which the ensemble

$$g_k,\ (h_{i,j}^{(k)})_{i \geq k+1}$$

agrees with

$$q_0(x) + x \cdot q_1(x),\ (p_{i,j}^{(k)})_{i \geq k+1} \tag{105}$$

from $L_k \times \prod_{i=k+1}^{r} (L_i^{(k)})^2$. Restricting to the support of the consistency measure, i.e.

$$A'_{\lambda_k} = \left\{ x \in \pi^{-1}(A_{\lambda_k}) \ :\ \delta_k(x) > 0 \right\},$$

we obtain a subset of weight $\mu_k(A'_{\lambda_k}) = \mu_k(\pi_X^{-1}(A_{\lambda_k})) = \mu_k^*(A_{\lambda_k})$, over which

$$g_{k-1,0} + \lambda_k \cdot g_{k-1,1} + \lambda_k^2 \cdot (h_{k,0} + \lambda_k \cdot h_{k,1}),\ (h_{i,j}^{(k)})_{i \geq k+1} \tag{106}$$

agrees with the ensemble from (105). Furthermore, since $\delta_k > 0$ on A'_{λ_k}, we have $\delta_k = \delta_{k-1}^*$ on A'_{λ_k}, and therefore $\mu_{k-1}^*(A'_{\lambda_k}) = \mu_k(A'_{\lambda_k}) \geq 1-\theta$. In words, for every good λ_k, the linear combination on the left-hand side of (106) agrees with a polynomial from L_k on a set A'_{λ_k} of μ_{k-1}^*-measure at least $1-\theta$, and on this set the liftings $(h_{i,j}^{(k)})_{i \geq k+1}$ agree with the lifted polynomials of respective degrees. We now apply the set-specific correlated agreement theorem, Theorem 28, to the collection of agreement sets

$$\mathcal{A} = \{A'_{\lambda_k}\}_{\lambda_k \in S}$$

stemming from all good $\lambda_k \in F$, denoted by the set S, and conclude that over one of the agreement sets $A \in \mathcal{A}$, the components of the random linear combinations jointly agree with polynomials from L_k, and we have agreement of the ensemble

$$g_{k-1,0}, g_{k-1,1}, h_{k,0}, h_{k,1},\ (h_{i,j}^{(k)})_{i \geq k+1}$$

with an element from $L^{(k)}$, over a set A of measure $\mu_{k-1}^*(A) \geq 1-\theta$. In other words,

$$(h_{i,j})_{i \geq k}, (\lambda_0, g_0, \ldots, \lambda_{k-1}, g_{k-1}) \in \mathcal{R}_k,$$

which shows the claim of the Lemma for $0 \leq k \leq r-1$.

The final reduction step $k = r$ for $\mathcal{R}_r \xrightarrow{\lambda_r} \mathcal{R}_{r+1}$ is proven in the same manner. The situation is even simpler as there are no further functions to be absorbed, and for this case the weighted correlated agreement theorem, Theorem 29, is sufficient to show the claimed soundness error. If an algorithm is able to respond on $\lambda_r \in F$ with a function $g_r \in L_r$ such that $\mu_r(D_r) \geq 1-\theta$, then the linear combination in (106) with $k = r$ coincides with $g_r \in L_r$ over the subset $A_\lambda = \{x \in D_r \,:\, \delta_r > 0\}$ of same measure $\mu_r(A_\lambda) = \mu_r(D_r)$. As before, $\delta^*_{r-1} = \delta_r$ on A_λ, and $\mu^*_{r-1}(A_\lambda) = \mu_r(A_\lambda) \geq 1-\theta$. From Theorem 29 we conclude that, if the probability of such λ_r is larger than ε_r as stated, we have correlated agreement of

$$g_{r-1,0}, g_{r-1,1}, h_{r,0}, h_{r,1}$$

with respective polynomials from L_r over a set A of weight $\mu^*_{r-1}(A) \geq 1-\theta$. This shows that

$$(h_{i,j})_{i \geq r}, (\lambda_0, g_0, \ldots, \lambda_{r-1}, g_{r-1}) \in \mathcal{R}_r,$$

and hence the claim of the Lemma for $k = r$. $\square$

We summarize our analysis in the following theorem. As discussed in the introduction, round-by-round soundness implies round-by-round knowledge soundness, with no change of the soundness errors.

Theorem 33 (Soundness multi-domain circle FRI)**.** *For proximity parameter $\theta \in \left[\frac{\delta}{2}, 1-\sqrt{1-\delta}\right)$ where $\delta = 1-\rho$ is the minimum distance of the circle codes. Protocol 3 is a round-by-round sound interactive oracle proof for*

$$\mathcal{R}_\theta := \left\{ (f_{i,j}) \in \prod_{i=0}^{r} (F^{D_i})^{M_i} \;:\; \begin{array}{c} \exists (p_{i,1}, \ldots, p_{i,M_i}) \in \mathcal{L}_i^{M_i}, 0 \leq i \leq r, \\ \exists A \subseteq D_0, |A|/|D_0| \geq 1-\theta, \\ (f^{(0)}_{i,1}, \ldots, f^{(0)}_{i,M_i})|_A = (p^{(0)}_{i,1}, \ldots, p^{(0)}_{i,M_i})|_A, \\ \textit{for all } i = 0, \ldots, r. \end{array} \right\},$$

with the round-wise soundness errors bounded by

1. *(Batching step, Protocol 3, 1a.)*

$$\varepsilon_{FRI,batch} = (M-1) \cdot \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta) + 1 \right) \cdot \frac{|D_0|}{|F|},$$

with $M = \max_i M_i$, and $\ell_{GS}(\theta)$ the Guruswami-Sudan list size bound for the circle code with minimum distance δ.

2. *(Folding steps, Protocol 3, 1b.) For folding round $k = 0, \ldots, r$,*

$$\varepsilon_{FRI,fold,k} = 3 \cdot \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot (1-\delta_{k+1}) + 1 \right) \cdot \frac{|S_k|}{|F|},$$

where $\ell_{GS}(\theta)$ is the Guruswami–Sudan list size bound for the line code over S_{k+1}, a Reed–Solomon code with minimum distance $\delta_{k+1} = 1-\rho+\frac{1}{|S_{k+1}|}$.

3. *(Query phase, Protocol 3, 2.)*

$$\varepsilon_{FRI,query} = (1-\theta)^s.$$

Proof. Follows from Lemma 3 and 4, and the fact that the success probability of a single query round is equal to $\mu_r(D_r) \geq 1-\theta$. □

Remark 34. Since the first folding step does not involve any previous commitments, one may even choose $\varepsilon_{FRI,fold,0} = \ell_{GS}(\theta) \cdot \left(\frac{2\ell_{GS}(\theta)^4}{3} \cdot \rho + 1\right) \cdot \frac{|S_0|}{|F|}$.

Remark 35 (Theorem 33 in UDR)**.** The entire analysis, Lemma 3 and 4, translates verbatim to the unique decoding regime, that is for distances θ up to $\delta/2$, where δ is the minimum distance of the circle codes. With the adjusted soundness errors for the correlated agreement theorems, Remark 32 and Remark 30, the obtained round-by-round errors are

1. (Batching step.) $\varepsilon_{FRI,batch} = (M-1) \cdot \frac{|D_0|}{|F|}$, with M as above,
2. (Folding steps.) $\varepsilon_{FRI,fold,k} = 3 \cdot \frac{|S_k|}{|F|}$, $k = 0, \ldots, r$,
3. (Query phase.) $\varepsilon_{FRI,query} = (1-\theta)^s$.

A.4 The complete circle IOPP

In this section we prove round-by-round soundness of Protocol 1, Theorem 15. We restrict to proximity parameters

$$\theta \in \left[0, 1 - \sqrt{1-\delta}\right),$$

where δ is the minimum distance of the circle codes, which covers both the Johnson regime as well as the unique decoding regime of the code.

For an ensemble of functions

$$\{f_{t,j}\} \in (\mathbb{F}_q^{D_1})^{M_1} \times \ldots \times (\mathbb{F}_q^{D_\tau})^{M_\tau}, \tag{107}$$

we write $\{f^*_{t,j}\}$ for their liftings to the largest evaluation domain D^* met in the course of the protocol, i.e. $D^* = G_{n+2\cdot\beta}$ with $n = \max_t n_t$, and we use the same notation for polynomials. Note that this is a larger domain than used in the cross-domain correlated agreement statements of Section 3.5. However, the agreement set density does not change and thus lifting to larger domains leads to an equivalent formulation of proximity.

Given the proximity parameter θ, the target of Protocol 1 is that the committed ensemble $\{f_{t,j}\}$ is θ-proximate to a solution of the flat AIR, in the sense of cross-domain correlated agreement: Their liftings to D^* agree on a joint set of density $\alpha = 1-\theta$ with the liftings of a collection of circle polynomials

$$\{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau}$$

that solves the flat AIR $\mathcal{A}$ on given inputs $\{\vec{I}_k\}$ and $\{\vec{O}_k\}$. In other words, the function ensemble $\{f_{t,j}\}$ belongs to the set

$$\mathcal{R}_{IOPP,\theta} = \left\{ \{f_{t,j}\} \; : \; \begin{array}{c} \exists \{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \\ d\left(\{p^*_{t,j}\}, \{f^*_{t,j}\}\right) < \theta, \\ \{p_{t,j}|_{H_t}\} \text{ satisfies (40) for } t = 1, \ldots, \tau, \\ \text{and (29) with (41) and (42)} \end{array} \right\}, \tag{108}$$

where d is the fractional Hamming distance over D^*. Recall that in the Johnson regime, we can bound the number of θ-proximate ensembles $\{p_{t,j}\}$ by

$$\ell(\theta) = \frac{\alpha - (1-\delta)}{\alpha^2 - (1-\delta)}, \tag{109}$$

the combinatorial Johnson bound for agreement parameter $\alpha = 1 - \theta$.[13] This bound is slightly smaller than the Guruswami–Sudan bound required in the analysis of circle FRI. For parameters below the unique decoding radius, we take $\ell(\theta) = 1$.

Logup fractional identity. In logUp the prover expresses the multi-set equality (29) of all used and yielded elements by means of multiplicity columns

$$\{p_{t,1}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q) \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)$$

subject to the identity

$$\begin{aligned} &\sum_{k=1}^{k_{out}} \frac{1}{X + O_k(Y)} + \sum_{t=1}^{\tau} \sum_{j=1}^{r_t} \sum_{P \in H_t} \frac{e_t(P)}{X + U_{t,j}(p_{t,1}(P), \ldots, p_{t,M}(P), Y)} \\ &\quad = \sum_{k=1}^{k_{in}} \frac{1}{X + I_k(Y)} + \sum_{t=1}^{\tau} \sum_{k=1}^{s_t} \sum_{P \in H_t} \frac{m_{t,k}(P)}{X + V_{t,k}(p_{t,1}(P), \ldots, p_{t,M}(P), Y)}, \end{aligned} \tag{110}$$

in the field of rational functions $\mathbb{F}_q(X, Y)$, where use and yield vectors are encoded as a polynomial in Y

$$U_{t,j} = \rho_{t,j} + U^{(t)}_{j,1}(p_{t,1}, \ldots, p_{t,M}) \cdot Y + \ldots + U^{(t)}_{j,\rho_{t,j}}(p_{t,1}, \ldots, p_{t,M}) \cdot Y^{\rho_{t,j}},$$

$$V_{t,k} = \sigma_{t,k} + V^{(t)}_{k,1}(p_{t,1}, \ldots, p_{t,M}) \cdot Y + \ldots + V^{(t)}_{k,\sigma_{t,k}}(p_{t,1}, \ldots, p_{t,M}) \cdot Y^{\sigma_{t,k}},$$

and likewise for the public inputs and outputs $\vec{I}_k$, $\vec{O}_k$, of the AIR. For simplicity we will work with the degree bound

$$r = \max_{t,j,k} \{\rho_{t,j}, \sigma_{t,k}\}. \tag{111}$$

[13] The interleaved code is a Reed–Solomon code with symbols over a finite extension of $\mathbb{F}_q$, and the same minimum distance δ.

Assuming that the multiplicities of each of the uses are less than the characteristic of the field $\mathbb{F}_q$, we may use this fractional identity to rephrase the target relation $\mathcal{R}_{IOPP,\theta}$ of the IOPP as

$$\mathcal{R}_0 = \left\{ \{f_{t,j}\} : \begin{array}{c} \exists \{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \\ d\left(\{p^*_{t,j}\}, \{f^*_{t,j}\}\right) < \theta, \\ \{p_{t,j}|_{H_t}\} \text{ satisfies (40) for } t = 1, \ldots, \tau, \\ \text{and the logUp identity (110)} \end{array} \right\}.$$

Round 1. In the first round, the logUp identity (110) over $\mathbb{F}_q(X, Y)$ is reduced to its validity at a single point $(\beta_0, \beta_1) \leftarrow\$ F^2$, i.e.

$$\begin{aligned} &\sum_{k=1}^{k_{out}} \frac{1}{X + O_k(\beta_0)} + \sum_{t=1}^{\tau} \sum_{j=1}^{r_t} \sum_{P \in H_t} \frac{e_t(P)}{\beta_0 + U_{t,j}(p_{t,1}(P), \ldots, p_{t,M}(P), \beta_1)} \\ &= \sum_{k=1}^{k_{in}} \frac{1}{X + I_k(\beta_0)} + \sum_{t=1}^{\tau} \sum_{k=1}^{s_t} \sum_{P \in H_t} \frac{m_{t,k}(P)}{\beta_0 + V_{t,k}(p_{t,1}(P), \ldots, p_{t,M}(P), \beta_1)}. \end{aligned} \tag{112}$$

Protocol 1 uses the centered running sums for proving (112), one for each table, which is translated into the existence of polynomials

$$\{\bar{u}_{t,j}, \bar{v}_{t,k}, \bar{w}_t\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{r_1+s_1+1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{r_\tau+s_\tau+1},$$

the values of which satisfy the domain identities

$$\bar{u}_{t,j}(P) \cdot (\beta_0 + U_{t,j}(p_{t,1}(P), \ldots, p_{t,M}(P), \beta_1)) = e_t(P), \tag{113}$$
$$\bar{v}_{t,k}(P) \cdot (\beta_0 + V_{t,k}(p_{t,1}(P), \ldots, p_{t,M}(P), \beta_1)) = m_{t,k}(P) \tag{114}$$

for all $P \in H_t$, and

$$\bar{w}_t(Q_{n_t} \cdot P) - \bar{w}_t(P) = -\frac{\nu_t}{N} - \sum_{j=1}^{r_t} \bar{u}_{t,j}(P) + \sum_{k=1}^{s_t} \bar{v}_{t,k}(P), \tag{115}$$

for every $P \in H_t$, where the table totals $\nu_t \in F$ sum up to

$$\sum_{k=1}^{k_{out}} \frac{1}{X + O_k(\beta_0)} + \sum_{t=1}^{\tau} \nu_t = \sum_{k=1}^{k_{in}} \frac{1}{X + I_k(\beta_0)}, \tag{116}$$

for the overall consistency of the communication channel. We write the set of functions committed by the prover again in chronological notation,

$$\{f_{t,M_t+1}, \ldots, f_{t,M_t+r_t+s_t+1}\} \in (F^{D_1})^{r_1+s_1+1} \times \ldots \times (F^{D_\tau})^{r_\tau+s_\tau+1}, \tag{117}$$

and we use the lazy notation $\{f_{t,j}\}$ also for all functions committed by the prover so far, that is (107), and (117). These, together with the claims $\{\nu_t\}$, are subject to

$$\mathcal{R}_1 = \left\{ \{f_{t,j}\}, \{\nu_t\} \ : \ \begin{array}{c} \exists \{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \\ \exists \{\bar{u}_{t,j}, \bar{v}_{t,k}, \bar{w}_t\} \in \mathscr{L}'_{n_1}(F)^{r_1+s_1+1} \times \cdots \times \mathscr{L}'_{n_\tau}(F)^{r_\tau+s_\tau+1} \\ d\left(\{p^*_{t,j}, \bar{u}^*_{t,j}, \bar{v}^*_{t,k}, \bar{w}^*_t\}, \{f^*_{t,j}\}\right) < \theta, \\ \text{and for every } t = 1, \ldots, \tau, \\ (40) \text{ and } (113), (114), (115) \text{ hold over } H_t, \\ \text{and } (116) \end{array} \right\}.$$

Lemma 5 (Soundness Round 1). *Round 1 of Protocol 1 is a randomized reduction from $\mathcal{R}_0$ to $\mathcal{R}_1$ with soundess error $\varepsilon \leq \ell(\theta) \cdot \frac{r \cdot \left(k_{in} + k_{out} + \sum_{t=1}^{\tau}(r_t+s_t)\cdot N_t\right)}{|F|}$, where $\ell(\theta)$ is the combinatorial list size bound from above.*

Proof. Given $\mathbf{m}_0 = \{f_{t,1}, \ldots, f_{t,M_t}\}$, and assume a (computationally possibly unbounded) algorithm which, with probability larger than ε in $(\beta_0, \beta_1) \leftarrow\$ F$, succeeds in serving a first-round ensemble

$$\mathbf{m}_1 = \{f_{t,M_t+1}, \ldots, f_{t,M_t+r_t+s_t+1}\}, \{\nu_t\}$$

as in (117) so that their concatenation of $\mathbf{m}_0$ and $\mathbf{m}_1$ belongs to $\mathcal{R}_1$. In other words, the joint ensemble has a θ-proximate solution to the domain identities (40), (113), (114), (115), while (116) holds. The latter domain identities are equivalent to the validity of the logUp identity spot-check (112) at $(X, Y) = (\beta_0, \beta_1)$. By the discussion at the beginning of this section, there are at most $\ell(\theta)$ such proximates. Thus by the pigeon-hole principle, at least one of them, say $\{p_{t,j}\}, \{\bar{u}_{t,j}, \bar{v}_{t,k}, \bar{w}_t\}$, are such that the $\{p_{t,j}\}$ satisfy the logUp identity (110) at more than

$$|F| \cdot r \cdot \left(k_{in} + k_{out} + \sum_{t=1}^{\tau}(r_t + s_t) \cdot N_t\right)$$

different $(\beta_0, \beta_1) \in F^2$. The degrees of the logUp identity are

$$\deg_X (110) \leq k_{in} + k_{out} + \sum_t (r_t + s_t) \cdot N_t,$$

$$\deg_Y (110) \leq r \cdot \left(k_{in} + k_{out} + \sum_t (r_t + s_t) \cdot N_t\right),$$

and hence we may conclude that (110) holds as a formal identity in $\mathbb{F}_q(X, Y)$. Moreover, since the flat constraints (40) are satisfied for every table, the proximate $\{p_{t,j}\}$ is as required by $\mathcal{R}_0$, proving that $\mathbf{m}_0 \in \mathcal{R}_0$. □

Remark 36. With domain separation via randomness, cf. Section 3.4, the individual degrees of the multivariate identities are smaller. This reduces the soundness error from Lemma 5, and it takes significantly less grinding than in the example parameters from Section 5.5, Table 5 and **??**.

Round 2. The cross-domain quotient is the linear combination of the table quotients

$$q_t = \frac{\mathsf{comp}_{t,\gamma}(p_{t,1}, \ldots, p_{t,M_t}, \bar{u}_{t,1}, \ldots, \bar{u}_{t,r_t}, \bar{v}_{t,1}, \ldots, \bar{v}_{t,s_k}, \bar{w}_t, \nu_t)}{v_{n_t}}, \tag{118}$$

each with its own composition polynomial

$$\begin{aligned}
&\mathsf{comp}_{t,\gamma}(p_{t,1}, \ldots, p_{t,M_t}, \bar{u}_{t,1}, \ldots, \bar{u}_{t,r_t}, \bar{v}_{t,1}, \ldots, \bar{v}_{t,s_t}, \bar{w}_t, \nu_t) \\
&\quad = \sum_{i=1}^{k_t} \gamma^{i-1} \cdot C_{t,i}(p_{t,1}, \ldots, p_{t,M_t}) \\
&\qquad + \sum_{j=1}^{r_t} \gamma^{k_t+j-1} \cdot (\bar{u}_{t,j} \cdot (\beta_0 + U_{t,j}(p_{t,1}, \ldots, p_{t,M}, \beta_1)) - e_t) \\
&\qquad + \sum_{k=1}^{s_t} \gamma^{k_t+r_t+k-1} \cdot (\bar{v}_{t,k} \cdot (\beta_0 + V_{t,k}(p_{t,1}, \ldots, p_{t,M}, \beta_1)) - m_{t,k}) \\
&\qquad + \gamma^{k_t+r_t+s_t} \cdot \left(T_{Q_{n_t}} \bar{w}_t - \bar{w}_t + \frac{\nu_t}{N_t} + \sum_{j=1}^{r_t} \bar{u}_{t,j} - \sum_{k=1}^{s_t} \bar{v}_{t,k} \right).
\end{aligned}$$

The prover commits to the values of

$$q = \sum_{t=1}^{\tau} \gamma^{\alpha_t} \cdot q_t \in \mathscr{L}'_{n+\beta}(F), \tag{119}$$

over the largest domain $D^* = G'_{n+2\cdot\beta}$. (Recall that the exponents e_t are chosen optimally, so that overall linear combination uses consecutive powers of γ.) We write the committed function as

$$f \in F^{D^*}, \tag{120}$$

append it to the already committed ensemble $\{f_{t,j}\}$, $\{\nu_t\}$ subject to

$$\mathcal{R}_2 = \left\{ \begin{matrix} \{f_{t,j}\} \\ \{\nu_t\} \\ f \end{matrix} : \begin{matrix} \exists \{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \\ \exists \{\bar{u}_{t,j}, \bar{v}_t, \bar{w}_t\} \in \mathscr{L}'_{n_1}(F)^{r_1+2} \times \cdots \times \mathscr{L}'_{n_\tau}(F)^{r_\tau+2} \\ \exists q \in \mathscr{L}'_{n+\beta}(F) \\ d\left(\{p^*_{t,j}, \bar{u}^*_{t,j}, \bar{v}^*_{t,k}, \bar{w}^*_t, q^*\}, \{f^*_{t,j}, f^*\} \right) < \theta, \\ \text{and the composition identity (119) holds with (118),} \\ \text{and (116)} \end{matrix} \right\}.$$

Again, the $\{f_{t,j}\}$ in the definition of $\mathcal{R}_2$ includes (107) and (117).

Lemma 6 (Soundness Round 2)**.** *Round 2 of Protocol 1 is a randomized reduction from $\mathcal{R}_1$ to $\mathcal{R}_2$ with soundess error $\varepsilon \leq \ell(\theta) \cdot \frac{\sum_{t=1}^{\tau} k_t + r_t + s_t}{|F|}$.*

Proof. Given $\{\mathbf{m}_0, \mathbf{m}_1\} = \{f_{t,j}\}, \{\nu_t\}$, and assume an algorithm which, with probability larger than ε in $\gamma \leftarrow\!\!\$\ F$, succeeds in serving a second-round function

$$\mathbf{m}_2 = f$$

as in (120) so that altogether $\{\mathbf{m}_0, \mathbf{m}_1, \mathbf{m}_2\}$ belongs to $\mathcal{R}_2$. Then $\{\mathbf{m}_0, \mathbf{m}_1\}$ has a θ-proximate ensemble of polynomials, for which

$$\sum_t \gamma^{\alpha_t} \frac{\mathsf{comp}_{t,\gamma}(p_{t,1}, \ldots, p_{t,M_t}, u_{t,1}, \ldots, u_{t,r_t}, v_{t,1}, \ldots, v_{t,s_t}, w_t, \nu_t)}{v_{n_t}} \tag{121}$$

is a polynomial $q = q_\gamma$ from $\mathscr{L}'_{n+\beta}(F)$. There are at most $\ell(\theta)$ such proximates, and by the pigeon-hole principle, at least one of them, say $\{p_{t,j}\}$, $\{\bar{u}_{t,j}, \bar{v}_{t,k}, \bar{w}_t\}$, the linear combination (121) belongs to $\mathscr{L}'_{n+\beta}(F)$ for more than

$$d_\gamma = \sum_{t=1}^{\tau} k_t + r_t + s_t$$

different $\gamma \in F$, where d_γ is the degree of (121) in γ. Use the Lagrange interpolation matrix for this set of γ's to argue that each fraction within each of the $_{t,\gamma}$ in (121) is a linear combination of the polynomials from $\mathscr{L}'_{n+\beta}$, and hence is divisible by the respective domain vanishing polynomial. This proves that for each $t = 1, \ldots, \tau$ the constraints (40), (113), (114), (115), hold over the respective domain H_t. In other words, the θ-proximate ensemble of polynomials attests membership of $\{\mathbf{m}_0, \mathbf{m}_1\}$ to $\mathcal{R}_1$. □

Round 3. This round is what is called the *algebraic linking step* in [BGKS20]. It reduces membership to $\mathcal{R}_2$, subject to the composite table identities (118), to the point-check relation at the out-of-domain query $Q \leftarrow\!\!\$\ C(F) \setminus C(\mathbb{F}_q)$,

$$\mathcal{R}_3 = \left\{ \begin{array}{c} \{f_{t,j}\}, \\ \{\nu_t\}, \\ f, \\ \{v_{t,j}, v\} \end{array} : \begin{array}{c} \exists\{p_{t,j}\} \in \mathscr{L}'_{n_1}(\mathbb{F}_q)^{M_1} \times \cdots \times \mathscr{L}'_{n_\tau}(\mathbb{F}_q)^{M_\tau} \\ \exists\{\bar{u}_{t,j}, \bar{v}_{t,k}, \bar{w}_t\} \in \mathscr{L}'_{n_1}(F)^{r_1+s_1+1} \times \cdots \times \mathscr{L}'_{n_\tau}(F)^{r_\tau+s_\tau+1} \\ \exists q \in \mathscr{L}'_{n+\beta}(F) \\ d\left(\{p^*_{t,j}, \bar{u}^*_{t,j}, \bar{v}^*_{t,k}, \bar{w}^*_t, q^*\}, \{f^*_{t,j}, f^*\}\right) < \theta, \\ \text{and the composition identity (119) with (118) holds at } Q, \\ \text{and (116) holds, and} \\ \{p_{t,j}(Q), \bar{u}_{t,j}(Q), \bar{v}_{t,k}(Q), \bar{w}_t(Q), \bar{w}_t(Q_{n_t+1} \cdot Q), q(Q)\} = \{v_{t,j}, v\} \end{array} \right\},$$

which now embraces also the third prover message, the evaluation claims $\{v_{t,j}, v\}$ for the polynomials at the requested points.

Lemma 7 (Soundness Round 3)**.** *Round 3 of Protocol 1 is a randomized reduction from $\mathcal{R}_2$ to $\mathcal{R}_3$ with soundness error $\varepsilon \leq \ell(\theta) \cdot \frac{1+(B+2)\cdot N}{|C(F) \setminus C(\mathbb{F}_q)|}$.*

Proof. As in the other rounds. The list size bound $\ell(\theta)$ together with the pigeonhole principle assure that, if $\{\mathbf{m}_0, \mathbf{m}_1, \mathbf{m}_2, \mathbf{m}_3\}$ belongs to $\mathcal{R}_3$ for more than $\ell(\theta) \cdot (1 + (B+2) \cdot N)$ different points Q, then at least one of the θ-proximates, say $\{p_{t,j}, \bar{u}_{t,j}, \bar{v}_t, \bar{w}_t, q\}$, satisfies

$$q \cdot \prod_k v_k = \sum_{t=1}^{\tau} \gamma^{\alpha_t} \cdot \mathsf{comp}^{(t)}_{t,\gamma}(p_{t,1}, \ldots, p_{t,M_t}, \bar{u}_{t,1}, \ldots, \bar{u}_{t,r_t}, \ldots) \cdot \prod_{k \neq n_t} v_k, \quad (122)$$

at more than

$$(1 + (B+2) \cdot N)$$

different points. (Identity (122) is (118) multiplied by the product of all vanishing polynomials over all used domains, which belongs to $\mathscr{L}_{n+1}(\mathbb{F}_q)$.) Both sides in (122) are circle polynomials of total degree at most $(B+2) \cdot N/2$, and we conclude that (122), and with it the composition identity (121) holds as a formal identity, showing that $\{\mathbf{m}_0, \mathbf{m}_1, \mathbf{m}_2\}$ belongs to $\mathcal{R}_2$. □

A.5 Beyond the Johnson radius?

Whether the soundness analysis can be extended to an even lazier testing regime, that is, with distances beyond the Johnson radius, is tied to the following properties:

1. Correlated agreement with acceptable soundness error, and in all the facets as required by our analysis of FRI. That is, with and without constraints, set-specific and weighted. A good notion that underpins all these variants is curve-decodability [GG25], or in a more elementary form *line-decodability*, as we explained in Section A.2.
2. Combinatorial list-decodability with small list sizes, for arguing the security of the protocol on top of FRI. That is, the number of θ-close polynomials at given proximity parameter θ should be asymptotically constant, i.e. independent of the domain size, and the constants should be reasonably small.

Correlated agreement and list-decodability are closely related, and in fact Crites and Stewart [CS25] and [BCH+25] show that the former always implies the latter. We cite the theorem of [CS25], which explicitly states the connection between soundness error and list sizes, with slightly modified bounds.

Theorem ([CS25])**.** *Assume that $\mathcal{C} = \mathsf{RS}(\mathbb{F}_q, D, k)$ satisfies correlated agreement (over lines) with soundness error $\varepsilon \leq \frac{1}{\gamma \cdot k}$ for some $\gamma > 1$, then $\mathcal{C}^+ = \mathsf{RS}(\mathbb{F}_q, D, k+1)$ is list-decodable at θ, with list sizes*

$$\ell(\theta) \leq \frac{\gamma}{\gamma - 1} \cdot \varepsilon q.$$

The main question is:

Can we expect these properties to hold beyond the Johnson radius?

Let us first explore their natural, information-theoretic limit.

List-decoding capacity. Over a noisy channel with error rate θ, the limit of list-decodability is where the rate of the code reaches the *q-ary capacity*

$$C_q = 1 - H_q(\theta), \tag{123}$$

with[14]

$$H_q(\theta) = \theta \log_q(q-1) - \theta \log_q(\theta) - (1-\theta)\log_q(1-\theta). \tag{124}$$

In our context, we look at it the reversed way. Given ρ the rate of the code, we ask for the maximum possible distance $\theta = r_E = r_E(\rho)$ so that

$$\rho = 1 - H_q(r_E),$$

and we call this radius the *Elias radius.* This information theoretic limit is known since the introduction of list decoding by Elias and Wozencraft [Eli57,Woz58], yet only lately entered the discussion of proximity gaps, foremost with Crites and Stewart in [CS25], and implicit in the counter examples of Diamond and Gruen [DG25]. The Elias radius lies within the interval

$$1 - \rho - \frac{1}{\log_2 q} \leq r_E < 1 - \rho,$$

which does not matter much for large fields. However, beyond that limit the list sizes grow exponentially in $n \cdot (\theta - r_E)$, making the Elias radius over the *characteristic* of the field the essential limit in practical applications [CS25], see also Fenzi and Sanso[FS25].

Subspace polynomials. Over binary fields, the main source of counter examples are subspace polynomials, which are sparse polynomials with their zeros guaranteed in the field. The examples from [BKR10] and their refinement in [BCH$^+$25, Section 5] show that in the case $q = 2^m$, Reed–Solomon codes can have bad list–decoding and correlated agreement properties outside the Johnson radius, way before list-decoding capacity. This basically rules out any distance beyond Johnson radius, even though these counter examples imply existential, yet not practical attacks.

. . . and prime fields. In prime fields however, the situation is much less clear. All known counter examples, including the most significant one from Krachun and Kazanin [KK25], seem only be able to tackle the issue near capacity,[15] and the reason for this might very well be the absence of (non-trivial) additive subspaces. In other words, there could be something special about prime fields that enables good list and curve-decodability properties beyond the Johnson radius. This hope is bolstered by the recent progress on random codes, which shows that *most* codes in fact are list- and curve-decodable up to list-decoding capacity. Let us quickly summarize these remarkable results.

[14] This is the q-ary entropy function. It measures the information rate (in terms of $\mathbb{F}_q$-symbols) of perfect random noise, which flips a symbol with probability θ, to any other symbol with equal probability.

[15] In the $\delta = \Omega(1)$ regime.

List-decodability of random Reed–Solomon codes. A modern line of works [RW13,ST20,GLS$^+$24,BGM23] and [GZ23,AGGZ25] shows that Reed–Solomon codes with a randomly sampled evaluation domains are list-decodable up to list-decoding capacity. We cite the most recent result, from Alrabiah, Gua, Guruswami and Zhang, who prove this fact under asymptotically moderate assumptions on the field size.

Theorem ([AGGZ25]). *Let $\eta \in (0,1)$ and q be a prime power such that $q \geq n + k2^{\frac{10}{\eta^2}}$. Then with probability at least $1 - 2^{-\frac{n}{\eta}}$, a randomly punctured Reed–Solomon code of block length n and rate $\rho = k/n$ over $\mathbb{F}_q$ has the following list-decoding property at $\theta = 1 - \rho - \frac{3}{2}\eta$.*

$$\ell(\theta) = |\{c \in \mathcal{C} \,:\, d(c, f) \leq \theta\}| \leq \frac{1-\rho}{\eta}$$

We note that in the theorem, the list-capacity condition $\rho < C_q$ is hidden in the choice of q.

Curve-decodability of random Reed–Solomon codes. Recently, Goyal and Guruswami [GG25] introduced the notion of curve-decodability (see Section A.2), and showed that random Reed-Solomon codes are curve-decodable up to list-decoding capacity, with relatively small threshold parameter.

Theorem ([GG25], Lemma 5.11). *For $\rho, \eta \in (0,1)$, $M \geq 1$, large enough n, and prime power[16] $q = O\left(n \cdot 2^{1/\eta^2}\right)$. Then with high probability a randomly punctured Reed–Solomon code $\mathcal{C} \subseteq \mathbb{F}_q^n$ of rate ρ is degree-M curve-decodable at distance $\theta = 1 - \rho - 2\eta$, with threshold*

$$a = O\left(\frac{1}{\eta^{2M+2}}\right) \cdot b,$$

for any concurrency number $b \geq 1$.

They moreover carve out the connection between list decodability and curve decodability, with the list sizes dominating the threshold for large concurrency numbers $b = \Omega(n)$.

Lemma 8 (Optimal curve-decodability for large concurrency numbers [GG25]). *For $\rho, \eta \in (0,1)$, $M \geq 1$, large enough n, and prime power $q = O\left(n \cdot 2^{1/\eta^2}\right)$. Then with probability at least $\frac{2}{3}$, a randomly punctured Reed–Solomon code $\mathcal{C} \subseteq \mathbb{F}_q^n$ of rate ρ is degree-M curve-decodable at distance $\theta = 1 - \rho - 2\eta$, with threshold*

$$a = b \cdot \frac{(1-\rho)}{\eta} + O\left(\frac{M^{M+2}}{\eta^{2M+3}}\right)$$

for any concurrency number $b \geq 1$.

[16] Again, the field size assures that θ is consistent with the limiting list-size capacity condition.

The proof of Lemma 8 is essentially that of [GG25, Theorem 3.4], which frames the statement in the context of *mutual correlated agreement* [ACFY24b], a 'global' agreement notion which is similarly useful for the analysis of protocols, in which the proximate polynomials are subject to constraints. As the dominance of list sizes in curve-decodability plays a role in our conjectures, we outline the proof.

Proof. We use curve-decodability with small threshold to bootstrap correlated agreement with distance loss η, and then use the list-size bound at distance $\theta+\eta$ to argue curve-decodability with the claimed larger threshold. For simplicity we restrict to $M = 1$. The general case does not bear any additional difficulties.

Combining the two above cited main results from [GG25] and [AGGZ25, Lemma 5.11] we obtain the following for a randomly punctured Reed–Solomon code $\mathcal{C}$. With large probability it is curve-decodable at distance $\theta + \eta$ with

$$a' = O\left(\frac{1}{\eta^4}\right) \cdot b, \quad b' = \left\lceil \frac{1}{\eta} \right\rceil,$$

and also list-decodable at distance $\theta + \eta$, with list size

$$\ell(\theta + \eta) \leq \frac{3}{2} \cdot \frac{1-\rho}{\eta}.$$

Now, assume any functions f_0, f_1 on the domain of the code, and a collection of codewords $\{p_z\}_{z\in S}$ with $d(f_0 + zf_1, p_z) \leq \theta$, and of size $|S| > a$, with a as claimed. We want to conclude that more than b of these proximate codewords p_z lie on a line in $\mathcal{C}$.

1. Since $a > a'$, the first property ensures that more than b' of the proximates lie on line in $\mathcal{C}$, spanned by some codewords P_0, P_1. This implies that

$$d((f_0, f_1), (P_0, P_1)) \leq \frac{b'+1}{b'}\,\theta \leq \theta + \eta,$$

   showing that (f_0, f_1) is $(\theta + \eta)$-close to the interleaved code $\mathcal{C}^2$.
2. By the second property, also $\mathcal{C}^2$ is list-decodable with the same list-size bound, and there are at most $\ell(\theta+\eta)$ words $(P_0^{(i)}, P_1^{(i)})$ from $\mathcal{C}^2$ with distance bounded by $\theta + \eta$. Each of them spans a line in $\mathcal{C}$.

Let $S' \subset S$ correspond to those proximates which do not lie on *any* of these lines spanned by the $(\theta + \eta)$-close words from $\mathcal{C}^2$. Note that $|S'| \leq a'$, because otherwise, we could again apply the first property and obtain a new codeword $(P_0', P_1') \in \mathcal{C}^2$ which is $(\theta + \eta)$-close to (f_0, f_1), but differs from all previous $(P_0^{(i)}, P_1^{(i)})$, contradicting the list size. Removing the few exceptions belonging to S', and applying the pigeon-hole principle on remaining proximates from $S \setminus S'$, yields the claim of the lemma. □

The counter example of Krachun and Kazanin. Most counter examples for prime fields [DG25,CS25] and [BCH⁺25] reflect the list-size growth when passing the information-theoretic limit, and hence apply only to the small interval between

$$r_E \leq \theta < 1 - \rho.$$

The only example that reaches *below* the Elias radius[17]is the recent one from Krachun and Kazanin [KK25], which shows that, when approaching list-decoding capacity, one cannot expect the same list-decoding and line-decoding properties as random Reed–Solomon codes have. For simplicity, we state their result for parameters which are a power of two. There, and in the following, $H(x) = -x \log x - (1-x)\log(1-x)$ is the binary entropy.

Theorem 37 ([KK25]). *For rate $\rho = 2^{-m}$, $m \geq 2$, integer $\tau = 2^t$, $t \geq 0$, and $\beta \geq \frac{\tau}{\rho} \geq 4$. Then for any sufficiently large block length $n = 2^{2^b}$ there is a prime $p \in [n^\beta, n^{2\beta}]$, with $\mathbb{F}_p$ containing a multiplicative subgroup H of order n, and such that the following holds for $\mathcal{C} = \mathsf{RS}[\mathbb{F}_p, H, \rho n + 1]$ at distance*

$$\theta = 1 - \rho - \eta, \quad \textit{with} \quad \eta = \frac{\rho}{\tau \log n}.$$

1. *Proximity gaps with distance loss smaller than $\theta^* = \frac{\rho}{\tau \log n}$ require more than*

$$a = 2^{H(2\rho)/(2\eta)} > n^\tau$$

linear combinations being θ-close to the code.

2. *There exists a function $f \in \mathbb{F}_p^H$ with*

$$\ell(\theta) \geq 2^{H(2\rho)/(2\eta)} > n^\tau$$

codewords which are θ-close to f.

Remark 38. By the lower bound on β, we have

$$\frac{1}{\log p} \leq \frac{1}{\beta \log n} < \frac{1}{2}\frac{\rho}{\tau \log n} = \frac{\eta}{2},$$

which shows that the distance θ is clearly below the Elias radius $r_E(\rho)$. The gap to Elias radius is at least $\eta/2$, and it approaches η the larger we choose β. Second, for the rates in question we have $H(2\rho) \leq 2H(\rho)$, which we will use to formulate the concrete bounds in our conjectures.

The construction behind Theorem 37 is as in [BCH⁺25, Section 7]. It considers the vanishing polynomials of subsets $S = \{x_1, \ldots, x_r\}$,

$$v_S(X) = \prod_{x \in S}(X - x) = X^r + (x_1 + \ldots + x_r) \cdot X^{r-1} + O(X^{r-2}),$$

[17] For rates $\rho = \Omega(1)$. The examples from [JH01] and [BCI⁺20, Section 7] do not fall in this regime.

where the points are taken from a small multiplicative subgroup

$$G \ll H,$$

of logarithmic size $|G| = m = \frac{\tau}{\rho} \cdot \log n$. If there are many distinct sums $\lambda = x_1 + \ldots + x_r$, then the line spanned by X^r and X^{r-1} has many polynomials of degree $\leq r-2$ with larger agreement than X^{r-1} admits.[18]The example is then lifted to the larger subgroup H by considering

$$v_S(X^{n/m}).$$

The main difference to [BCH+25] is that Krachun and Kazanin do not rely on any conjecture on the number of distinct sums.

Lemma 9 ([KK25]). *Let G be a multiplicative subgroup of a prime field $\mathbb{F}_p$, of size $|G| = m$. Assume that $p > \varphi(m)^{\varphi(m)}$, where φ is the Euler totient function. Then for any integer $1 \leq r \leq \varphi(m)/2$,*

$$|\{x_1 + \ldots + x_r \,:\, x_1, \ldots, x_r \in G \text{ distinct }\}| \geq \binom{\varphi(m)}{r}.$$

The proof of the lemma uses the fact that the first $\varphi(m) - 1$ powers of a primitive m-th root of unity are linearly independent over the integers, and elaborates that this does not change modulo the large prime, at least not if the coefficients are from $\{-1, 0, +1\}$. (The concrete bound for p stems from bounding the absolute value of the resultant of the m-th cyclotomic polynomial $\Phi_m(X)$ with a polynomial that expresses a collision of two r-sums.)

Note that Lemma 9 already implies the claimed properties, if one does not care much about the size of the prime: In the construction, take any prime $p > \varphi(m)^{\varphi(m)}$ satisfying

$$p = 1 \bmod n,$$

for our chosen $n = 2^{2^b}$ and $m = \frac{\tau}{\rho} \log n = \tau 2^b$, all powers of two.[19] We then have $\varphi(m) = m/2$, and

$$\binom{\varphi(m)}{r} = 2^{(H(2\rho)+o(1))\cdot m/2},$$

[18] The construction greatly simplifies if one is only interested in the list sizes. All r-subsets $S \subset G$ yield different vanishing polynomials, and the concrete form of the second-highest order coefficient does not play a role. In this case, the polynomial $X^{r \cdot n/m}$ has more than

$$\ell(\theta) \geq \binom{m}{r} = 2^{(H(\rho)+o(1))\cdot m} = 2^{(H(\rho)+o(1))/\eta}$$

polynomials of degree at most $(r-1) \cdot n/m$, which agree on a set of density r/m, also for rate $\rho = 1/2$. This is a larger bound as we stated in the theorem.

[19] By Dirichlet's theorem, any arithmetic progression contains infinitely many primes.

using the well-known entropy approximation for the binomial coefficient. This yields the bounds claimed in the theorem, for a prime of exponential order in n. For achieving the poly n assumption, Krachun and Kazanin refine the argument the argument behind Lemma 9.

Two plausible conjectures. Motivated by the above results on random codes, and with [KK25] in mind, we conjecture that over *prime fields* $\mathbb{F}_p$ Reed–Solomon codes have acceptable list- and curve-decodability properties up to the information theoretic limit, the *Elias radius* $r_E = r_E(\rho)$, where

$$\rho = 1 - H_p(r_E)$$

holds. Recall that

$$1 - \rho - \frac{1}{\log_2 p} \leq r_E < 1 - \rho,$$

and the lower bound of the interval is a quite good approximation for not too small rates. Although list- and curve-decodability are closely related, we split our expectations into two conjectures.

List-decodability up to list-capacity.

Conjecture 1. *For any $\rho \in (0, 1)$, there exist relatively small constants $c_1, c_2 \geq 1$ such that every Reed–Solomon code defined over a prime field $\mathbb{F}_p$ with rate ρ, is (combinatorial) list-decodable up to the Elias radius $r_E = r_E(\rho)$, with list sizes*

$$\ell(\theta) \leq c_1 \cdot 2^{c_2 \cdot H(\rho)/\eta},$$

at distance $\theta = 1 - \rho - \eta$, and $H(\rho)$ is the binary entropy function.

By relatively small constants, we expect $c_1 = c_2 = O(1)$ if the rate ρ stays away from zero (or one). To take the example of Krachun and Kazanin into account, it suffices to choose $c_1 = c_2 = 1$, and we use these to determine the security parameters in Section 5.5. However, we stress the fact that the

Line-decodability up to list-capacity. The second conjecture guarantees correlated agreement up to the Elias radius. For simplicity, we restrict ourselves to *line-decodability*, which is degree-M curve-decodability as in Definition 24, with $M = 1$.

Conjecture 2. *Any Reed–Solomon code with domain of definition D in a prime field $\mathbb{F}_p$, is line-decodable up to the Elias radius $r_E = r_E(\rho)$, with threshold*

$$a = \ell(\theta) \cdot n + o(n),$$

for concurrency number $n = |D|$, and where $\ell(\theta)$ is as in Conjecture 1. The same is true if the alphabet of the code is from an extension field of $\mathbb{F}_p$.

We note that the specific form of the threshold a is motivated by the above discussed technique from Goyal and Guruswami, Lemma 8, which shows that for large concurrency numbers, the list sizes essentially determine the threshold. For our example parameters in Section 5.5 we neglected the little-oh term.

Why line-decodability? Our pivot for line-decodability over curve-decodability is deliberate. The line case captures already the core difficulty of the problem, yet it is flexible enough to serve essentially everything that could be wished for: In practice, higher degree combinations can be throughout replaced by *multilinear* ones, at minor implementation costs, and correlated agreement for multilinear combinations can be derived from the line case in a black-box manner. (The example parameters in Section 5.5 rely on this replacement.)

As we explained in Section A.2, line-decodability implies proximity gaps in all facets as needed in our security analysis of multi-table FRI. Set-specific correlated agreement (Theorem 28), weighted correlated agreement (Theorem 29) as well as correlated agreement under linear constraints (Theorem 31), they all hold with a soundness error

$$\varepsilon \leq \ell(\theta) \cdot \frac{n}{|F|} + o\left(\frac{n}{|F|}\right),$$

for $M = 1$, and by a factor $\log_2 M$ larger in their multilinear variants.

For similar, if not the same reasons, line-decodability can be used to underpin the security of a wide range of other low-degree tests, considering the above adaption to multilinear combinations: Basefold [ZCF23], WHIR [ACFY24b] (as demonstrated in [Hab24a]), STIR [ACFY24a], Blaze [BCF$^+$24] and Tensor-Switch [BFRW25], to give some examples. We also point out, that the property implies (the line case of) *mutual correlated agreement* [ACFY24b], an agreement notion preferred by some cryptographers for its simplicity. We refer to [Hab25] and [GG25] for details.

B Circle FFT algorithms

B.1 Bit-reverse order on twin-cosets.

In this section we describe the bit-reverse order on canonical cosets G'_n, $n = 1, \ldots, n_{max}$, the group structured domains of the circle FFT. The canonical cosets G'_n form a chain of domains under repeated application of the group squaring map π, and so do their projections onto the x-axis, $S_{n-1} = \pi_x(G'_n)$, the line domains of half the size.

$$\begin{tikzcd}
G'_{n_{max}} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & G'_{n_{max}-1} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & G'_{n_{max}-2} \arrow[r, "\pi"] \arrow[d, "\pi_x"] & \cdots\cdots \arrow[r, "\pi"] & G'_1 \arrow[d, "\pi_x"] \\
S_{n_{max}-1} \arrow[r, "\pi"] & S_{n_{max}-2} \arrow[r, "\pi"] & S_{n_{max}-3} \arrow[r, "\pi"] & \cdots\cdots \arrow[r, "\pi"] & S_0
\end{tikzcd}$$

We assume a square-mapping-consistent choice of $Q_n \in C(\mathbb{F}_q)$ the generators of the cyclic subgroups G_n of size 2^n, see Section 1.2, and we consider each canonical coset G'_n as the twin-coset

$$G'_n = Q_{n+1} \cdot G_n = Q_{n+1} \cdot G_{n-1} \cup Q_{n+1}^{-1} \cdot G_{n-1},$$

where the two half-sized cosets are disjoint, $Q_{n+1} \cdot G_{n-1} \cap Q_{n+1}^{-1} \cdot G_{n-1} = \varnothing$, since $Q_{n+1}^2 = Q_n$ is not contained in G_{n-1}. Both half-sized cosets are mapped to one another under the inverse map $J(x, y) = (x, -y)$, and both are mapped by π_x onto S_{n-1} in a one-to-one manner.

The *bit-reverse order on G_n'* corresponds to the bit-reverse order on each of the half-sized cosets, chosen in a J-consistent manner. That is, for $(i_1, \ldots, i_n) \in \{0,1\}^n$, we set

$$\begin{aligned} Q'(i_1, i_2, \ldots, i_n) &= \left(Q_{n+1} \cdot Q_{n-1}^{i_n + \ldots + i_3 \cdot 2^{n-3} + i_2 2^{n-2}}\right)^{1-2i_1} \\ &= \left(Q_{n+1} \cdot Q_{n-1}^{i_n} \cdot \ldots \cdot Q_2^{i_3} \cdot Q_2^{i_1}\right)^{1-2i_1}. \end{aligned} \tag{125}$$

The least significant bit i_1 chooses between the two cosets $Q_{n+1} \cdot G_{n-1}$ and $Q_{n+1}^{-1} \cdot G_{n-1}$, and the remaining bits $i_2, \ldots, i_n$ correspond to the bit-reverse of their natural cyclic enumeration defined by their respective generator $Q_{n-1}^{\pm 1}$.

Note that by definition,

$$J(Q'(i_1, i_2, \ldots, i_n)) = Q'(1 - i_1, i_2 \ldots, i_n). \tag{126}$$

and since the generators are chosen π-consistent,

$$\begin{aligned} \pi(Q'(i_1, i_2, \ldots, i_n)) &= \left(Q_{n+1} \cdot Q_{n-1}^{i_n} \cdot \ldots \cdot Q_2^{i_3} \cdot Q_1^{i_2}\right)^{(1-2i_1)\cdot 2} \\ &= \left(Q_n \cdot Q_{n-2}^{i_n} \cdot \ldots \cdot Q_1^{i_3} \cdot Q_0^{i_2}\right)^{(1-2i_1)} = Q'(i_1, i_3, \ldots, i_n), \end{aligned} \tag{127}$$

since $Q_0 = (1, 0)$ is the group unit. Over the line domain S_{n-1} we define the bit-reverse order on by taking the x-coordinate

$$x'(i_2, \ldots, i_n) = Q'(0, i_2, \ldots, i_n)_x = Q'(1, i_2, \ldots, i_n)_x.$$

The in-place FFT relies on the following two orthogonal partitions of G_n': The sets

$$U_{i_{k+1}, \ldots, i_n} = \left\{Q'(i_1, \ldots, i_k, i_{k+1}, \ldots, i_n) \,:\, (i_1, \ldots, i_k) \in \{0,1\}^n\right\},$$

for $(i_{k+1}, \ldots, i_n) \in \{0,1\}^{n-k}$ partition G_n' into the fibers of the chained projection map $\pi^{k-1} \circ \pi_x : G_n' \longrightarrow S_{n-k}$, i.e. the preimages of $x'(i_{k+1}, \ldots, i_n) \in S_{n-k}$. On the other hand, the sets

$$V_{i_1, \ldots, i_k} = \left\{Q'(i_1, \ldots, i_k, i_{k+1}, \ldots, i_n) \,:\, (i_{k+1}, \ldots, i_n) \in \{0,1\}^n\right\},$$

for $(i_1, \ldots, i_k) \in \{0,1\}^k$, are a partition of G_n' into sections of the chained projection $\pi^{k-1} \circ \pi_x : G_n' \longrightarrow S_{n-k}$, i.e. sets which are mapped onto S_{n-k} in a one-to-one and onto manner. The latter partition is used to host the outputs of the k-th layer of the FFT, which are a result of the initial values over the complementary partition $U_{i_{k+1}, \ldots, i_n}$, with $(i_{k+1}, \ldots, i_n)$ ranging over $\{0,1\}^{n-k}$, the fibers of the chained projection map.

B.2 Evaluation and interpolation FFT

We arrange the twiddle factors of the entire chain of domains $G'_n \xrightarrow{\pi_x} S_{n-1} \xrightarrow{\pi} S_{n-2} \xrightarrow{\pi} \ldots \xrightarrow{\pi} S_0$ in a single vector $t \in \mathbb{F}_q^{2^n}$, via

$$t_{[i_1,\ldots,i_n]} = \begin{cases} Q'(0, i_2, \ldots, i_n)_y & i_1 = 0, \\ Q'(0, i_{k+1}, \ldots i_n)_x & (i_1, \ldots, i_k) = (1, \ldots, 1). \end{cases} \tag{128}$$

The evaluation FFT over $F \geq \mathbb{F}_q$ has given the twiddle factors $t \in \mathbb{F}_q^{2^n}$ and the coefficients $(c_0, \ldots, c_{2^n-1}) \in F^{2^n}$ of a circle polynomial $p(x, y)$ with respect to the circle FFT basis $\mathscr{B}_n = \{b_i(x, y)\}$. That is,

$$p(x, y) = \sum_{i=0}^{2^n-1} c_i \cdot b_i(x, y),$$

where $b_i(x, y) = y^{i_1} \cdot v_1(x)^{i_2} \cdot v_2(x)^{i_3} \cdot \ldots \cdot v_{n-1}(x)^{i_n}$. It iteratively modifies the vector c "in-place", and eventually ends up with the values of p over G'_n in bit-reverse order, that is

$$v_{[i_1,\ldots,i_n]} = p(Q'(i_1, \ldots, i_n)).$$

See Algorithm 2.

Algorithm 2 Evaluation FFT, with precomputed twiddles $t \in \mathbb{F}_q^{2^n}$ as above. Given vector $c \in F^{2^n}$ the coefficients of a circle polynomial $p(x, y)$ w.r.t. the FFT basis $\mathscr{B}_n$. Outputs $(v_0, \ldots, v_{2^n-1}) \in F^n$, the values of p over G'_n in bit-reverse order.

---

function FFT$((c_0, \ldots, c_{2^n-1}), (t_0, \ldots, t_{2^n-1}))$
    for $k = n, \ldots, 1$ **do**
        for $(i_1, \ldots, i_{k-1}, i_{k+1}, \ldots, i_n) \in \{0, 1\}^{n-1}$ **do**
            $(w_0, w_1) \leftarrow (c_{[i_1,\ldots,i_{k-1},0,i_{k+1},\ldots,i_n]}, c_{[i_1,\ldots,i_{k-1},1,i_{k+1},\ldots,i_n]})$
            $w_1 \leftarrow w_1 \cdot t_{[1,\ldots,1,0,i_{k+1},\ldots,i_n]}$
            $(w_0, w_1) \leftarrow (w_0 + w_1, w_0 - w_1)$
            $(c_{[i_1,\ldots,i_{k-1},0,i_{k+1},\ldots,i_n]}, c_{[i_1,\ldots,i_{k-1},1,i_{k+1},\ldots,i_n]}) \leftarrow (w_0, w_1)$
    return $(c_0, \ldots, c_{2^n-1})$

The interpolation FFT over $F \geq \mathbb{F}_q$ has given the inverse twiddle factors $t' \in \mathbb{F}_q^{2^n}$, $t'_{[i_1,\ldots,i_n]} = t^{-1}_{[i_1,\ldots,i_n]}$, and the values $v \in F^{2^n}$ of a function $f : G'_n \longrightarrow F$ in bit-reverse order, that is

$$v_{[i_1,\ldots,i_n]} = f(Q'(i_1, \ldots, i_n)),$$

for every $(i_1, \ldots i_n) \in \{0, 1\}^n$. It computes

$$c = (c_0, \ldots, c_{2^n-1}) \in F^{2^n},$$

the *non-normalized* coefficients of the circle polynomial $p(x, y) \in \mathscr{L}'_n(F)$ with $p(Q'(i_1, \dots, i_n)) = v_{[i_1,\dots,i_n]}$. That is,

$$p(x, y) = 2^{-n} \cdot \sum_{i=0}^{2^n-1} c_i \cdot b_i(x, y). \tag{129}$$

See Algorithm 3. Normalization of the coefficients can be avoided if the application allows a corresponding modification, or can be integrated into the last iteration of the FFT.

Algorithm 3 Scaled interpolation FFT over $F \geq \mathbb{F}_q$, with precomputed inverse twiddles $t' \in \mathbb{F}_q^{2^n}$ as above. Given $v \in F^{2^n}$ the values of a function $f : G'_n \to F$, $v_{[i_1,\dots,i_n]} = f(Q'(i_1, \dots, i_n))$. Outputs $(c_0, \dots, c_{2^n-1}) \in F^n$, the non-normalized coefficients of the interpolating circle polynomial.

function IFFT$((v_0, \dots, v_{2^n-1}), (t'_0, \dots, t'_{2^n-1}))$
  for $k = 1, \dots, n$ **do**
    for $(i_1, \dots, i_{k-1}, i_{k+1}, \dots, i_n) \in \{0, 1\}^{n-1}$ **do**
      $(w_0, w_1) \leftarrow (v_{[i_1,\dots,i_{k-1},0,i_{k+1},\dots,i_n]}, v_{[i_1,\dots,i_{k-1},1,i_{k+1},\dots,i_n]})$
      $(w_0, w_1) \leftarrow \left(w_1 + w_0, t'_{[1,\dots,1,0,i_{k+1},\dots,i_n]} \cdot (w_1 - w_0)\right)$
      $(v_{[i_1,\dots,i_{k-1},0,i_{k+1},\dots,i_n]}, v_{[i_1,\dots,i_{k-1},1,i_{k+1},\dots,i_n]}) \leftarrow (w_0, w_1)$
  return $(v_0, \dots, v_{2^n-1})$

Remark 39. A tweak of Algorithm 2 allows for efficient evaluation of polynomials from $\mathscr{L}'_n(F)$ over a larger domain $G'_{n+\beta}$. One simply starts the algorithm at $k = n$ (instead of $k = n + \beta$) on the 2^β-fold concatenation of the coefficient vector,

$$(c_0, \dots, c_{2^n-1} \| \dots \| c_0, \dots, c_{2^n-1}).$$

The latter corresponds to the output after the first β layers on the zero-padded coefficient vector.

Remark 40 (Implementation)**.** The choice for reverse-bit order in S-two is due to cache-friendliness of Reed-Solomon encoding, which is performed on each trace column separately. Interpolation is the first operation, and memory locality of the pairs to be modified is crucial for performance, which is given in the first layers of Algorithm 3. To maintain locality in all layers of the FFT, we transpose the data array after the first half layers: Upper and lower half of bit indices are swapped, changing the order of the output. Output normalization by the domain size is done via an extra layer of multiplications. The subsequent evaluation FFTs operate on the transposed array for the first half of layers, again acting locally, then revert the transposition, and continue the remaining layers in the regular order.